$D = \prod(z - r_i)$ and $N(z) = \sum_i w_i z \prod_{j \neq i}(z - r_j)$, as in (4); at a retained $r_i$ the denominator equals $-\kappa N(r_i) = -\kappa w_i r_i \prod_{j \neq i}(r_i - r_j) \neq 0$ (memory rates are positive because $\eta_i < 1$), so no root of $D$ cancels. The numerator $\kappa z D(z)$ has no roots other than 0 and the $r_i$, so these are the only possible cancellations. The remaining numerator factor $z$ cancels if and only if the denominator vanishes at 0; since $N(0) = 0$ the denominator there equals $-aD(0)$, which is nonzero for $a > 0$ and zero for $a = 0$. Hence the McMillan degree, which is the dimension of a minimal realization, is $1 + |\mathrm{supp}(M) \cap (0, 1)|$ for $a' \neq 0$ and $|\mathrm{supp}(M) \cap (0, 1)|$ for $a' = 0$ (an atom at rate 0, a cohort with $\eta = 1$, contributes the constant $M(\{0\})$ to $G$ and no pole, so it shifts $a$ to $a' = a + \kappa M(\{0\})$ and the argument above applies with $a'$) (numerically, the rank of the Hankel matrix of $(h_j)_{j \geq 1}$ is $K + 1$ and $K$ respectively for $K = 1, 2, 3$). For (ii), direct algebra gives

$$G_\delta(z) - G_0(z) = -\frac{qz(z-1)\delta^2}{(z - 1 + \bar{\eta})\Big((z - 1 + \bar{\eta})^2 - \delta^2\Big)}.$$

The denominator is the product of the three linear factors $z - r$ with $r \in \{1 - \bar{\eta}, 1 - \bar{\eta} \pm \delta\}$, all in $[\epsilon, 1 - \epsilon]$ by assumption; on $|z| = 1$ each satisfies $|z - r| \geq 1 - r \geq \epsilon$, so $|G_\delta - G_0| \leq 2q\delta^2/\epsilon^3$ there. Uniform stability gives $M_H = \sup |H_\theta| < \infty$, and the identity $H_\delta - H_0 = z^{-1} H_\delta H_0 (G_\delta - G_0)$, whose factor $z^{-1}$ has modulus one on the circle, gives an $\ell_2$ gain at most $C_{\mathcal{F}}\delta^2$. The mean difference of a power-constrained probe has squared norm at most $C_{\mathcal{F}}^2 \delta^4 PT$; the equal-noise KL is at most $C_{\mathcal{F}}^2 \delta^4 PT/(2\sigma_\nu^2)$, and Pinsker's inequality with total variation at least $1 - 2\alpha$ gives (9).

**Proposition 3.3.** Equal maps have the same fixed-point set; uniqueness or the declared common selection yields (i). For (ii), put $d = |\kappa_A - \kappa_B|$. The fixed-point equations and the two Lipschitz bounds give $d \leq L_K(L_\Gamma d + \epsilon)$. Since $1 - L_K L_\Gamma > 0$, rearrangement proves the displayed bound. The scalar inequality is machine checked in `p1_endogenous_contraction_sensitivity`; the economic derivation and uniformity of the covariance map are assumptions, not consequences of that algebra. A discontinuous threshold pricing rule need not preserve a small discrepancy. □

**Theorem 5.1.** In either positive economy $J \geq 0$ and $\|J\|_\infty \leq 1 - \epsilon(1 - S) < 1$, so the response is nonnegative with $\max h = \kappa$ and $\sum h = \kappa/(1 - S)$. From the display above, $|G_\delta - G_0| \leq 2q\delta^2/\epsilon^3$ on the circle; combined with the resolvent identity and positivity this gives $\|H_\delta - H_0\|_\infty \leq K_* \delta^2$, and Parseval bounds $\|h^\delta - h^0\|_2$ by the same constant. Young's inequality gives $\|\mu_\delta - \mu_0\|_2 \leq K_* \delta^2 \|v\|_1$, and Gaussian KL with Pinsker gives (12). For (ii), the mean difference is a lower-triangular convolution matrix times $v$; each column is a shifted

copy of a prefix of $d_T$, so $\|\mu_\delta - \mu_0\|_2 \le V\|d_T\|_2$ with equality for a pulse at $t = 0$. For two known equal-covariance Gaussian laws the minimax error is $\Phi(-\|\mu_\delta - \mu_0\|_2/(2\sigma))$, attained by the midpoint likelihood-ratio test. For (iii), the merged measures have equal mass and their first moments differ by $-q\delta^2$, which enters $h_2$ with factor $\kappa^2$ while $h_0, h_1$ coincide; a pulse $A$ separates the third-observation means by $A\kappa^2 q\delta^2$. Part (iv) follows from the $\ell_2$ gain bound, and (v) from superposition and $\max_j h_j = \kappa$, which give $\sup_t |\lambda x_t| \le |\lambda|\kappa V$, attained by a single pulse.

**Lemma S24.1.** After the uncontrolled update, the pre-probe displacement at step $j$ is $\sum_{i<j} h_{j-i} v_i$; average linear-book execution adds $\kappa v_j/2$. Summing signed execution costs gives $\frac{1}{2} v^\top T_h v$. Each kernel $\lambda^{|i-j|}$ is a positive-definite AR(1) covariance for $0 \le \lambda < 1$, and a nonnegative mixture with positive total weight is positive definite. For a round trip the fundamental cash term cancels.

**Theorem S24.2.** Let $U_t = \sum_{j<t} v_j$. Positive responses give $0 \le x_t \le \kappa U_t$ and $I_t = -U_t$. Summing cash and marking inventory, $B - E_t = U_t x_t - \sum_{j<t} v_j x^0_{j+1} - \frac{\kappa}{2}\sum_{j<t} v_j^2 + cU_t$. Its quadratic part $v^\top A_t v$ has diagonal $h_{t-1-i} - \kappa/2$ and off-diagonal $\frac{1}{2}(h_{t-1-i} + h_{t-1-j} - h_{j-i})$ for $i < j$; monotonicity of $h$ bounds every entry by $\kappa/2$. Hence $B - E_t \le \kappa U_t^2/2 + cU_t \le \kappa V^2/2 + cV$, while position value is at most $V(P_* + \kappa V)$; the stated inequality guarantees margin. A single pulse attains both adverse bounds at its first response, so no larger allowance is robust. Solving the quadratic gives (S17). The quantity requirement $V_B \gtrsim \delta^{-2}$ then forces $B \gtrsim \delta^{-4}$, and the pulse $A$ funded with $(\mu + \frac{1}{2})\kappa A^2 + (c + \mu P_*)A$ attains it.

**Proposition S24.3.** The initial cash receipt is common to both alternatives and the dealer then holds $-A$. At time three $E_{3,r} = B + \kappa A^2/2 - cA - A^2 h_{r,2}$, so $E_{3,\delta} - E_{3,0} = A^2\kappa^2 q\delta^2 > 0$.

**Lemma S24.4.** Marking residual inventory at $P_*$ gives $E_\infty = B - \frac{1}{2} v^\top T_h v - c\sum |v_t| \ge 0$. For each mode $\lambda \le r$ the Toeplitz kernel has spectral density $(1 - \lambda^2)/(1 - 2\lambda\cos\omega + \lambda^2) \ge (1 - r)/(1 + r)$, so $v^\top T_h v \ge m\|v\|_2^2$ and $\|v\|_2^2 \le 2B/m$. The transfer bound gives mean-difference squared norm at most $K_*^2\delta^4\|v\|_2^2$, hence the KL bound; Pinsker gives (S18). For one pulse, cash becomes $B_A - A(P_* + \kappa A/2) - cA$ and inventory $A$; at later marks $E_t = \mu A P_* + A^2 h_{t-1}$ and $\mu|I_t|P_t = \mu A P_* + \mu A^2 h_{t-1}$, so margin holds for $\mu \le 1$.

**Proposition S9.1.** Parts (i) and (ii) follow from (S2) with $\Delta h = \Theta(\delta)$ for a rate shift and $\Theta(\delta^2)$ for symmetric splitting, and from additivity of KL over independent pulse cycles. Part (iii) solves the session constraint for $\delta$; substituting $d_T(\delta) \approx \delta^2 \tilde{d}_T$ gives the fourth-root approximation.

**Theorems S9.2 and S9.3.** For a flat round trip, telescoping the cash account gives $L_i(v) = \frac{1}{2}v^\top T_{h_i} v = \frac{1}{2\pi}\int g_i(\omega)|V(e^{i\omega})|^2 d\omega$, with $g_i > 0$ by Lemma 2.3. Parseval gives $\|(H_1 - H_0)v\|_2^2 = \frac{1}{2\pi}\int |\Delta\hat{H}|^2|V|^2 \le \sup_\omega(|\Delta\hat{H}|^2/g_i)L_i(v)$, and the Neyman–Pearson requirement $\|(H_1 - H_0)v\|_2 \ge 2\sigma z_{1-\alpha}$ gives (S6). For adaptive policies, Doob's optional stopping theorem applied to the log-likelihood-ratio martingale and the data-processing inequality give $\mathbb{E}_0[\Lambda_\tau] \ge \mathrm{kl}(\alpha, 1-\beta)$, while $\mathbb{E}_0[\Lambda_\tau] = \mathbb{E}_0 \sum_t (\Delta\mu_t)^2/(2\sigma^2) \le \sup(|\Delta\hat{H}|^2/g_0)\mathbb{E}_0[L_0]/(2\sigma^2)$, which gives (S8).

*Part (iv).* Let $G_\theta$ be the $T \times T$ lower-triangular Toeplitz section of $H_\theta/\kappa$, so the observation covariance is $\Sigma_\theta = \sigma_w^2 G_\theta G_\theta^\top + \sigma_\nu^2 I$. A finite section of a bounded analytic symbol has norm at most its supremum on the circle, so $\|G_\theta\| \le \sup_\omega |H_\theta|/\kappa \le \bar{M}$ because $|H_\theta(e^{i\omega})| \le \sum_j h_j = \kappa/(1-S)$. The inverse of a lower-triangular Toeplitz section is the section of the reciprocal series; $\kappa/H_\theta$ is analytic on and outside the unit circle because $H_\theta$ is minimum phase (Theorem 6.4), and Lemma 2.3 gives $\mathrm{Re}\, H_\theta(e^{i\omega}) = \sum_\ell \omega_\ell(1-\lambda_\ell\cos\omega)/(1-2\lambda_\ell\cos\omega+\lambda_\ell^2) \ge \sum_\ell \omega_\ell/(1+\lambda_\ell) \ge \kappa/2$, so $\|G_\theta^{-1}\| \le 2$ and $\Sigma_\theta \succeq \sigma_{\mathrm{eff}}^2 I$ (the smallest singular value of $G_\theta$ is at least 0.522 in 762 random positive economies and sections). The Gaussian divergence splits into a mean and a covariance part. The mean part is $\frac{1}{2}\Delta\mu^\top\Sigma_1^{-1}\Delta\mu \le \|(H_1 - H_0)v\|_2^2/(2\sigma_{\mathrm{eff}}^2)$. For the covariance part, the eigenvalues $\lambda_k$ of $A = \Sigma_1^{-1/2}\Sigma_0\Sigma_1^{-1/2}$ are at least $c = \sigma_\nu^2/(\sigma_\nu^2 + \sigma_w^2\bar{M}^2)$, and $\lambda - 1 - \log\lambda \le (\lambda-1)^2/\lambda$, so it is at most $\|A - I\|_F^2/(2c) \le \|\Sigma_0 - \Sigma_1\|_F^2/(2c\,\sigma_{\mathrm{eff}}^4)$. Writing $\Sigma_0 - \Sigma_1 = \sigma_w^2(\Delta G\, G_0^\top + G_1\Delta G^\top)$ and $\|\Delta G\|_F^2 \le T\|h^\delta - h^0\|_2^2/\kappa^2 \le TK_*^2\delta^4/\kappa^2$ (proof of Theorem 5.1) gives a covariance part of at most $C_w T\delta^4$. The data-processing inequality turns errors $(\alpha, \beta)$ into $D_{\mathrm{KL}} \ge \mathrm{kl}(\alpha, 1-\beta)$, and part (ii) bounds $\|(H_1 - H_0)v\|_2^2$ by $\sup_\omega(|\Delta\hat{H}|^2/g_i)L_i(v)$, which gives (S7).

**Theorem S9.4.** Write a feasible flat probe as $v = Eu$ and expand $u = \sum_j c_j e_j$ in the $G_i$-orthonormal eigenbasis, so $L_i(v) = \sum_j c_j^2$ and the separation constraint reads $\sum_j \lambda_j c_j^2 \ge c_{\mathrm{req}}$. Write the top-two component as $(c_{L-2}, c_{L-1}) = r(\cos\theta, \sin\theta)$ with $r \ge 0$; replacing $\theta$ by $\theta + \pi$ flips $d$ and $s_\theta$ and leaves (S9) unchanged, so $\theta \in [0, \pi)$ suffices. Put $L_0(\theta) = c_{\mathrm{req}}/\lambda(\theta)$ and $D = L_i(v) - L_0(\theta)$. The constraint gives $r^2\lambda(\theta) \ge c_{\mathrm{req}} - \sum_{j \le L-3}\lambda_j c_j^2$, hence

$$D = r^2 + \sum_{j \le L-3} c_j^2 - L_0(\theta) \;\ge\; \sum_{j \le L-3}\Big(1 - \frac{\lambda_j}{\lambda(\theta)}\Big)c_j^2 \;\ge\; \Big(1 - \frac{\lambda_{L-3}}{\lambda(\theta)}\Big)\sum_{j \le L-3} c_j^2 \;\ge 0,$$

and $r^2 \ge L_0(\theta) - \frac{\lambda_{L-3}}{\lambda(\theta)}\sum_{j \le L-3} c_j^2 \ge L_0(\theta)\big(1 - \gamma(\theta)D\big)$. By Hölder's inequality with the sign vector $s_\theta$, $c_{\mathrm{fee}}\|Eu\|_1 \ge c_{\mathrm{fee}} r\|Ed(\theta)\|_1 + c_{\mathrm{fee}}\sum_{j \le L-3} c_j\, s_\theta^\top E e_j$, and Cauchy–Schwarz with weights $1 - \lambda_j/\lambda(\theta)$ bounds the second term below by $-\sqrt{D\,W_\perp(\theta)}$. Using $r \ge \sqrt{L_0(1-\gamma D)_+} \ge \sqrt{L_0}\,(1-\gamma D)_+$ and the fact that the fee is nonnegative, with

$x = \sqrt{D}$,

$$L_i(v) + c_{\text{fee}}\|v\|_1 \;\geq\; L_0(\theta) + x^2 + \Big[F_{\text{dom}}(\theta)(1-\gamma(\theta)x^2)_+ - \sqrt{W_\perp(\theta)}\,x\Big]_+ \;\geq\; L_0(\theta) + \Psi(\theta).$$

Minimizing over the unknown $\theta$ gives (S10), and $\Psi \geq 0$, $L_0(\theta) \geq c_{\text{req}}/\lambda_{L-1}$ give the last inequality. On the piece $\gamma x^2 \leq 1$ with $\gamma F_{\text{dom}} < 1$ the bracket is a convex quadratic in $x$ with minimum $F_{\text{dom}} - W_\perp/(4(1-\gamma F_{\text{dom}}))$, which gives the closed form.

**Corollary S9.5.** Since $\Delta h_t = O(\delta^2)$, the variance difference $\sigma^2_{1,t} - \sigma^2_{0,t} = \sigma^2 \xi[(h_1 + h_0) * v]_t[\Delta h * v]_t = O(\delta^2)$, and the variance term of the Gaussian KL, $\frac{1}{2}\sum(x - 1 - \log x)$ with $x = \sigma^2_{0,t}/\sigma^2_{1,t}$, is quadratic in that difference. Convex impact only increases dissipation.

**Proposition S7.1.** Part (i): for every alternative $\theta'$ the Gaussian likelihood ratio has second moment $\exp(\|h(\theta') - h(\theta)\|^2/\sigma_h^2)$, so the Hammersley–Chapman–Robbins inequality gives $(\mathbb{E}_{\theta'}\hat{w}_k - \mathbb{E}_\theta \hat{w}_k)^2 \leq \mathrm{Var}_\theta(\hat{w}_k)\,(e^{\|h(\theta')-h(\theta)\|^2/\sigma_h^2} - 1)$; put $\theta' = \theta + ta$, use local unbiasedness, divide by $t^2$ and let $t \to 0$ to get $a_k^2 \leq \mathrm{Var}_\theta(\hat{w}_k)\|Ja\|^2/\sigma_h^2$, and choose $a = F_{\text{unit}}^{-1} e_k$. Part (iii): for $K = 2$ the columns of $J$ for the two rates become collinear as $\Delta\eta \to 0$; expanding to third order, the smallest singular value of $J$ is of order $(\Delta\eta)^3$, which gives the upper bound $[F^{-1}]_{kk} \leq \sigma_{\min}(J)^{-2} = O((\Delta\eta)^{-6})$. The matching lower bound for the weight coordinate does not follow from the singular value alone; it uses the exact inverse row of the $4 \times 4$ confluent Vandermonde matrix in moment coordinates together with the bounded conditioning of the map from moments to the observed taps on the fixed-centre family.

**Theorem S25.1.** Expanding $M_m(\theta_1) - M_m(\theta_0)$ under (S19) gives zero for $m \leq 2$ and $4\tau p_- p_+ (\Delta\eta)^3$ for $m = 3$. The map from moments to impulse responses is analytic with nonsingular derivative at the common limit, so $\|\Delta h\|_2 = \Theta(\tau(\Delta\eta)^3)$. Le Cam's two-point lemma with $N$ independent Gaussian experiments gives (S21). For the permutation-invariant loss, the identity matching of $\theta_0$ and $\theta_1$ costs $2\tau + (p_+ - p_-)\Delta\eta = 2\tau(1 + \Delta\eta/q)$ per atom and the swapped matching costs $\max(2p_+, 2p_-)\Delta\eta = (1 + 2\tau/q)\Delta\eta$; when both are at least $2\tau$, any estimate within $\tau$ of one economy in $d_\infty$ is farther than $\tau$ from the other by the triangle inequality, and the two-point argument is unchanged. For (iii), the upper bound is Theorem 6.2; the Prony–Hankel determinant identity follows from $g_m = w_1\eta_1 r_1^m + w_2\eta_2 r_2^m$.

**Theorem 6.1.** Let $(s^A, \omega^A)$ and $(s^B, \omega^B)$ be the $K$-point Gauss and Gauss–Radau rules for the uniform weight on $[-1, 1]$, normalized to unit mass. Both are positive and exact for polynomials of degree $2K - 2$, and their moments of degree $2K - 1$ differ by $g_K \neq 0$. With $r = r_c + \Delta\eta\, s$ and weights $W\omega$, the raw moments $\sum w_i r_i^m$ are polynomials in $\Delta\eta$ that agree

below degree $2K-1$, so the kernels differ by $O((\Delta\eta)^{2K-1})$ with nonzero leading coefficient; analyticity of the kernel-to-response map gives $\|h^A - h^B\|_2 \asymp (\Delta\eta)^{2K-1}$, and Le Cam's lemma gives the bound. For the $W_1$ version, the two merged measures are the images of the fixed normalized rules under $s \mapsto r_c + \Delta\eta\, s$ scaled to mass $W$, so their distance is $W\Delta\eta$ times the distance $2c_K > 0$ between the rules, and the same two-point argument applies with the metric loss.

**Lemma S17.1.** Let $L = 2n-2$ and let $p^*$ be the best uniform approximation of $|x|$ on $[-1,1]$ by polynomials of degree $L$, with error $E = E_L(|x|)$. By Chebyshev's equioscillation theorem (Rivlin, 1981) there are points $t_0 < \cdots < t_{L+1}$ at which $|x| - p^*(x)$ equals $\pm E$ with alternating signs. The divided-difference weights $w_i = 1/\prod_{j\neq i}(t_i - t_j)$ alternate in sign and satisfy $\sum_i w_i p(t_i) = 0$ for every polynomial $p$ of degree at most $L$; multiplying by $-1$ if necessary, $w_i$ has the sign of $|t_i| - p^*(t_i)$ for every $i$. Exactly $n$ weights are positive and $n$ negative, and the positive and negative parts have equal mass because constants are annihilated. Normalizing each part to unit mass and mapping $x \mapsto \rho(x+1)/2$ gives $n$-atom probability measures $\pi_A$ (positive part) and $\pi_B$ (negative part) on $[0,\rho]$ with equal moments of orders $0,\dots,L$. For the 1-Lipschitz function $f(r) = |r - \rho/2| = (\rho/2)|x|$,

$$\int f\, d(\pi_A - \pi_B) = \frac{\rho}{2}\int \big(|x| - p^*(x)\big)\, d(\pi_A - \pi_B) = \frac{\rho}{2}(E + E) = \rho E,$$

because $p^*$ integrates to the same value under both measures, $|x| - p^* = E$ on the support of $\pi_A$ and $-E$ on that of $\pi_B$. By Kantorovich–Rubinstein duality $W_1(\pi_A, \pi_B) \geq \rho E \geq c_*\rho/L$.

**Theorem S17.2.** Write $K(z) = \sum_m k_m z^m$, $D = 1 - az - \kappa z K$ and $H = \kappa/D$, which is the generating function of the recursion $h_t = a h_{t-1} + \kappa \sum_m k_m h_{t-1-m}$ with $h_0 = \kappa$. Since $k_m \geq 0$ and $\sum_m k_m = \int (1-r)^{-1} d\mu \leq W/(1-\rho)$, the $\ell_1$ norm of $az + \kappa z K$ is at most $s < 1$, so $\|H\|_1 \leq \kappa/(1-s)$ and $\|1/H\|_1 = \|D\|_1/\kappa \leq (1+s)/\kappa$.

(i) Let $\mu_A = W\pi_A$ and $\mu_B = W\pi_B$ with $\pi_A, \pi_B$ from Lemma S17.1; both lie in $\mathcal{M}(\rho, W)$. With $\nu = \mu_A - \mu_B$, $\sum_m (k_m^A - k_m^B) z^m = \int (1 - rz)^{-1} d\nu(r) = z^{2n-1} \int r^{2n-1}(1-rz)^{-1} d\nu(r)$, because the first $2n-1$ terms of the geometric series integrate to zero. On $|z| = 1$ the modulus is at most $2W\rho^{2n-1}/(1-\rho)$, so Parseval gives $\|k^A - k^B\|_2 \leq 2W\rho^{2n-1}/(1-\rho)$. Next, $H_A - H_B = z(K_A - K_B) H_A H_B$, so by Young's inequality $\|h^A - h^B\|_2 \leq \|H_A\|_1 \|H_B\|_1 \|k^A - k^B\|_2 \leq C_h \|k^A - k^B\|_2$. The Kullback–Leibler divergence between the $N$-sample Gaussian experiments is $N\|h^A - h^B\|_2^2/(2\sigma^2) \leq (N/\sigma^2)\, 2C_h^2 W^2 \rho^{2(2n-1)}/(1-\rho)^2$, which is at most one as soon as $2(2n-1)\log(1/\rho) \geq \Lambda_N - 2\log(1/\rho)$. Take $n$ the smallest integer with this property and $n \geq 2$; then $2n - 2 < \Lambda_N/(2\log(1/\rho))$ when $\Lambda_N > 4\log(1/\rho)$. Since $d$ is a norm on signed measures and $d(\mu_A, \mu_B) = W\, W_1(\pi_A, \pi_B) \geq W c_* \rho/(2n-2) > 2c_*\rho W \log(1/\rho)/\Lambda_N$,

the Bretagnolle–Huber form of Le Cam's lemma (Tsybakov, 2009, Lemma 2.6 and Theorem 2.2) gives $\max_{j\in\{A,B\}} P_j(d(\hat{\mu},\mu_j) \geq c_*\rho W \log(1/\rho)/\Lambda_N) \geq \frac{1}{4}e^{-1}$.

(ii) *Approximation.* Let $f$ be 1-Lipschitz on $[0,\rho]$ with $f(0) = 0$, so $|f| \leq \rho$. The function $g(x) = f(\rho(x+1)/2)$ is $(\rho/2)$-Lipschitz on $[-1,1]$, and Jackson's theorem (Rivlin, 1981) gives a polynomial $p$ of degree $L$ with $\sup|g-p| \leq (\pi/2)(\rho/2)/(L+1) \leq \rho$, hence $\sup|p| \leq 2\rho$. Its Chebyshev coefficients satisfy $|b_j| \leq 2\sup|p| \leq 4\rho$. The polynomial $q(r) = p(2r/\rho - 1)$ approximates $f$ with the same error, and its monomial coefficients satisfy $\sum_j |q_j| \leq \sum_{j\leq L} |b_j|\,\|T_j(2r/\rho-1)\|_{\text{coef}} \leq 4\rho\sum_{j\leq L} B(\rho)^j \leq \rho B(\rho)^{L+1}$, where $\|\cdot\|_{\text{coef}}$ is the sum of absolute monomial coefficients, $\|T_j\|_{\text{coef}} \leq (1+\sqrt{2})^j$, substituting $y = \alpha r - 1$ multiplies the coefficient sum of $y^i$ by at most $(1+\alpha)^i$, and $B(\rho)-1 \geq 4$. On the event $\max_{j\leq L}|\hat{k}_j - k_j| \leq \delta$ the true $\mu$ attains objective at most $\delta$, so the minimizer $\hat{\mu}$ satisfies $\max_{j\leq L}|\int r^j d\hat{\mu} - k_j| \leq 2\delta$ by the triangle inequality. *Moment comparison.* If $\max_{j\leq L}|\int r^j d\hat{\mu} - k_j| \leq 2\delta$, then $\hat{\mu}$ has mass at most $W + 2\delta$ and

$$\int f\,d(\hat{\mu}-\mu) = \int (f-q)\,d(\hat{\mu}-\mu) + \sum_{j\leq L} q_j\Big(\int r^j d\hat{\mu} - k_j\Big) \leq \frac{\pi\rho(W+\delta)}{2(L+1)} + 2\delta\rho B(\rho)^{L+1}.$$

*Kernel recovery.* From $\kappa z K = 1 - az - \kappa/H$, the kernel values $k_0,\ldots,k_L$ are the coefficients of $(1-az-\kappa/H)/(\kappa z)$, which depend on $h_0,\ldots,h_{L+1}$ only. Work in the algebra of power series modulo $z^{L+2}$ with the $\ell_1$ norm, which is submultiplicative. If the averaged response is $\hat{H} = H + E$ and $\|1/H\|_1\|E\|_1 \leq \frac{1}{2}$, then $1/\hat{H} = (1/H)(1+E/H)^{-1}$ and $\|1/\hat{H} - 1/H\|_1 \leq 2\|1/H\|_1^2\|E\|_1$, so $\max_j|\hat{k}_j - k_j| \leq \delta \equiv 2((1+s)/\kappa)^2(L+2)\max_{t\leq L+1}|E_t|$. Each $E_t$ is Gaussian with variance $\sigma^2/N$, so with probability at least $1-\beta$, $\max_t|E_t| \leq \sigma\sqrt{2\log(2(L+2)/\beta)/N}$. The true $\mu$ is feasible for the linear program with value at most $\delta$, so the minimizer satisfies the moment condition above. *Choice of $L$.* With $L+1 = \lfloor \log(N/\sigma^2)/(4\log B(\rho))\rfloor$, $B(\rho)^{L+1} \leq (N/\sigma^2)^{1/4}$, so $\delta B(\rho)^{L+1} = O((\sigma^2/N)^{1/4}L\sqrt{\log(L/\beta)})$ and $\delta/(L+1)$ are $o(1/\log(N/\sigma^2))$, while $\pi\rho W/(2(L+1)) \leq 2\pi\rho W\log B(\rho)/(\log(N/\sigma^2) - 4\log B(\rho))$. For $N/\sigma^2$ larger than a threshold depending on $(\rho, W, a, \kappa, \beta)$ the side condition $\|1/H\|_1\|E\|_1 \leq \frac{1}{2}$ holds on the event and the total is below $3\pi\rho W\log B(\rho)/\log(N/\sigma^2)$, uniformly over $\mathcal{M}(\rho, W)$. The final statement combines (i) and (ii).

**Theorem 6.5.** For stationary Gaussian processes with spectral densities $f_j \propto |H_j(e^{i\omega})|^2$, the divergence rate is

$$D = \frac{1}{4\pi}\int_{-\pi}^{\pi}\big(\varrho - 1 - \log\varrho\big)\,d\omega, \qquad \varrho = f_1/f_0 = |H_1/H_0|^2.$$

By the resolvent identity, $H_1/H_0 - 1 = z^{-1}H_1\Delta G$ with $\Delta G = G_1 - G_0$, and uniform stability bounds $|H_1|$ on the circle, so $u \equiv \varrho - 1 = 2\,\mathrm{Re}(z^{-1}H_1\Delta G) + O(|\Delta G|^2)$. Write $G(z) = \sum_k w_k\phi(z,\eta_k)$, with $w_k$ the cohort weights (any factor of $\eta_k$ is absorbed in $\phi$), where $\phi$ is analytic in $\eta$ on a neighbourhood of $[\eta_{\min},\eta_{\max}]$ uniformly for $|z| = 1$ (its pole $z = 1-\eta$ stays inside the disk). Expanding $\phi$ around $\bar\eta$, $\Delta G = \sum_{j\ge 0}\frac{\partial_\eta^j\phi(z,\bar\eta)}{j!}\,\Delta\tilde M_j$, where $\tilde M_j = \sum_k w_k(\eta_k - \bar\eta)^j$. For the pair, $\Delta\tilde M_0 = \Delta\tilde M_1 = 0$ and $\Delta\tilde M_2 = q\delta^2$; for the moment-matched economies, $\Delta\tilde M_j = 0$ for $j \le 2$ and $\Delta\tilde M_3 = 4\tau p_- p_+(\Delta\eta)^3$ by (S20); for the $K$-cohort Gauss and Gauss–Radau economies with rates $\bar\eta + \Delta\eta\, s$ and weights $W\omega$, both rules integrate polynomials of degree $2K-2$ exactly, so $\Delta\tilde M_j = W(\Delta\eta)^j\sum_k(\omega_k^A(s_k^A)^j - \omega_k^B(s_k^B)^j)$ vanishes for $j \le 2K-2$ and equals $\pm W g_K(\Delta\eta)^{2K-1} \ne 0$ at $j = 2K-1$. Higher terms are smaller by a factor of the spacing: $|\Delta\tilde M_j| \le 2W(\Delta\eta)^j$, and analyticity of $\phi$ in $\eta$ on a disk of radius $\varrho > 0$ around $\bar\eta$, uniformly on the circle, gives $|\partial_\eta^j\phi|/j! \le C\varrho^{-j}$, so for $\Delta\eta < \varrho/2$ the tail beyond the first nonzero term is $O((\Delta\eta)^{j_0+1})$. Hence $\sup_\omega|u| = O(\delta^2)$, $O(\tau(\Delta\eta)^3)$, respectively $O((\Delta\eta)^{2K-1})$, and $u = c(\omega)\epsilon + o(\epsilon)$ with $\epsilon = \delta^2$, $\tau(\Delta\eta)^3$ or $(\Delta\eta)^{2K-1}$ and $c(\omega)$ proportional to $2\,\mathrm{Re}\big(z^{-1}H_0\,\partial_\eta^{j_0}\phi(\cdot,\bar\eta)\big)/j_0!$, $j_0 = 2$, 3 or $2K-1$. The function $c$ does not vanish identically: $f = z^{-1}H_0\partial_\eta^j\phi$ is analytic outside the disk including at infinity, where it vanishes, and is not identically zero; an analytic function on the exterior whose real part vanishes on the circle is an imaginary constant, which here would have to be zero. For $|u| \le \frac12$, $u^2/4 \le u - \log(1+u) \le u^2$, so $D \asymp \int c(\omega)^2 d\omega\,\epsilon^2$, which gives (i)–(iii). The sample requirements need a finite-$T$ bound that is uniform in the separation, which we obtain directly. The first $T$ observations are $N(0,\Sigma_j)$ with $\Sigma_j$ the $T\times T$ Toeplitz matrices of the spectral densities $f_j = \sigma^2|H_j(e^{i\omega})|^2$. On the compact, uniformly stable family the transfer functions have no zeros or poles on the circle (Theorem 6.4), so $0 < \underline f \le f_j \le \bar f < \infty$ with constants that do not depend on the separation. A Toeplitz matrix has eigenvalues within the range of its symbol, so $\lambda_{\min}(\Sigma_0) \ge \underline f$ and $\|\Sigma_1 - \Sigma_0\| \le \sup_\omega|f_1 - f_0| = \sup_\omega f_0|u| \le \bar f\sup_\omega|u|$. With $A = \Sigma_0^{-1/2}\Sigma_1\Sigma_0^{-1/2}$, whose eigenvalues lie in $[\underline f/\bar f, \bar f/\underline f]$, the inequality $\lambda - 1 - \log\lambda \le (\lambda-1)^2/\lambda$ gives

$$D_{\mathrm{KL}}(P_1^T\|P_0^T) = \tfrac12\sum_k(\lambda_k - 1 - \log\lambda_k) \le \frac{\bar f}{2\underline f}\,\|A - I\|_F^2 \le \frac{\bar f}{2\underline f}\,T\Big(\frac{\bar f}{\underline f}\Big)^2\sup_\omega|u|^2 \le C\,T\epsilon^2,$$

with $C$ independent of $T$ and of the separation, because $\sup_\omega|u| = O(\epsilon)$ uniformly. The Bretagnolle–Huber inequality (Tsybakov, 2009, Lemma 2.6) then gives the stated lower bounds on $T$ for every $T$, not only asymptotically. For a fixed pair, $T^{-1}D_{\mathrm{KL}}(P_1^T\|P_0^T) \to D$ and Stein's lemma (one error fixed) and the Chernoff bound (both errors vanishing) give the error exponents of the likelihood-ratio test, of the same order in the separation; that statement is pointwise in the pair.

**Theorem 6.4.** In the delay variable $\lambda = z^{-1}$, $H = \kappa \prod_i (1 - r_i\lambda) / \big[(1 - a\lambda) \prod_i (1 - r_i\lambda) - \kappa\lambda \sum_i w_i \prod_{j \neq i} (1 - r_j\lambda)\big]$, which is (4) divided through by $z^{K+1}$; its nonstructural zeros are the memory rates, which do not cancel under the nondegenerate assumptions of Theorem 6.4. In the z-plane there is also the structural timing zero at zero: the numerator is proportional to $z \prod_i (z - r_i)$. Stability puts the poles inside the disk. A rational spectral density $\sigma^2 |H(e^{i\omega})|^2$ has a unique minimum-phase factor up to a unimodular constant, which is $H$; Theorem 3.1 then gives the merged measure.

**Proposition 7.1.** With tagged flow, $u_{i,t} = r_i u_{i,t-1} + w_i x_t$ holds exactly and involves only cohort $i$, so $\tilde{u}_{i,t} = r_i \tilde{u}_{i,t-1} + w_i x_t + e_{i,t}$ with the MA(1) error $e_{i,t}$. Prices are independent of the measurement noise, so $\mathbb{E}[Z_t e_{i,t}] = 0$ and, by the ergodic theorem for the stationary linear price process, the IV estimator is consistent when $\Sigma_i = \mathbb{E}[Z_t(\tilde{u}_{i,t-1}, x_t)]$ is nonsingular. Since $u_{i,t-1} = w_i \sum_{s \geq 0} r_i^s x_{t-1-s}$, $\mathbb{E}[x_{t-1} u_{i,t-1}] = w_i A_0$ and $\mathbb{E}[x_t u_{i,t-1}] = w_i A_1$, which gives $\Sigma_i$. The price is a nonnegative mixture of geometric responses to its innovation (Lemma 2.3), so $\gamma_x(s) = \sum_\ell c_\ell \lambda_\ell^s$ with $c_\ell = \sigma_\xi^2 \kappa^{-2} \omega_\ell \sum_m \omega_m / (1 - \lambda_\ell \lambda_m) \geq 0$; substituting, $\det \Sigma_i = w_i(\gamma_x(0) A_0 - \gamma_x(1) A_1) = w_i \sum_{\ell,m} c_\ell c_m (1 - \lambda_\ell \lambda_m)/(1 - r_i \lambda_m)$, each term positive. By summation by parts, $\sum_{t=1}^T Z_t e_{i,t} = \sum_{t=1}^T (Z_t - r_i Z_{t+1}) \varepsilon_{i,t} + O_P(1)$. Since $(\varepsilon_{i,t})$ is i.i.d. with mean zero and variance $\sigma_f^2$ and independent of the whole stationary price process, the summands $m_t = (Z_t - r_i Z_{t+1}) \varepsilon_{i,t}$ form a stationary ergodic martingale difference sequence with respect to $\mathcal{F}_t = \sigma((x_s)_{s \in \mathbb{Z}}, \varepsilon_{i,s} : s \leq t)$, with $\mathbb{E}[m_t m_t^\top] = \sigma_f^2 [(1 + r_i^2)\Gamma_0 - r_i(\Gamma_1 + \Gamma_1^\top)] = \Omega_i$, so the Billingsley–Ibragimov martingale central limit theorem gives (15). On a uniformly stable family the modes and weights lie in a compact set with $\lambda_\ell \leq \bar{\lambda} < 1$, so with $w_i \geq w_{\min} > 0$ we get $\det \Sigma_i \geq w_{\min}(1 - \bar{\lambda}^2) \gamma_x(0)^2 > 0$ and $\|\Omega_i\|$ is bounded, uniformly in the spacing.

**Proposition 7.3.** (i) The current-price regressor column is unchanged, whereas the lagged true flow column scales by $\alpha_{i,m}$; valid independent reporting noise has zero population covariance with the instruments. Multiplicativity of determinants gives the stated factor. Summing shares at least $\underline{\alpha}$ gives $M_i \underline{\alpha} \leq 1$, checked in `p1_sybil_share_count_bound`. For a fixed finite collection the summed estimation errors remain of order $T^{-1/2}$ under the maintained joint central limit assumptions; this asymptotic step is not newly formalized here. (ii) Subtract the cost lower bound from the uniform gross-profit upper bound. Once the former exceeds the latter, expected net profit is negative, as checked in `p1_camouflage_unprofitable`. The cost coercivity and variance-to-difficulty link are explicit hypotheses, not consequences of fairness alone. (iii) Divide the scaled weight by $1 - r_i \geq \eta_{\min}$; without that bound, taking $1 - r_i = \gamma_i$ can keep $q_i = w_i$ constant. □

**Theorem S12.1.** With $J_\theta$ upper triangular the eigenvalues are fixed. The stationary covariance solves $P = J_\theta P J_\theta^\top + \mathrm{diag}(s_1, s_2)$, and $P_{11}$ is affine in $s_2$ with positive slope, so for each $\theta$ a unique $s_2(\theta) \geq 0$ gives $P_{11} = V$ when $V$ exceeds the value at $s_2 = 0$. The one-step price variance is $s_1$ throughout, and $e_1^\top J_\theta e_2 = \theta$ gives $\sup_k \|e_1^\top J_\theta^k\| \geq |\theta|$.

**Proposition S27.1.** (i) $\mathcal{L}_{\text{disc}} < \mathcal{L}_{\text{price}}$ is $F + Mc < M\ell G$. (ii) $T_p$ and $T_t$ are decreasing in $\Delta\eta$, with $T_t \geq \bar{T}$ exactly when $\Delta\eta \leq \Delta\eta_t$ and $T_p \geq \bar{T}$ exactly when $\Delta\eta \leq \Delta\eta_p$. On $\Delta\eta \leq \Delta\eta_t$ both minima equal $\bar{T}$ and $G = 0$. On $(\Delta\eta_t, \Delta\eta_p]$, $G = \bar{T} - c_t(\Delta\eta)^{-2}$, which increases to $G^* = \bar{T} - T_t(\Delta\eta_p) > 0$. On $\Delta\eta > \Delta\eta_p$, $G = c_p(\Delta\eta)^{-(4K-2)} - c_t(\Delta\eta)^{-2} = (\Delta\eta)^{-2}\left[c_p(\Delta\eta)^{-(4K-4)} - c_t\right]$; the bracket is positive exactly when $\Delta\eta < \Delta\eta_0 = (c_p/c_t)^{1/(4K-4)}$, and $G'(\Delta\eta) = (\Delta\eta)^{-3}\left[2c_t - (4K-2)c_p(\Delta\eta)^{-(4K-4)}\right] < 0$ whenever $c_p(\Delta\eta)^{-(4K-4)} > c_t$, because $4K - 2 > 2$. So $G$ decreases wherever it is positive beyond $\Delta\eta_p$, and is non-positive beyond $\Delta\eta_0$. Every superlevel set of $G$ at a positive level is therefore an interval (the negative tail, where $G$ rises back toward zero from below, never reaches a positive level), and the level $(F/M + c)/\ell > 0$ lies above $G = 0$, so the set where (i) holds is an interval contained in $(\Delta\eta_t, \Delta\eta_0)$ that contains $\Delta\eta_p$ whenever it is non-empty. (iii) The interval is non-empty if and only if the level is below the maximum, $F/M + c < \ell G^*$, that is $M > M^*$; the level decreases in $M$, so the superlevel set grows. □

# S2 Notation

| Symbol | Meaning |
|---|---|
| $x_t$ | pricing error (price minus fundamental) |
| $a$ | price persistence (anchor), $0 \leq a < 1$ |
| $\kappa$ | price impact per unit of executed quantity |
| $q_i,\ \eta_i,\ r_i$ | cohort weight, adaptation rate, memory rate $r_i = 1 - \eta_i$ |
| $w_i = q_i\eta_i$ | merged weight |
| $M = \sum_i w_i\delta_{r_i}$ | merged rate–weight measure |
| $k_m,\ G(z)$ | feedback kernel and its transform |
| $H(z),\ h_j$ | closed-loop transfer function and impulse response, $h_0 = \kappa$ |
| $J,\ S$ | state transition matrix and $S = a + \kappa\sum q_i$ |
| $v_t,\ V$ | executed probe quantity and total quantity $\sum \lvert v_t\rvert$ |
| $d_T$ | impulse-response difference between hypotheses, first $T$ taps |
| $\sigma$ | observation-noise standard deviation |
| $\delta,\ \Delta\eta$ | half-separation of a known pair; spacing of unknown cohorts |

| Symbol | Meaning |
|---|---|
| $K_*$ | transfer-difference constant $2\kappa^2 q/[\epsilon^3(1-S)^2]$ |
| $T_h$ | Toeplitz matrix $(h_{\lvert i-j\rvert})$ of execution cost |
| $g_i(\omega)$ | dissipation kernel $\mathrm{Re}\,\hat{H}_i(e^{i\omega}) - \kappa/2$ |
| $B,\ \mu,\ c,\ \phi$ | capital, maintenance-margin fraction, proportional cost, taker fee |
| $V_{\text{eff}}$ | per-trade execution capacity $\min(V_{\max}, D_{\max}/\kappa)$ |
| $\hat{Q},\ \hat{S}$ | reduced-form loop gain and persistence indices |

# S3 Further results

## S3.1 A finite-window price-surveillance bound

**Theorem S3.1** (Finite-window abstention)**.** *Observe $x_t = sx_{t-1}+w_t$, $w_t \sim N(0,\sigma^2)$, $x_0 = 0$, $t \le T$, with $s \in (0,2)$. Let a possibly randomized rule $\Psi$ output Stable, Unstable or Abstain, with $\sup_{s<1} P_s(\Psi = \text{Unstable}) \le \alpha$ and $\sup_{s\ge 1} P_s(\Psi = \text{Stable}) \le \alpha$, $\alpha < 1/2$. Choose $0 < c_0 < 1$ with $c_0 e^{c_0}/(2\sqrt{2}) < 1 - 2\alpha$. On $B_T = (1 - c_0/T, 1 + c_0/T)$,*

$$\inf_{s\in B_T} P_s(\Psi = \text{Abstain}) \ \ge\ 1 - 2\alpha - \frac{c_0 e^{c_0}}{2\sqrt{2}} > 0,$$

*and for every finite $T$ the minimax worst-case error of binary rules over $s \in (0,2)$ is at least $1/2$.*

*Proof.* Pair $s_0 \in (1-c_0/T, 1)$ with $s_1 = s_0 + c_0/T$. Then $D_{\mathrm{KL}}(P_{s_0}\|P_{s_1}) = \frac{(s_1-s_0)^2}{2\sigma^2}\sum_t \mathbb{E}x_{t-1}^2 \le c_0^2 e^{2c_0}/4$, so Pinsker bounds total variation by $c_0 e^{c_0}/(2\sqrt{2})$. Error control gives $P_{s_0}(\text{Stable}) \le \alpha + \text{TV}$ and $P_{s_0}(\text{Unstable}) \le \alpha$, which yields the abstention bound. For binary rules, continuity in total variation and Le Cam's two-point bound give worst-case error tending to $1/2$. □

The interval $B_T$ is an abstention layer that any error-controlled monitor must respect in the scalar Gaussian case; richer disclosure can remove it.

## S3.2 Passive distinction of a known pair in finite samples

**Proposition S3.2** (Passive distinction with zero controlled quantity)**.** *Keep the known pair, but add independent uniform innovations $w_t \in [-b, b]$ of variance $s_w^2 = b^2/3$ at each price update, observe $\tilde{x}_t = x_t + \nu_t$ with $\nu_t \sim N(0,\sigma^2)$, use independent three-step reset blocks and*

*no probes, with $P_* > 3b$. Let $b_1 = a + \kappa q\eta$, $b_{2,0} = b_1^2 + \kappa q\eta(1-\eta)$ and $b_{2,\delta} = b_{2,0} - \kappa q\delta^2$. In each block the covariance of the first and third observations differs by $\Delta = \kappa q s_w^2 \delta^2$. With*

$$R = \max_{r\in\{0,\delta\}} \Big\{ (s_w^2 + \sigma^2)\big[s_w^2(1 + b_1^2 + b_{2,r}^2) + \sigma^2\big] + s_w^4 b_{2,r}^2 \Big\},$$

*the midpoint test of the average product has worst-case error at most $\alpha$ whenever $n \geq 4R/(\alpha\Delta^2)$ blocks are observed: zero controlled quantity and order $\delta^{-4}$ blocks.*

The Chebyshev bound is conservative but the order is real. With $b = 0.03$, observation s.d. 0.01 and $\delta = 0.10$, the bound asks for $3.9 \times 10^7$ blocks at $\alpha = 0.1$; in simulation the midpoint test still errs 32–38% of the time with $2 \times 10^5$ blocks. Independent resets and known innovation laws are extra conditions. The proposition does not show that anonymous public records identify participants; it shows that passive excitation can accumulate memory information while the probe account stays untouched.

*Proof.* In each block $x_1 = w_1$, $x_2 = b_1 w_1 + w_2$ and $x_3 = b_{2,r} w_1 + b_1 w_2 + w_3$, so $|x_t| \leq 3b$. Observation noise changes variances but not the covariance $s_w^2 b_{2,r}$. The product variance is the product of marginal variances plus the squared covariance plus a fourth-cumulant term $b_{2,r}^2(\mathbb{E}w^4 - 3s_w^4) \leq 0$ for uniform innovations, so it is at most $R$. Chebyshev bounds each midpoint error by $4R/(n\Delta^2)$. □

## S3.3 Blind inversion of the closed loop

1. *Impulse decoupling:* $g_t = h_{t+1} - a h_t$.
2. *Minimal realization:* form the Hankel matrix $H_{ij} = g_{i+j-1}$ and run Kung's eigensystem realization with a truncated SVD.
3. *Transmission zeros:* compute the zeros of the realization, which under known $(\kappa, a)$ isolate the roots of the denominator of $G$.
4. *Weights:* with recovered rates, project $u_t = (x_t - a x_{t-1})/\kappa$ on the reconstructed filters $y_{k,t-1}$.

Without noise each step is exact: the realization has the order of the true system, its nonstructural transmission zeros are the memory rates (Theorem 3.1); the known timing zero is discarded, and the projection returns the weights, so the procedure removes the pole-shift bias of fitting poles directly to the price response. With noise, per-cohort recovery degrades as rates move together at the rate given by Theorems 6.1 and 6.2, which is why we do not report it as an estimator of individual cohorts.

## S3.4 Probing protocols

Table S2 compares four probing protocols at observation horizon $T = 4$ per cycle.

Table S2: Probing protocols at observation horizon $T = 4$ per cycle

| Protocol | $\delta$ | Executed vol. | Max trade | Error (SE) | Capital |
|---|---|---|---|---|---|
| A: single pulse (unconstrained) | 0.03 | 131.5 | 131.5 | 0.047 (0.003) | 6,985.91 |
| | 0.05 | 47.3 | 47.3 | 0.054 (0.004) | 921.73 |
| | 0.08 | 18.5 | 18.5 | 0.053 (0.004) | 146.73 |
| | 0.12 | 8.2 | 8.2 | 0.044 (0.003) | 31.45 |
| B: repeated capped pulses | 0.03 | 2,768.8 | 6.2 | 0.050 (0.003) | 7,035.75 |
| | 0.05 | 362.5 | 6.2 | 0.054 (0.004) | 923.88 |
| | 0.08 | 56.2 | 6.2 | 0.047 (0.003) | 146.00 |
| | 0.12 | 12.5 | 6.2 | 0.038 (0.003) | 34.88 |
| C: sequential test, capped | 0.03 | 1,410.2 | 6.2 | 0.045 (0.003) | 3,585.14 |
| | 0.05 | 195.6 | 6.2 | 0.038 (0.003) | 500.00 |
| | 0.08 | 35.9 | 6.2 | 0.032 (0.003) | 94.30 |
| | 0.12 | 10.5 | 6.2 | 0.011 (0.002) | 29.85 |
| D: generalized eigenvector | 0.03 | 2,359.7 | 3.7 | 0.050 (0.003) | 2,632.77 |
| | 0.05 | 314.8 | 3.7 | 0.048 (0.003) | 353.67 |
| | 0.08 | 49.2 | 3.7 | 0.046 (0.003) | 57.53 |
| | 0.12 | 9.9 | 3.8 | 0.045 (0.003) | 13.56 |

2,000 replications per hypothesis and cell. Log–log capital slopes: $-3.90$, $-3.84$, $-3.47$, $-3.81$. Protocols A and B reproduce the earlier version exactly. C and D are re-implemented because the earlier code was not retained: C is a Wald test with capped pulses; D repeats the top generalized eigenvector of a flat $(T + 1)$-step probe, scaled to a maximum trade of 3.75, and is cheaper than the earlier D.

## S3.5 Coordinate-weighted output

For $J = \begin{pmatrix} a & \kappa q \\ \eta a & 1-\eta+\eta\kappa q \end{pmatrix}$ with $a = 0.5$, $\kappa = 0.8$, $q = 0.2$, $\eta = 0.3$ ($\rho(J) = 0.8224$, non-normal), the price output has $\sup_t \|e_x^\top J^t\| = 1$. The weighted output $C_w = (1, \lambda/\mathrm{MMR}) = (1, 160)$ with $\lambda = 0.8$ and $\mathrm{MMR} = 0.005$ has $\sup_t \|C_w J^t\| = \|C_w\| = 160.003$, attained at $t = 0$, then 122.32, 98.32, 80.79 . The factor 160 is a static coordinate weight; $C_w z$ is not a margin quantity because $z$ contains no inventory. Margin must be computed on a cash-and-inventory ledger.

## S3.6 Real-noise injection: requirements at each spacing

Table S3 gives the requirements behind Table S23. $N^*$ is read off the grid $2^{j/2}$, so neighbouring values differ by a factor $\sqrt{2}$, and a ratio of $\sqrt{2}$ or $1/\sqrt{2}$ between the two arms is one grid step. Entries above 4,096 use the central-limit draw described in Supplement S16.

Table S3: Real-noise injection: number of experiments $N^*$ at each spacing

| | $\Delta\eta$ | Price, real | Price, Gaussian | Tagged, real | Tagged, Gaussian |
|---|---|---|---|---|---|
| $K = 2$ | 0.500 | 64 | 64 | 724 | 1,024 |
| | 0.400 | 362 | 362 | 362 | 512 |
| | 0.320 | 1,448 | 1,024 | 362 | 362 |
| | 0.250 | 5,793 | 5,793 | 256 | 256 |
| | 0.200 | 16,384 | 23,170 | 256 | 362 |
| | 0.160 | 65,536 | 92,682 | 256 | 256 |
| | 0.125 | 370,728 | 370,728 | 256 | 256 |
| $K = 3$ | 0.60 | 23,170 | 16,384 | | |
| | 0.50 | 65,536 | 65,536 | | |
| | 0.40 | 741,455 | 741,455 | | |
| | 0.32 | 8,388,608 | 5,931,642 | | |
| | 0.25 | 94,906,266 | 94,906,266 | | |

Real noise: 30,219 one-minute log-price changes of Binance BTCUSDT on 21 days (the 6th of each month, January 2025 to September 2026), kurtosis 19.4. Tagged requirements were computed for $K = 2$ only.

## S3.7 Numerical stress tests of results that are not machine-checked

Four results were proved on paper but not in Lean when these tests were run (Section 6 and Proposition 7.1); two of them, Theorem 6.2 and the general-memory theorem (Theorem S17.2, whose lower bound Lean proves with a different moment-matched pair than the one checked here), have since been machine-checked in Lean. We stress-tested each across randomly drawn parameter settings. These are numerical checks, not proofs and not pre-specified tests; no prediction failed (Table S4).

## S3.8 Probing a running market without resets

The probing results count independent experiments that each start from the reset state, which no outside participant can arrange in a live market. As a numerical check, not a proof and not a pre-specified test, we drop the resets. The observer sends a known i.i.d. $\pm 1$ probe into one long stationary run of the closed loop of Table S23 ($K = 2$, $a = 0.5$, $\kappa = 0.8$, total weight 0.08, $r_c = 0.6$), which is also driven by its own price innovations $w_t \sim N(0, \sigma_w^2)$; prices are observed with noise $\nu_t \sim N(0, \sigma_\nu^2)$, $\sigma_\nu = 10^{-3}$ as in the reset experiments. The groups' state is never reset or known: its free response is part of the noise. The observer estimates the response taps $h_1, \ldots, h_4$ by least squares of the observed price on the current and 200 lagged probe values, then applies the same Prony estimator and success rule as Table S35 (every weight within 25% of the smallest true weight in at least 75% of 200 replications). Because

Table S4: Numerical stress tests (numerical checks, not proofs)

| Result | What was checked | Outcome |
|---|---|---|
| Upper bound, Theorem 6.2 | Prony's requirement $N^*$ for random separated configurations (15 for $K = 2$, 12 for $K = 3$); slope in $\Delta\eta$ and whether $N^*(\Delta\eta)^{4K-2}/\sigma^2$ grows as the spacing shrinks | Slopes $-6.72$ to $-5.79$ ($K = 2$, theory $-6$) and $-11.01$ to $-9.29$ ($K = 3$, theory $-10$); the normalized constant changes by a median factor of 1.05 and 0.94 from the widest to the narrowest spacing (at most 2.1); no configuration failed |
| Passive data, Theorem 6.5 | Exact Whittle divergence rate between two moment-matched economies, 10 random settings for each $K$ | Slopes 6.19–6.34, 10.27–10.40 and 14.33–14.54 for $K = 2, 3, 4$ (theory 6, 10, 14): the divergence shrinks at least as fast as the theorem requires |
| General memory, Lemma S17.1 | Alternation pairs on $[0, \rho]$ for 12 random $\rho \in [0.3, 0.99]$ and $n = 2, \ldots, 6$: equal low moments, $W_1 \geq \rho E_{2n-2}$, kernel distance bound | 60 of 60 cases pass |
| Value of disclosure, Theorem 7.2 | Tagged and price-only requirements over the spacing grid, $K = 2, 3, 4$ | Tagged requirement does not grow as the spacing shrinks (it varies by factors 2.8, 11.3 and 5.7 over the grid); price-only slopes $-6.01$ and $-9.47$ for $K = 2, 3$; for $K = 4$ our Prony implementation fails numerically at every spacing, so no price-only comparison |

Gaussian observation noise, $\sigma = 10^{-3}$ (prices) and $\sigma_f = 0.02$ (tagged flow), $a = 0.5$, $\kappa = 0.8$, total weight 0.08, 200 replications per point, accuracy: every weight within 25% of the smallest true weight in at least 75% of replications. Moment-matched economies for the passive check: binomial weights on equally spaced nodes (the pair used in the Lean proof of Theorem 6.1). Seed 20261031.

the probe is white and independent of the noise, the least-squares error is asymptotically Gaussian with covariance $\Gamma_e/T$, where $\Gamma_e$ is the autocovariance of the regression residual; the required number of periods $T^*$ is computed from this law, which we checked against 120 simulated runs of $10^5$ periods at $\Delta\eta = 0.5$ (standard deviation of the tap estimates 5.44 to $6.05\times10^{-6}$ against $5.35\times10^{-6}$ predicted).

Table S5 shows that the order survives: the number of periods grows with fitted slopes -6.26 and -6.22 in $\Delta\eta$, against -5.87 for reset experiments and $-6$ in theory. The constant is larger, because the market's own innovations and the groups' unknown state add noise: with $\sigma_w = \sigma_\nu$ a running market needs 1.4 to 2.8 times as many periods as the reset benchmark needs experiments, and with $\sigma_w = 3\sigma_\nu$ four to six times as many (the grid of $T$ and $N$ has steps of $\sqrt{2}$). Since every reset experiment itself uses at least $2K + 1$ periods, the running market is not slower in calendar time. What the check does not show is that a live prober can trade a $\pm1$ sequence every period at the prices assumed, or that other traders ignore it; those are the strategic-response caveats of Section 2.1.

Table S5: Probing with and without resets: required reset experiments $N^*$ and required periods $T^*$ of one running market

| $\Delta\eta$ | Reset experiments $N^*$ | Running market $T^*$ $\sigma_w = \sigma_\nu$ | Running market $T^*$ $\sigma_w = 3\sigma_\nu$ |
|---|---|---|---|
| 0.5 | 91 | 128 | 362 |
| 0.4 | 362 | 512 | 1,448 |
| 0.32 | 1,024 | 2,048 | 5,793 |
| 0.25 | 5,793 | 8,192 | 32,768 |
| 0.2 | 23,170 | 32,768 | 131,072 |
| 0.16 | 65,536 | 185,364 | 370,728 |
| Fitted slope in $\Delta\eta$ | $-5.87$ | $-6.26$ | $-6.22$ |

Numerical check, not pre-specified. $K = 2$ Gauss-node economy of Table S23; i.i.d. $\pm1$ probe; least squares on 200 probe lags; Prony estimator and success rule as in Table S35.

# S4 Extended related literature

**Earlier statement of the contribution.** The two paragraphs below are the previous version's comparison and contribution statements, kept verbatim; Table 1 replaces them in the main text.

**Comparison with existing identification research.** Clustered exponential recovery already explains ill-conditioning and separation-dependent rates (Batenkov et al., 2021; Katz et al., 2023); those exponents are not claimed as a general new discovery. Data informativity

already distinguishes information sufficient for a control task from unique system identification (van Waarde et al., 2020); the broad distinction is likewise not new. Our candidate contribution is their explicit combination in this market feedback model: the public-response equivalence class, an execution/capital interpretation under a specified probing contract, and a two-population supervisory decision whose feasible targeted action differs despite close public responses. The all-$K$ action extension needs an additional weight-gap construction. Classical system identification (Ljung, 1999) provides the observation/estimation framework; a price-only record here is neither exogenous order flow nor a revealed portfolio preference. Novelty must be judged on these model-specific results against the cited theorems, not on AI terminology or a claim to have invented supervision without identification. Task-optimal exploration already supplies quantitative, task-dependent lower bounds and efficient experimental designs in linear systems (Wagenmaker et al., 2021). Accordingly, neither quantitative task-dependent learning nor choosing information for a decision is claimed as new here. Our comparison must turn on the latent population map, the financial execution/capital contract, and the specified class intervention; no general dominance over task-optimal exploration is asserted.

**Contribution.** The candidate contribution is the market-model connection between a public-response equivalence class, the sample and financial resources needed for finer identification, and a specified supervisory action. Equal-memory groups are exactly inseparable; distinct but nearby groups face a finite-noise difficulty, not permanent nonidentification. Proposition 7.4 establishes a two-group separation example and a conditional general-$K$ extension. The statistical clustering exponent and the distinction between task-relevant information and full identification have precedents; the precise comparison below states what is inherited and what requires model-specific proof. Historical position-report evidence is descriptive and does not establish the action model or the necessity of granular reporting. Formal receipts cover particular typed statements; semantic correspondence and independent proof review remain separate obligations.

**Algorithms, AI and supervision.** Recent work in finance asks how learning algorithms and AI agents behave in markets: reinforcement-learning market makers and informed traders can sustain wide spreads or collusive outcomes (Colliard et al., 2022; Dou et al., 2025), prices formed against Q-learners have distinctive dynamics (Banerjee and Szydlowski, 2026), and algorithmic markets can be manipulated (Tremacoldi-Rossi, 2025). Traders who differ in horizon or speed are studied in equilibrium by Cespa and Vives (2015) and Biais et al. (2015), and Crouzet et al. (2020) model investors who trade at different frequencies.

These papers model or simulate the algorithms. We take the supervisor's side and ask what can be learned about them from market data, and what disclosure is worth. A related literature measures the cost of regulation and of regulatory reporting (Trebbi et al., 2024); our dollar comparison with the Consolidated Audit Trail is a statement about the benefit side: what the audit trail buys that prices cannot.

**Price formation and impact.** Our price equation is a discrete linear impact model in the tradition of Kyle (1985), Glosten and Milgrom (1985) and Hasbrouck (1991), in which order flow moves prices and prices revert toward value. Transient impact and the cost of trading follow Almgren and Chriss (2001), Obizhaeva and Wang (2013), Gatheral (2010) and Bouchaud et al. (2004); Gatheral (2010) shows that absence of dynamic arbitrage requires a positive-definite decay kernel, Alfonsi et al. (2012) that excluding transaction-triggered manipulation requires convex, nonincreasing decay, and Alfonsi et al. (2016) extend positive definiteness to several assets. Positive definiteness is the property Lemma S24.1 delivers in our cohort economy. Bouchaud et al. (2018) document order splitting and long memory in order flow, which is the main confounder in Supplement S13. Relative to this literature we treat the impact kernel not as a primitive but as the closed-loop output of adaptive traders, and ask what it reveals about them.

**Algorithmic trading and feedback.** Algorithms that react to recent prices can amplify or dampen moves. Brunnermeier and Pedersen (2009) and Shleifer and Vishny (1992) study liquidity spirals and fire sales, and Kirilenko et al. (2017) document the behaviour of automated intermediaries in the 2010 Flash Crash. Sornette et al. (2023) show that non-normal interactions can generate transient bubbles. Our contribution is on the measurement side: we show what a closed-loop observer can and cannot learn about the feedback, and that standard reduced-form indices are dominated by filtering and timing.

**Learning algorithms and what can be inferred about them.** Empirical game-theoretic analysis treats the choice among simulated trading strategies as a game (Wellman, 2006). Empirically, algorithmic trading has been found to improve liquidity and price discovery on average (Hendershott et al., 2011; Brogaard et al., 2014). These studies simulate the algorithms or observe them through regulatory data; we ask what an outside observer can infer about them from market data alone, and answer in samples and in capital. Our stance is that of partial identification (Manski, 2003): when the structure is not identified, characterize what the data do determine, here the merged measure and, through it, the intensity-weighted distribution of the desks' information quality (Proposition S5.1).

**Control without identification.** We do not claim that acting on a system without identifying it is new. Moallemi et al. (2008) study strategic execution against an observer who learns from prices, which bears directly on our assumption that cohorts do not react to being observed, and Park and Van Roy (2015) study execution that learns price impact while trading, with regret guarantees; our object is different, the latent trader cohorts behind the impact rather than the aggregate impact coefficients. Active identification of linear systems has matching bounds and optimal experiment design (Wagenmaker and Jamieson, 2020), and task-directed exploration asks what must be learned for a decision (Wagenmaker et al., 2021); our information matrix differs because the unknowns are zeros of a closed-loop response and the input is paid for in trading losses. van Waarde et al. (2020) characterize when control is possible without unique system identification. Our results concern a specific market observation contract and its costs in executed quantity, capital and money.

**Closest work.** The closest literature is super-resolution and moment-based mixture estimation, which gives rates for recovering clustered point sources and mixing measures from noisy moments (Moitra, 2015; Heinrich and Kahn, 2018; Batenkov et al., 2021; Liu and Ammari, 2022; Wu and Yang, 2020; Kong and Valiant, 2017). Our $K$-cohort rate is, in that language, the square of the amplitude condition number $(\Delta\eta)^{-(2K-1)}$ for $K$ clustered nodes (Heinrich and Kahn, 2018; Wu and Yang, 2020), and we do not claim the exponent as new. What changes when the mixture sits inside a closed-loop financial market is fourfold (Table S6): first, in mixture estimation and open-loop system identification the unknown rates are locations or poles of the observed law, whereas in a closed-loop market the cohorts feed back into the price, making the memory rates transmission *zeros* of the observable transfer function $H(z)$ that must first be extracted from the loop (Theorem 3.1); with $a$ and $\kappa$ known this extraction is a smooth invertible map, so it decides which method is valid (pole fitting on prices is not) rather than the rate; second, passive price observations face at least the same $(\Delta\eta)^{-(4K-2)}$ sample-complexity barrier as active probing, uniformly in the spacing, because $H(z)$ is minimum phase and the Whittle likelihood inherits the moment gap (Theorems 6.4 and 6.5); third, in a market statistical sample size translates into executed quantity $\Theta(\delta^{-2})$ and round-trip trading loss or margin capital $\Theta(\delta^{-4})$ via the self-impact Toeplitz quadratic form $\frac{1}{2}v^\top T_h v$ (Theorems 5.1–S9.4); and fourth, class-tagged disclosure breaks the feedback loop cohort by cohort and cuts the sample requirement by $(\Delta\eta)^{-(4K-4)}$ (Proposition 7.1 and Theorem 7.2). Beyond the $K$-cohort case, we prove matching lower and upper bounds of exact logarithmic order $\Theta(1/\log N)$ for general completely monotone memory (Theorem S17.2) and price the resulting disclosure savings against audit-trail budgets. Several ingredients are known in other settings, and we claim only their closed-loop,

passive and financial forms: the $\delta^{-4}$ law for telling one component from two closely spaced ones is classical in resolution analysis (Shahram and Milanfar, 2005); logarithmic rates are known for mixtures of exponentials (Rebafka and Roueff, 2015) and for general mixing distributions, by the same moment-matching and polynomial-approximation technique (Wu and Yang, 2020); and labelled samples are known to be far more informative than unlabelled ones in mixture problems (Castelli and Cover, 1996).

**Estimating heterogeneous-agent models from prices.** A large literature models markets as populations of trader types that switch between forecasting rules (Brock and Hommes, 1998) and estimates such models from price data. Estimates of the parameters that separate the types are often imprecise: Amilon (2008) reports that the estimated model reproduces properties of returns but that its fit is generally poor, Franke and Westerhoff (2012) compare switching mechanisms by the method of simulated moments, and Kukacka and Barunik (2017) find the switching coefficient statistically insignificant by simulated maximum likelihood while the belief parameters are significant. These papers take the identification of the types from prices as given. Our results suggest one reason the type-separating parameters are hard to pin down: around a steady state, prices determine only the merged measure, and separating types whose memories differ by $\Delta\eta$ takes a number of samples of order $(\Delta\eta)^{-(4K-2)}$ for $K$ types (Theorems 3.1 and 6.1). The models these papers estimate are nonlinear, so this is a statement about their linearization, not a claim that their estimates are wrong; it does give a quantity, the separation of the types' memories, that bounds the precision one can expect before estimating.

**Closed-loop system identification.** Engineering has long studied identification under feedback (Ljung, 1999; Forssell and Ljung, 1999). There the object is the plant, the difficulty is that feedback correlates the input with the noise, and the remedy is external excitation that makes the experiment informative. Our problem is the reverse in two ways. The object is the feedback law (the traders), not the plant (the price), and the memory rates enter as zeros of the observed transfer function. And external excitation does not remove the difficulty: with a probe the merged measure is identified exactly (Theorem 3.1), so the barrier is not qualitative identifiability but the conditioning of the map from memories to responses, which we measure in samples, quantity and capital. In finance, Leach and Madhavan (1993) show that a market maker can gain by setting prices to learn about demand, so learning through trading has an equilibrium counterpart; our probing observer is a reduced-form version of that idea without the equilibrium. In econometric terms, as $\Delta\eta \to 0$ the cohort parameters pass from strong to weak identification (Andrews and Cheng, 2012), and our rates state that

Table S6: Comparison with finite-mixture estimation and open-loop identification

| Feature | Finite-mixture estimation (Heinrich and Kahn, 2018; Wu and Yang, 2020) | Open-loop exponential fitting (Roy and Kailath, 1989; Batenkov et al., 2021) | Closed-loop market identification (this paper) |
|---|---|---|---|
| Where rates live | Component locations / moments of density | Poles of impulse response | Transmission zeros of closed-loop $H(z)$ (Thm 3.1) |
| Active vs. passive data | i.i.d. draws only | Designed input vs. output spectrum | At least the same $(\Delta\eta)^{-(4K-2)}$ order, uniformly, via the minimum-phase Whittle gap (Thm 6.5) |
| Economic unit of cost | Sample size $N$ | Signal-to-noise ratio | Quantity $\Theta(\delta^{-2})$, trading loss and capital $\Theta(\delta^{-4})$ via $\frac{1}{2}v^\top T_h v$ (Thms 5.1–S9.4) |
| Effect of class tagging | Observed component labels ($\Theta(1)$) | Separate sensor per mode | Breaks closed loop cohort-by-cohort; saves $\Theta((\Delta\eta)^{-(4K-2)})$ at fixed accuracy, $\Theta((\Delta\eta)^{-(4K-4)})$ for rates (Thm 7.2) |

passage in samples.

**Inferring trader types from market data.** Market microstructure has a tradition of inferring hidden trader populations from public data. The probability of informed trading (Easley et al., 1996) fits a mixture model to daily counts of buys and sells to estimate the share of informed orders, and what it captures has been debated (Duarte and Young, 2009). With regulatory account data, Kirilenko et al. (2017) classify traders by their holding and trading patterns. Our results speak to the case where types differ in how long they remember prices: from prices alone the separation is costly, and account or class records of the kind Kirilenko et al. (2017) use remove most of the cost (Proposition 7.1).

**Statistical identification.** The identification results use two-point testing arguments (Le Cam, 1986; Tsybakov, 2009). The moment-matching construction of Section 6 is the dynamical analogue of the slow rates in finite-mixture estimation (Manole and Ho, 2022) and of super-resolution of near-colliding sources (Liu and Ammari, 2022; Batenkov et al., 2021); the general-memory result combines best-approximation duality (Bernstein, 1914; Rivlin, 1981) with the moment comparisons used by Kong and Valiant (2017) and Wu and Yang (2020), and its logarithmic rate parallels supersmooth deconvolution (Fan, 1991); recovery of exponentials connects to ESPRIT and matrix-pencil methods (Roy and Kailath, 1989; Hua and

Table S7: Closest results and how this paper differs

| Existing result | Observations | Target | Conclusion | Difference from this paper |
|---|---|---|---|---|
| Mixture estimation (Heinrich and Kahn, 2018; Wu and Yang, 2020) | i.i.d. draws from a $K$-component mixture | Mixing measure | Rates that slow sharply as components merge; $n^{-1/(4K-2)}$ for Gaussian location mixtures | Same exponent; here the measure is observed through a feedback loop, as zeros of $H$, actively or passively, and the cost is also priced in quantity and capital |
| Super-resolution and Prony (Moitra, 2015; Batenkov et al., 2021; Katz et al., 2023) | Noisy Fourier or exponential samples | Point-source locations | Stability governed by Vandermonde conditioning and node separation; Prony attains the min-max bound for clustered exponents | Real nodes extracted from a closed loop; our matching upper bound (Theorem 6.2) adapts this known result |
| Closed-loop system identification (Ljung, 1999; Forssell and Ljung, 1999) | Input–output data under feedback | The plant | Consistency requires informative (externally excited) experiments | Target is the feedback law (the traders); excitation gives exact identification of $M$, and the barrier is conditioning, measured in samples and capital |
| Heterogeneous-agent estimation (Brock and Hommes, 1998; Franke and Westerhoff, 2012; Kukacka and Barunik, 2017) | Prices and returns | Type and switching parameters | Type-separating parameters often imprecise | Bounds that precision by the types' memory separation (in the linearization) |
| Probability of informed trading (Easley et al., 1996) | Daily buy and sell counts | Share of informed orders | Mixture estimate of trader population | Types that differ in memory; price-only versus class-tagged data |
| Data informativity (van Waarde et al., 2020) | Finite data, often noiseless | Whether a property is determined | Qualitative conditions | Quantitative minimax costs under noise |

Sarkar, 1990). Minimum-phase spectral factorization and the Whittle likelihood are classical (Whittle, 1953). What is new here is the mapping from these statistical rates into executed quantity and money through a financially specified account, the passive equivalence, and the value of class records; that the relevant rates are zeros, not poles, of the observable transfer function matters for which estimator is valid, not for the rate.

**Non-normal dynamics and supervision.** Transient growth of stable non-normal systems is classical (Trefethen and Embree, 2005; Kreiss, 1962). We use it to show that supervision through eigenvalues and second moments of prices is blind to transient risk (Theorem S12.1), which motivates the identification-free control of the companion paper. On the regulatory side, pre-trade risk controls for broker-dealers with market access are governed by SEC Rule 15c3-5 (SEC, 2010), and participant-level audit trails are the natural disclosure regime for Proposition 7.1. A large literature studies how pre- and post-trade transparency affects liquidity and trading costs (Pagano and Röell, 1996; Boehmer et al., 2005; Bessembinder et al., 2006); we ask a different question, what disclosure lets a supervisor learn about the algorithms that trade, and give the answer in samples. Crypto venues, whose market structure is studied by Makarov and Schoar (2020), are a natural laboratory for this question because they publish class-level positioning every five minutes, while the CFTC publishes it weekly for regulated futures.

Partial fractions, Gaussian testing, convolution inequalities and positive-system gains are established tools. Our contribution is to connect them into a quantitative chain from memory resolution, through probing resources, to money, with matching orders, explicit constants and a supervisory interpretation, and then to measure the relevant quantities in data.

# S5 What prices hide, in economic terms

The worked example states the barrier in signal-to-noise ratios. That is the general picture. Write $s = \sigma_\zeta^2/\sigma_\nu^2$ for a desk's signal-to-noise ratio, so that by (7) its memory rate is $1 - \eta(s)$ with

$$\eta(s) = \tfrac{1}{2}\Big(\sqrt{s^2+4s} - s\Big), \qquad \eta'(s) = \tfrac{1}{2}\Big(\frac{s+2}{\sqrt{s^2+4s}} - 1\Big) > 0,$$

and $\eta$ is increasing and concave. With no inventory penalty, the first-order condition of the Remark on exponential memory gives the trading intensity $q_i = \beta_i/\lambda_{\mathrm{tc},i}$: what the desk expects to earn per unit of belief, over what it pays to trade.

**Proposition S5.1** (Identification in economic primitives)**.** *Consider $K$ Kalman desks (Lemma 2.1(i)) with signal-to-noise ratios $s_i \in [\underline{s}, \bar{s}] \subset (0, \infty)$ and intensities $q_i = \beta_i/\lambda_{\mathrm{tc},i}$,*

*and write* $c_- = \eta'(\bar{s})$ *and* $c_+ = \eta'(\underline{s})$.

*(i) Prices and aggregate order flow depend on the desks only through the measure*

$$\sum_i q_i \, \eta(s_i) \, \delta_{1-\eta(s_i)}.$$

*Desks are identified through their signal-to-noise ratios and intensities only; expected returns and trading costs enter only through their ratio, and two populations with the same intensity-weighted distribution of signal-to-noise ratios are observationally equivalent.*

*(ii) Fix a centre* $s_c$ *and let the ratios lie in* $s_c + \Delta s\,[-1,1] \subset [\underline{s}, \bar{s}]$. *Recovering the intensity-weighted distribution of signal-to-noise ratios to Wasserstein accuracy of order* $\Delta s$, *or its weights* $q_i\eta(s_i)$, *from* $N$ *noisy reset experiments with worst-case error probability below* $1/4$ *requires* $N \geq c\,\sigma^2(\Delta s)^{-(4K-2)}$; *on separated configurations, Prony's method of Theorem 6.2 composed with* $\eta^{-1}$ *achieves this order. The constants are those of Theorems 6.1 and 6.2 multiplied by powers of* $c_-$ *and* $c_+$.

**In words.** What prices reveal about Kalman-filter desks is the mix of how good their information is (their signal-to-noise ratios), weighted by how hard they trade. Expected returns and trading costs matter only through their ratio. Pinning down that mix carries the same steep cost as separating memory rates.

*Proof.* (i) By Lemma 2.1(i) desk $i$ is a cohort with memory rate $1-\eta(s_i)$ and weight $q_i$, so Theorem 3.1 applies, and $\beta_i, \lambda_{\mathrm{tc},i}$ enter the economy only through $q_i$. (ii) On $[\underline{s}, \bar{s}]$, $\eta$ is a $C^1$ bijection onto its image with $c_- \leq \eta' \leq c_+$. Lower bound: take the two economies of Theorem 6.1 with centre $r_c = 1-\eta(s_c)$ and rate spacing $\Delta\eta = c_-\Delta s$. By the mean value theorem their gains are images of ratios in $s_c + \Delta s\,[-1,1]$, so they are two populations of Kalman desks in the class, and the bound $N \geq c\,\sigma^2(\Delta\eta)^{-(4K-2)} = c\,c_-^{-(4K-2)}\sigma^2(\Delta s)^{-(4K-2)}$ applies to recovering their weights. Because $\eta$ is $c_+$-Lipschitz, the Wasserstein distance between the two distributions of ratios is at least $1/c_+$ times that between the two distributions of rates, which Theorem 6.1 bounds below by a constant times $\Delta\eta = c_-\Delta s$. Upper bound: ratios in $s_c + \Delta s\,[-1,1]$ with spacing at least $\gamma\Delta s$ have gains in $\eta(s_c) + c_+\Delta s\,[-1,1]$ with spacing at least $c_-\gamma\Delta s$, a class of the form required by Theorem 6.2 with rescaled separation $\gamma c_-/c_+$; Prony's method recovers the rates and weights at that theorem's rate, and $\eta^{-1}$, which is $1/c_-$-Lipschitz, maps rate errors to ratio errors. □

The private information a supervisor would most like to know, how good each desk's information is, is exactly what prices hide. In the worked example the two desks' ratios

differ by a factor of almost two, and separating them already costs hundreds of thousands of dollars of probing capital. The proposition takes intensities to vary independently of information quality, because they also reflect costs and capital. The next proposition asks what happens when the economics ties them together.

**Proposition S5.2** (Intensity tied to information quality)**.** *Let each desk's intensity be a known function of its signal-to-noise ratio, $q = \varphi(s)$ with $\varphi$ positive and $C^2$ on $[\underline{s}, \bar{s}]$ (for a mean–variance desk, $\varphi(s)$ increases with $s$), and let $n_i \geq 0$ denote the mass of desks with ratio $s_i$ (a divisible population, not an integer desk count). Write $s(\eta)$ for the inverse gain map and $w(\eta) = \varphi(s(\eta))\,\eta$ for the weight per desk as a function of its gain.*

*(i)* Unknown numbers of desks. *The merged measure is $\sum_i n_i\, w(\eta(s_i))\, \delta_{1-\eta(s_i)}$. Every positive measure with finitely many atoms in $[1-\eta(\bar{s}), 1-\eta(\underline{s})]$ is of this form, with $n_i = m_i / w(\eta_i)$ for an atom of mass $m_i$ at $1-\eta_i$. So when the numbers of desks of each type are not known, tying intensity to information quality removes no economy, and every lower bound of Proposition S5.1(ii) holds unchanged.*

*(ii)* Known numbers of desks. *If the $n_i$ are also known, the set of economies is smaller, but the single-versus-pair problem keeps its order: one type at gain $\bar{\eta}$ with $2n$ desks and two types at $\bar{\eta} \pm \delta$ with $n$ desks each have taps that differ by $n\,\delta^2\,\partial_\eta^2[w(\eta)(1-\eta)^m]_{\bar{\eta}} + o(\delta^2)$, so separating them needs $N \geq c\,\sigma^2\delta^{-4}$ reset experiments, the order the symmetric pair has when intensities vary freely.*

*(iii)* Self-confirming configurations. *Suppose the ratio each desk perceives is the one implied by the closed-loop dynamics it trades in, through a map $T$ from configurations $(s_1, \ldots, s_K)$ to perceived ratios that is continuous on a box $\prod_i [\underline{s}, \bar{s}]$ and maps it into itself. Then a configuration with $T(s) = s$ exists, and parts (i)–(ii) and Proposition S5.1 apply at it.*

*Proof.* (i) Desk type $i$ is a cohort with gain $\eta_i = \eta(s_i)$, memory rate $1-\eta_i$ and total weight $n_i\varphi(s_i)$, so it contributes $n_i\varphi(s_i)\eta_i = n_i w(\eta_i)$ to the atom at $1-\eta_i$. Since $w > 0$, any atom mass $m_i \geq 0$ is attained with $n_i = m_i / w(\eta_i)$. (ii) Write $g_m(\eta) = w(\eta)(1-\eta)^m$; the tap difference is $n[g_m(\bar{\eta}+\delta) + g_m(\bar{\eta}-\delta) - 2g_m(\bar{\eta})] = n\delta^2 g_m''(\bar{\eta}) + o(\delta^2)$, with $\sum_m \sup|g_m''|^2 < \infty$ on gains bounded away from zero, so the Euclidean distance between the two tap vectors is $O(n\delta^2)$ and the Kullback–Leibler divergence over $N$ experiments is $O(Nn^2\delta^4/\sigma^2)$; Le Cam's two-point argument gives the bound. (iii) is Brouwer's fixed-point theorem. □

So the barrier does not depend on intensities varying freely: it survives any known link between intensity and information quality as long as the supervisor does not also know how

many desks of each kind there are, and even then the simplest separation problem keeps its $\delta^{-4}$ order. Whether the full $(\Delta\eta)^{-(4K-2)}$ exponent survives with known numbers of desks is open. Part (iii) shows only that self-confirming configurations exist under continuity; it does not derive the map $T$ from a model of the desks' learning.

# S6 Natural excitation and the cost of ambiguity

Natural price innovations can also separate the pair with zero controlled quantity: an exact finite-sample test on independent three-step blocks needs order $\delta^{-4}$ blocks (Online Supplement, Proposition S3.2), the same order as the passive Whittle bound of Section 6. Non-identification also has a price on the quote side: when the relevant set of economies includes costlier ones, a robust dealer widens its quote, which the companion paper analyses (Niroula et al., 2026); cohort non-identification alone does not establish that the worst relevant cost exceeds the informed one.

# S7 Local information: the Cramér–Rao benchmark

**Proposition S7.1** (Cramér–Rao benchmark)**.** *Let $\theta \in \Theta \subset \mathbb{R}^{2K}$ collect the weights $w_1, \dots, w_K$ and rates $\eta_1, \dots, \eta_K$, with $\Theta$ open. Each of $N$ reset probes of size $V_{\text{probe}}$ records the taps $h_0, \dots, h_T$ with independent $N(0, \sigma_p^2)$ errors, so the averaged, rescaled record is $Y = h(\theta) + \sigma_h Z$ with $Z \sim N(0, I_{T+1})$ and $\sigma_h = \sigma_p/(V_{\text{probe}}\sqrt{N})$. Let $J_{tj} = \partial h_t/\partial\theta_j$ and $F_{\text{unit}} = J^\top J$, and assume $T \geq 2K$ and that $J$ has full column rank at $\theta$, so that $F_{\text{unit}}$ is invertible.*

*(i) The Fisher information of $Y$ is $I(\theta) = \sigma_h^{-2} F_{\text{unit}}$. Every estimator $\hat{w}_k(Y)$ with finite variance at $\theta$ that is unbiased for $w_k$ on a neighbourhood of $\theta$ satisfies $\mathrm{Var}_\theta(\hat{w}_k) \geq \sigma_h^2 [F_{\text{unit}}^{-1}]_{kk}$. No further regularity is needed: the bound follows from the Hammersley–Chapman–Robbins inequality by letting the alternative approach $\theta$ along a line, without differentiating under the integral.*

*(ii)* Benchmark interpretation. *For an estimator that attains this variance and is approximately normal, a 95% interval of half-width* tol *needs*

$$N \geq N_{\text{CR}} \equiv \left(\frac{1.96}{\text{tol}}\right)^2 \left(\frac{\sigma_p}{V_{\text{probe}}}\right)^2 [F_{\text{unit}}^{-1}]_{kk}. \tag{S1}$$

*This is a sample-size benchmark, not a lower bound on the interval width of every procedure.*

*(iii) For $K = 2$, fix the horizon $T \geq 4$, known $a$ and $\kappa$ with $\kappa_{\min} \leq |\kappa| \leq \kappa_{\max}$, and weights $0 < w_i \leq w_{\max}$. For all rates with $0 \leq \eta_2 < \eta_1 \leq 1$, writing $\Delta\eta = \eta_1 - \eta_2$, $c\,(\Delta\eta)^{-6} \leq [F_{\text{unit}}^{-1}]_{kk} \leq C\,(\Delta\eta)^{-6}$ for both weight coordinates $k$, where $c, C > 0$ depend only on $T$, $|a|$, $\kappa_{\min}$, $\kappa_{\max}$ and $w_{\max}$ (not on $\Delta\eta$ or on where the rates lie). Hence $N_{\text{CR}} = \Theta((\Delta\eta)^{-6})$. The constants grow with $T$; a bound uniform in $T$ is not claimed. That an estimator attains this order is a separate statement, given by Theorem 6.2; the minimax lower bound of the same order is Theorem S25.1.*

**In words.** Part (i) bounds the variance of every estimator that is unbiased near the truth; part (ii) turns it into the benchmark number of test trades an efficient estimator needs to pin down one group's trading weight to a stated precision. Each extra group with a nearby memory raises the number sharply, as Table S9 shows.

At the critical noise level maximum likelihood attains the bound: its RMSE for the hardest weight is between 0.88 and 1.22 times the Cramér–Rao value and every weight is within tolerance in 93% to 98% of replications (Table S9), so the critical noise levels, not estimator failure, are what make close cohorts expensive. Under ordinary surveillance budgets ($N \leq 50$) unbiased estimators cannot resolve close weights.

# S8 A hypothetical scale illustration: disclosure in dollars

*This subsection is a hypothetical scale illustration of an outside prober's shadow loss, not a public cost–benefit calculation.*

Theorem 7.2 states the value of class-tagged disclosure in samples. Theorems S9.2 and 6.1 let us translate that value into a flow of round-trip trading losses for an outside participant. Two adjustments make the figures exact:

(i) *Exact $K$-cohort moment-gap constant.* Rather than extrapolating from Cramér–Rao ratios, we evaluate the spectral round-trip trading loss bound of Theorem S9.2,

$$L_{\text{rt}}^{*}(K, \Delta\eta) \;=\; \frac{4\sigma^2 z_{1-\alpha}^2}{\sup_{\omega}\Big(|\Delta\hat{H}_{K,\Delta\eta}(e^{i\omega})|^2 / g_0(\omega)\Big)},$$

directly for the $K$-point Gauss versus Gauss–Radau economies of Theorem 6.1 and Table S33, whose moments below order $2K-1$ coincide and whose $(2K-1)$-th moment gap has the exact constant $\Delta_2 = 0.222222$, $\Delta_3 = 0.053333$, and $\Delta_4 = 0.013061$ at spacing $\Delta\eta$ (with $\Delta\eta = 2\delta$ for the symmetric known pair).

(ii) *Flow against flow.* A single probe loss $L^*_{\mathrm{rt}}$ is a one-time certification cost, whereas an audit trail is an ongoing annual expenditure. Because algorithm populations turn over, we report the annualised round-trip trading loss $52 \times L^*_{\mathrm{rt}}$ under weekly renewal (matching the CFTC weekly reporting cycle of Supplement S18). For orientation only, we also record official audit-trail figures of two different kinds; we do not divide one by the other, because a prober's trading loss is a private transfer to other traders calibrated to a Bitcoin perpetual, while an audit trail's budget is a resource cost for equities and options. The first is CAT LLC's own operating budget: \$248.8 million for 2025 in the November 2024 filing (SEC Release No. 34-101901, Dec. 12, 2024, p. 4), revised to \$228.3 million in May 2025 and reduced further later in 2025, and the revised 2026 Financial and Operating Budget of 31 March 2026, with total expenses of \$147,044,869, which we use (CAT LLC, 2026). The second is historical, ex ante estimates of industry-wide compliance costs, which are not operating budgets and cover a different scope: the SEC's 2016 CAT NMS Plan Approval Order estimate of \$2.4 billion in initial industry build costs and \$1.7 billion per year in ongoing industry costs (SEC, 2016, Release No. 34-79318, Nov. 15, 2016, 81 FR 84696 at 84801, p. 363); and the SEC's Rule 613 proposing and adopting releases estimating \$4.0 billion initial and \$2.1 billion annual ongoing costs (SEC, 2012, Release No. 34-62174, May 26, 2010, 75 FR 32556 at 32598, 32600, pp. 147, 153; Release No. 34-67457, July 18, 2012, 77 FR 45722 at 45790–45795, pp. 220–238).

The figures are a shadow cost of learning by trading, the price an outside participant would pay; a supervisor without an audit trail does not probe, and its cost is detection delay, which Proposition S27.1 models. The cost results require a positive economy: with signed weights the execution form $\frac{1}{2}v^\top T_h v$ can be indefinite and a round trip can earn money, so none of the dollar figures applies there. Probing cost is a lower bound on the cost of learning from prices by trading, not on passive learning and not a social cost. It is also a lower bound for deterministic probes only: sequential probing can lower the expected loss by a factor of up to 0.49 (Theorem S9.3). Only the observation noise is estimated from data; the structural parameters are illustrative, and the constant is sensitive to them. At $\delta = 0.05$ the numerically evaluated lower bound of Theorem S9.2 is 358.8, 179.0, 61.1 and 13.9 par units for $a = 0$, 0.25, 0.5 and 0.7, and 3.9, 19.6, 61.1 and 305.2 for $\bar{\eta} = 0.2$, 0.3, 0.4 and 0.6; at $\Delta\eta = 0.20$ instead of 0.10 the unknown-pair figures fall by a factor of about 64. The Gauss–Radau pair is used in the orientation that gives the larger, still valid, lower bound. The known-pair rows span a rate width of $2\delta = \Delta\eta$, whereas the $K = 2$ Gauss–Radau rows on $r_c + \Delta\eta[-1, 1]$ span $\frac{4}{3}\Delta\eta$. Within that interpretation, Table S10 shows how rapidly the polynomial exponent $(\Delta\eta)^{-(4K-2)}$ and the shrinking Gauss–Radau moment-gap constant

$\Delta_K$ turn the comparison in favour of disclosure. For a known pair at coarse separation $\Delta\eta = 0.20$ ($\delta = 0.10$), weekly renewal of price-only probing costs \$0.32 million per market per year at Roll noise and \$7.79 million at one-second noise. At $\Delta\eta = 0.10$ ($\delta = 0.05$) the unknown $K = 2$ Gauss–Radau pair costs \$14.74 million (Roll) or \$356.44 million (one-second) per market per year; with $K = 3$ unknown cohorts at $\Delta\eta = 0.20$ the figures rise to \$41.83 million and \$1.01 billion, and at $K = 3$, $\Delta\eta = 0.10$ or $K = 4$, $\Delta\eta = 0.20$ they run to billions. For orientation only, CAT LLC's revised 2026 budget is \$147.0 million; as explained above, the two numbers measure different things. The renewal frequency is an assumption: annual figures scale with it, so monthly renewal divides them by 4.3 and renewal on each of 252 trading days multiplies them by 4.8.

# S9 Account-level capital

Lemma S24.4 gives the order but not the constant. This section computes what a trader actually needs under a calibrated linear-impact environment, then converts it to Bitcoin dollars.

**Numerical values.** Decimal optima and brackets below are numerical evaluations of the stated bounds.

## S9.1 Observation design and the tri-region partition

An observer distinguishes two candidate mechanisms with rates $\theta_0$ and $\theta_1 = \theta_0 + \delta$ over $N$ periods, submitting probes $V = (V_1, \dots, V_N)^\top$ that generate inventory $I_t = \sum_{s \le t} V_s$. Under contract $I_0$ (active probing, Gaussian observation noise) $p_t = (h_i * V)_t + \epsilon_t$, $\epsilon_t \sim N(0, \sigma^2)$, so

$$D_{\mathrm{KL}}(P_0^{\otimes N} \| P_1^{\otimes N}) = \frac{1}{2\sigma^2} \sum_{t=1}^{N} \Big( (h_1 * V)_t - (h_0 * V)_t \Big)^2. \tag{S2}$$

The observer minimizes capital $C(N, V) = \sum_t (\lambda V_t^2 + \phi |V_t|) + \max_t \mu p_0 |I_t|$ subject to (i) separation $D_{\mathrm{KL}} \ge \frac{1}{2}(z_\alpha + z_\beta)^2$, (ii) caps $|V_t| \le V_{\mathrm{eff}}$ and $|I_t| \le I_{\max}$, and (iii) a solvency budget $\max_t \mathrm{Loss}_t(V) \le B$. Here $V_{\mathrm{eff}} = \min(V_{\max}, D_{\max}/\kappa)$ combines resting depth and a price-displacement limit.

**Proposition S9.1** (Tri-region partition and finite-session resolution floor)**.** *Under observation horizon $T$ per pulse, $(\delta, B)$ space splits into three regions.*

*(i)* Feasible single pulse. *For $\delta \ge \delta_{\min}(T)$ and $B \ge C^*(\delta)$ one pulse within capacity suffices. For a simple rate shift ($\eta_1 = \eta_0 + \delta$) the response difference is $\Theta(\delta)$ and $C^* = \Theta(\delta^{-2})$;*

*for symmetric splitting the first-order variation cancels, the difference is* $\Theta(\delta^2)$ *and* $C^* = \Theta(\delta^{-4})$.

*(ii)* Sequential bounded probing. *For* $\delta < \delta_{\min}(T)$ *the required quantity exceeds* $V_{\text{eff}}$. *Capped pulses each supply divergence* $V_{\text{eff}}^2\|d_T\|^2/(2\sigma^2)$, *so*

$$N_{\text{pulses}}(T,\delta) = \left\lceil \frac{4\sigma^2 z_\alpha^2}{V_{\text{eff}}^2 \|d_T\|_2^2} \right\rceil. \tag{S3}$$

*(iii)* Infeasible with isolated pulses. *If the budget cannot finance sequential costs, peak inventory breaches* $I_{\max}$, *or a session deadline binds, the finite-session floor* $\delta_{\min}(T_{\text{session}})$ *is the root of* $V_{\text{eff}}\|d_{T_{\text{cycle}}}(\delta)\|_2 = 2\sigma z_\alpha/\sqrt{\lfloor T_{\text{session}}/T_{\text{cycle}} \rfloor}$, *with small-*$\delta$ *approximation* $\delta_{\min} = \Theta(T_{\text{session}}^{-1/4})$.

**In words.** A single test trade works only when the groups are far enough apart and the account has enough capital. Closer groups need many capped test trades, and a longer observation window after each trade makes every test trade more informative.

At the calibration $\kappa = 0.8$, $q = 0.2$, $\sigma = 0.01$, $V_{\max} = 50$, $D_{\max} = 5$ ($V_{\text{eff}} = 6.25$), the single-tap floor is $\delta_{\min}(3) = \sqrt{2\sigma z_\alpha/(V_{\text{eff}}\kappa^2 q)} = 0.2028$; extended observation lowers it to 0.1376 ($T = 4$), 0.0973 ($T = 10$), 0.0914 ($T = 20$) and 0.0905 ($T = 100$). At $\delta = 0.05$ the simplified one-leg benchmark (cost $\frac{1}{2}\kappa V_{\text{eff}}^2 + \phi P_* V_{\text{eff}} = 15.875$ per pulse plus initial margin 3.125) needs 271 pulses and 4,305.25 par units with $T = 3$, 58 pulses and 923.88 with $T = 4$, but only 12 pulses and 193.63 with $T = 20$ and 11 pulses and 177.75 with $T = 100$; optimizing the pulse size (6.06 rather than 6.25) lowers the $T = 20$ figure to 182.03. Most of the saving comes from observing the transient response rather than a single tap (Figure S1).

These are costs within an isolated-pulse design class, not minimum capital; the next subsection replaces the formula by a ledger.

## S9.2 Account-level capital and the gap to the lower bound

We run each candidate design through an explicit synthetic ledger at reference price $P_* = 100$: fills at the average linear-book price, taker fees of 4 basis points, marks at the prevailing price, and 0.5% maintenance margin. Initial capital is $C_0 = \max_t(\text{margin}_t - \text{equity}_t)$, maximized over both hypotheses, and the test uses the full price path including unwind legs.

Table S11 gives three conclusions. First, twelve buys that are never unwound fail the 5% target once the test uses the joint price path (divergence $4.95 < 5.41$) and end with 75 units of inventory. Second, if every unwind leg is itself informative, five or six round trips meet the target with zero terminal inventory. Third, pulses are not optimal: a generalized-eigenvector

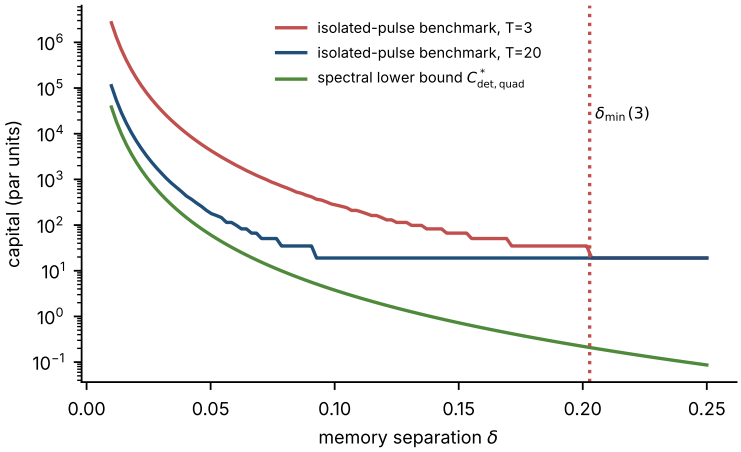



Figure S1: Capital against separation: isolated-pulse benchmarks with three and twenty observations per pulse, and the spectral lower bound of Theorem S9.2. The dotted line is the single-tap resolution floor $\delta_{\min}(3) = 0.2028$. Recomputed in this revision.

probe, which maximizes divergence per unit of fee-free round-trip cost, needs about 65 par units with maximum trade 1.24 and maximum displacement 2.23, well inside limits.

For flat fee-paying policies $v \in \mathbb{R}^L$ with $\sum_t v_t = 0$, the capital problem is

$$\min_{v} \max_{i\in\{0,1\}} L_i(v) + \sum_t \phi P_* |v_t| + \max_t \mu P_* |I_t| \quad \text{s.t.} \quad \|(H_1 - H_0)v\|_2 \geq 2\sigma z_{1-\alpha}, \tag{S4}$$

where $L_i(v) = \frac{1}{2} v^\top T_{h_i} v$ is the exact Toeplitz round-trip loss. Zero-padding embeds horizon $L$ into $L+1$ at identical cost and information, so the minimum is non-increasing in $L$.

**Theorem S9.2** (Spectral dissipation lower bound)**.** *Let $\hat{H}_i(z) = \sum_{j\geq 0} h_{i,j} z^{-j}$, $\Delta\hat{H} = \hat{H}_1 - \hat{H}_0$, and define the dissipation kernel*

$$g_i(\omega) = \operatorname{Re} \hat{H}_i(e^{i\omega}) - \kappa/2 > 0 \qquad \forall \omega \in [-\pi, \pi]. \tag{S5}$$

*(i)* Toeplitz energy identity. *For any flat round trip, $L_i(v) = \frac{1}{2\pi} \int_{-\pi}^{\pi} g_i(\omega) |V(e^{i\omega})|^2 d\omega$.*

*(ii)* Lower bound. $\|(H_1 - H_0)v\|_2^2 \leq \sup_\omega \frac{|\Delta\hat{H}(e^{i\omega})|^2}{g_i(\omega)} L_i(v)$, *so every deterministic round trip with error at most $\alpha$ has*

$$L_i(v) \;\geq\; 4\sigma^2 z_{1-\alpha}^2 \inf_\omega \frac{g_i(\omega)}{|\Delta\hat{H}(e^{i\omega})|^2} \;\equiv\; C^*_{\text{det,quad}}(\delta). \tag{S6}$$

*(iii)* Bracket. *At $\delta = 0.05$, $C^*_{\text{det,quad}} = 61.14$ at $\omega^* = 0.2526$; adding the unavoidable fee $\phi P_* \cdot 2\sigma z / \sup_\omega |\Delta\hat{H}|$ gives the numerical lower estimate $C_{\text{det}} \geq 61.41$. A feasible 240-*

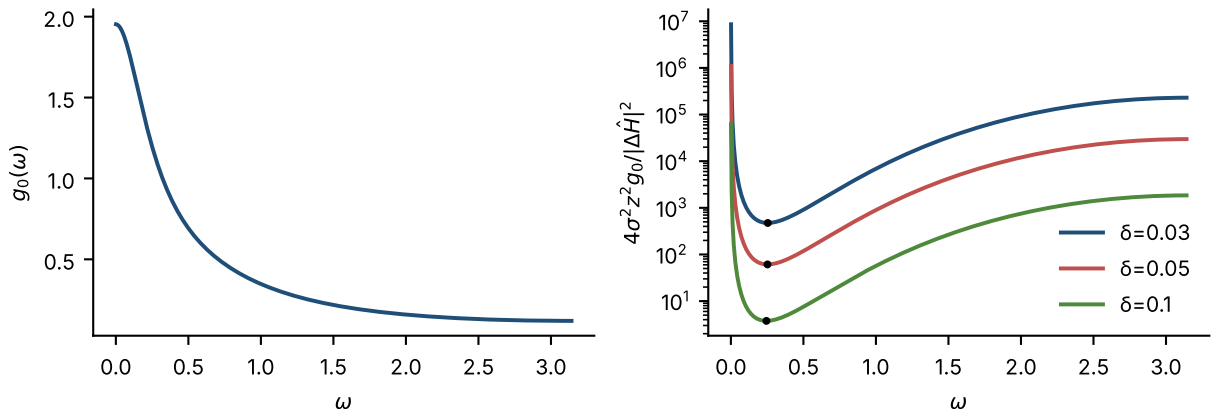



Figure S2: Left: the dissipation kernel $g_0(\omega)$ is strictly positive. Right: the capital ratio in (S6); dots mark its minimum, near $\omega^* \approx 0.25$ for every separation. Recomputed in this revision.

*step sequence attains 64.48, so the fee-paying optimum lies in* $[61.41, 64.48]$.

*(iv)* Process noise. *Let the price also receive the innovations* $w_t \sim N(0, \sigma_w^2)$ *of* (1)*, independent of observation noise* $\nu_t \sim N(0, \sigma_\nu^2)$*, so that from the reset state the observer sees* $\tilde{x}_t = (h_i * v)_t + \kappa^{-1}(h_i * w)_t + \nu_t$*,* $t < T$*. For the known pair of Theorem 5.1, put* $\sigma_{\text{eff}}^2 = \sigma_\nu^2 + \sigma_w^2/4$*,* $\bar{M} = 1/(1-S)$ *and* $C_w = 2\sigma_w^4 \bar{M}^2 K_*^2(\sigma_\nu^2 + \sigma_w^2 \bar{M}^2)/(\kappa^2 \sigma_\nu^2 \sigma_{\text{eff}}^4)$*. Every deterministic round trip whose test has errors* $\alpha$ *and* $\beta$ *under the two hypotheses satisfies*

$$L_i(v) \;\geq\; 2\sigma_{\text{eff}}^2 \big[\mathrm{kl}(\alpha, 1-\beta) - C_w T \delta^4\big]_+ \inf_\omega \frac{g_i(\omega)}{|\Delta\hat{H}(e^{i\omega})|^2}, \tag{S7}$$

*with* kl *as in Theorem S9.3. In-loop noise therefore leaves the* $\delta^{-4}$ *capital order intact for every horizon* $T \leq \mathrm{kl}(\alpha, 1-\beta)/(2C_w\delta^4)$*; at horizons of order* $\delta^{-4}$ *and longer, the noise itself separates the pair without a probe (Theorem 6.5).*

**In words.** This turns the order into a dollar floor. No round trip that tells the two candidate markets apart at 5% error can lose less than this amount, and a concrete probe comes within 5% of it.

Positivity of $g_i$ follows from Lemma 2.3: $g_i(\omega) = \frac{1}{2}\sum_\ell \omega_\ell \frac{1-\lambda_\ell^2}{1-2\lambda_\ell \cos\omega + \lambda_\ell^2} > 0$. We compute $C^*_{\text{det,quad}}$ across separations (Table S12): 61.14 at $\delta = 0.05$, with $\delta^4 C^*_{\text{det,quad}}$ between $3.67 \times 10^{-4}$ and $3.84 \times 10^{-4}$, confirming the $\delta^{-4}$ law. The fee term, the feasible upper bounds and the adaptive bound of Theorem S9.3 are reported in the same table.

The infimum over $\omega$ is evaluated on a fine frequency grid; a local refinement agrees to four digits (61.1377 at $\omega^* = 0.2526$), and the reported constant is the larger of the two

hypotheses' bounds. "Certified" here means a numerical evaluation of the stated bound, not an interval-arithmetic enclosure.

**Theorem S9.3** (Adaptive and sequential probing)**.** *Maintain a common-variance conditional Gaussian likelihood for the causal reset observations. Require finite expected total squared conditional mean separation, finite expected execution loss, and the stopped likelihood/Kullback–Leibler identity used below. Let* $(v_t)_{t\le\tau}$ *be a non-anticipating policy adapted to that filtration, with an almost surely finite stopping time and* $\sum_{t\le\tau} v_t = 0$*, followed by a terminal test* $\psi_\tau \in \{0,1\}$ *with* $P_0(\psi_\tau = 1) \le \alpha$ *and* $P_1(\psi_\tau = 0) \le \beta$*. Then*

$$\mathbb{E}_0[L_0(v)] \;\ge\; 2\sigma^2\,\mathrm{kl}(\alpha, 1-\beta)\inf_{\omega}\frac{g_0(\omega)}{|\Delta\hat{H}(e^{i\omega})|^2}, \tag{S8}$$

*where* $\mathrm{kl}(\alpha,1-\beta) = \alpha\log\frac{\alpha}{1-\beta} + (1-\alpha)\log\frac{1-\alpha}{\beta}$*. The adaptive lower-bound constant divided by the deterministic lower-bound constant is the factor* $\mathrm{kl}(\alpha,1-\beta)/(2z_{1-\alpha}^2) \approx 0.4897$ *at* $\alpha = \beta = 0.05$*, under the stated likelihood and execution assumptions. Comparing two lower bounds alone does not bound the achievable savings relative to an optimal deterministic policy.*

**In words.** The stated likelihood argument supplies an expected-loss lower bound for adaptive probing. It requires an integrable stopped likelihood ratio and valid conditional Gaussian innovations; almost-sure finite stopping alone does not justify optional stopping. The ratio of lower-bound constants is not a bound on achieved percentage savings.

At $\delta = 0.05$ the adaptive bound is 29.94, or 30.22 with the minimal fee. Pairing it with the feasible deterministic 64.48 gives a numerical ratio of at least 0.464. In 20,000 simulated sequential probability-ratio tests, which split the optimal probe into ten copies of equal information, mean loss was 39.5 under $H_0$, between the two bounds, but losses vary with the stopping time: the 95th and 99th percentiles of required account capital were 91.7 and 128.4, and funding a trader with only the mean loss gives a 37.5% insolvency rate before the test ends. Expected-loss savings are not solvency savings.

**Theorem S9.4** (Finite-horizon Rayleigh–Hölder duality)**.** *For a horizon of* $L \ge 4$ *trading periods, let* $E \in \mathbb{R}^{L\times(L-1)}$ *be the first-difference basis of flat probes (*$\sum_t v_t = 0$*),* $Q_i = \frac{1}{2}T_{h_i}$ *(so that* $L_i(v) = v^\top Q_i v$*),* $A_{\mathrm{diff}} = (H_1 - H_0)^\top(H_1 - H_0)$*, and let* $G_i = E^\top Q_i E \succ 0$ *and* $M = E^\top A_{\mathrm{diff}} E \succeq 0$ *be the reduced execution-loss and information matrices, with* $G_i$*-orthonormal generalized eigenvectors* $e_j$ *and eigenvalues* $0 \le \lambda_1 \le \cdots \le \lambda_{L-3} < \lambda_{L-2} \le \lambda_{L-1}$*; put* $c_{\mathrm{req}} = 4\sigma^2 z_{1-\alpha}^2$ *and* $c_{\mathrm{fee}} = \phi P_*$*. For* $\theta \in [0,\pi)$ *let* $d(\theta) = \cos\theta\, e_{L-2} + \sin\theta\, e_{L-1}$*,* $\lambda(\theta) =$

$\lambda_{L-2}\cos^2\theta + \lambda_{L-1}\sin^2\theta$, $s_\theta = \mathrm{sign}(Ed(\theta))$, *and*

$$F_{\mathrm{dom}}(\theta) = c_{\mathrm{fee}}\sqrt{\frac{c_{\mathrm{req}}}{\lambda(\theta)}}\,\|Ed(\theta)\|_1, \qquad W_\perp(\theta) = c_{\mathrm{fee}}^2 \sum_{j\le L-3} \frac{(s_\theta^\top E e_j)^2}{1-\lambda_j/\lambda(\theta)}, \qquad \gamma(\theta) = \frac{\lambda(\theta)\,\lambda_{L-3}}{(\lambda(\theta)-\lambda_{L-3})\,c_{\mathrm{req}}}. \tag{S9}$$

*Then every flat probe meeting the separation constraint of* (S4) *satisfies*

$$L_i(v) + c_{\mathrm{fee}}\|v\|_1 \;\ge\; \min_\theta\left\{\frac{c_{\mathrm{req}}}{\lambda(\theta)} + \Psi(\theta)\right\} \;\equiv\; C_{\mathrm{RH}}(L) \;\ge\; \frac{c_{\mathrm{req}}}{\lambda_{L-1}}, \tag{S10}$$

$$\Psi(\theta) = \min_{x\ge 0}\left\{x^2 + \left[F_{\mathrm{dom}}(\theta)\big(1-\gamma(\theta)x^2\big)_+ - \sqrt{W_\perp(\theta)}\,x\right]_+\right\} \;\ge 0,$$

*a piecewise-quadratic minimization in closed form; when* $\gamma F_{\mathrm{dom}} < 1$ *and the minimizer has* $\gamma x^2 \le 1$, $\Psi \ge \max\big(0, F_{\mathrm{dom}} - W_\perp/(4(1-\gamma F_{\mathrm{dom}}))\big)$, *which is what the bound uses.*

**In words.** This is a computable lower bound on the cost of a finite-horizon probe, fees included; equality with the optimum is not asserted. It confirms that the dollar floor of Theorem S9.2 is not an artefact of assuming an infinitely long probe.

The bound is computable for any horizon. Table S13 reports its quadratic part $c_{\mathrm{req}}/\lambda_{L-1}$, which decays toward the infinite-horizon value 61.14 at rate $O(L^{-1})$, the full bound $C_{\mathrm{RH}}(L)$, and the best feasible fee-paying probe we find at each horizon. At $L = 80$ the full bound is 64.72 against a best probe of 64.76 on the same loss-plus-fee objective (Table S13 reports 65.02 for ledger capital, which also includes margin), so the dual is essentially tight at short horizons; at longer horizons the fee term of the dual weakens and the bracket widens to about 5%. The fee of unpadded full-support eigenvectors grows as $\Theta(\sqrt{L})$, which is why zero-padded or $L_1$-localized probes are needed at long horizons.

## S9.3 Robustness: heteroskedastic impact and probing protocols

**Corollary S9.5** (Invariance under impact-induced heteroskedasticity)**.** *Suppose probing inflates innovation variance as* $\sigma_{i,t}^2(v) = \sigma^2(1+\xi[(h_i * v)_t]^2)$ *and slippage scales as* $\kappa_{\mathrm{eff}} = \kappa(1+\xi_0|v_t|)$. *For capacity-bounded probes,* $D_{\mathrm{KL}}(P_0^v\|P_1^v) \le \frac{1+O(\xi^2 V_{\mathrm{eff}}^2)}{2\sigma^2}\|(H_1-H_0)v\|_2^2 = O(\|v\|_2^2\delta^4)$, *and convex impact only raises dissipation. Required capital remains* $\Theta(\delta^{-4})$.

Four probing protocols (unconstrained single pulse, repeated capped pulses, a sequential test with dynamic throttling, and a predetermined generalized-eigenvector trajectory) all show capital slopes near $-4$ in $\log\delta$: $-3.90$, $-3.84$, $-3.47$ and $-3.81$ over $\delta \in [0.03, 0.12]$,

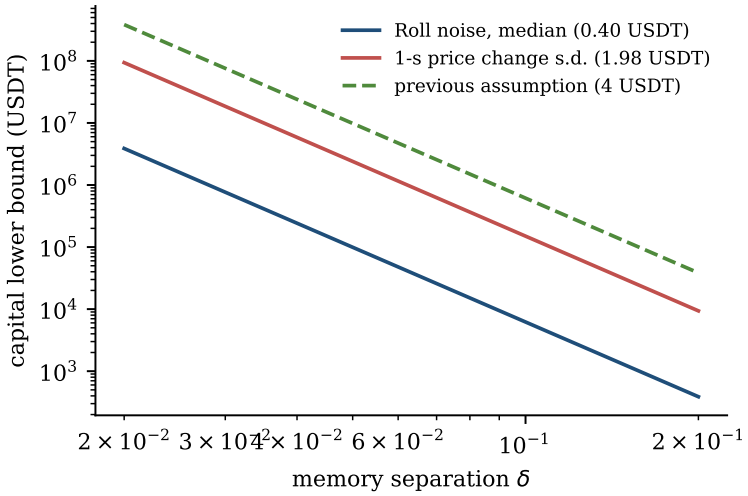



Figure S3: Capital lower bound against memory separation for three observation-noise calibrations.

where the sequential slope is flattened by the 1.7 pulses needed at $\delta = 0.12$ (Appendix Table S2). At equal observation horizon, sequential stopping saves 14–49% of capital relative to repeated capped pulses, but most of the apparent saving from the single-tap baseline comes from observing the transient response, not from adaptivity.

## S9.4 From par units to Bitcoin dollars

Capital scales as $\sigma^2\delta^{-4}$, so the dollar magnitude hinges on the observation noise. Treating the 4 USDT ten-second price volatility as observation noise would give 9.8 million USDT at $\delta = 0.05$, but that conflates price volatility with measurement error. We now estimate observation noise from Binance BTCUSDT one-second closes on January 10–20, 2023, with the Roll covariance estimator $\hat{\sigma} = \sqrt{-\mathrm{cov}(\Delta p_t, \Delta p_{t-1})}$ (Roll, 1984). Daily estimates range from 0.09 to 0.77 USDT, median 0.40 USDT.

At realistic noise, separating rates that differ by 0.05 costs about $9.9 \times 10^4$ USDT, which is not prohibitive for a well-capitalized desk (Table S14). The barrier becomes large only for finer separation (3.9 million USDT at $\delta = 0.02$) and, through Theorem 6.1, when several cohorts are unknown. Two factors we do not model would raise these figures: walking several levels of a real book costs more than the linear top-of-book impact assumed here, and impact-induced heteroskedasticity raises the noise a probe must overcome. The capital figures are therefore lower bounds for the stated model.

# S10 Near-colliding rates: a uniform rate for the merged measure

Theorem 6.2 needs the rates to stay $\gamma$-separated and every weight to stay away from zero. Neither restriction can be dropped for weight recovery: two atoms of weights $w$ and $W-w$ at rates $r$ and $r+\Delta$ and two of weights $W-w$ and $w$ at the same rates have merged measures that converge to the same measure as $\Delta \to 0$, so no estimator recovers the individual weights uniformly over near-colliding economies. What can be recovered uniformly is the merged measure itself, in the permutation-invariant Wasserstein distance.

**Theorem S10.1** (Uniform rate without separation)**.** *Fix $K \geq 1$, a memory bound $\bar{r} < 1$ and $0 < m_{\min} \leq B$. Let $\mathcal{M}_K$ be the merged measures with at most $K$ atoms in $[0, \bar{r}]$ and total mass $k_0 \in [m_{\min}, B]$, with no separation condition and no lower bound on individual weights, and write $\bar{M} = M/k_0$ for the normalized memory distribution. From $N$ reset experiments as in Theorem 6.2, let $\hat{M}$ be any element of $\mathcal{M}_K$ whose first $2K$ kernel values are closest in the sup norm to the estimated ones. There is $C$ depending only on $(K, \bar{r}, a, \kappa, B)$ such that, with probability at least $1-\beta$,*

$$W_1(\hat{\bar{M}}, \bar{M}) \;\leq\; C\Big(\frac{\sigma\sqrt{\log(4K/\beta)}}{m_{\min}\sqrt{N}}\Big)^{1/(2K-1)} \qquad \text{uniformly over } \mathcal{M}_K.$$

*Hence $N = O(\sigma^2 m_{\min}^{-2}\epsilon^{-(4K-2)})$ experiments give accuracy $\epsilon$, and with the Wasserstein statement of Theorem 6.1 the minimax rate for the merged memory distribution is $N^{-1/(4K-2)}$, uniformly over near-colliding economies.*

**In words.** Even with no lower limit on how close the groups' memories are or how small a group is, the combined memory distribution can still be estimated, and no faster rate is possible. What is lost when groups nearly coincide is the ability to tell them apart, not the ability to describe the combined picture.

*Proof.* As in the proof of Theorem 6.2, the kernel values $k_0, \ldots, k_{2K-1}$ are a Lipschitz function of the first $2K+1$ responses on the bounded set of responses the class produces, so a union bound gives $\max_{t<2K} |\bar{k}_t - k_t| \leq e = C_1\sigma\sqrt{\log(4K/\beta)/N}$ with probability at least $1-\beta$ (the Lipschitz constant is for a bounded neighbourhood of the class; when the noise is too large for that, the right-hand side of the theorem exceeds $\bar{r} \geq W_1$ once $C$ is enlarged, so the bound holds trivially). The class is compact and the moment map continuous, so a closest element $\hat{M}$ exists, and $\max_t |k_t(\hat{M}) - k_t(M)| \leq 2e$ by the triangle inequality. Since all atoms lie in $[0,1]$, $|k_t| \leq k_0$, and the normalized moments $m_t = k_t/k_0$ satisfy

$|m_t(\hat{M}) - m_t(M)| \le 2e/k_0 + (|\hat{k}_t|/\hat{k}_0)(2e/k_0) \le 4e/m_{\min}$. The affine map $r \mapsto 2r/\bar{r} - 1$ carries both normalized measures to distributions on $[-1, 1]$ with at most $K$ atoms, multiplies $W_1$ by $2/\bar{r}$ and the differences of moments of order below $2K$ by at most $(1 + 2/\bar{r})^{2K-1}$. The moment comparison inequality for $K$-atomic distributions on $[-1, 1]$ (Wu and Yang, 2020), $W_1 \le c_K \delta^{1/(2K-1)}$ when the first $2K - 1$ moments differ by at most $\delta$, then gives the bound with $C = c_K(\bar{r}/2)(1 + 2/\bar{r})(4C_1)^{1/(2K-1)}$. □

The theorem separates two questions that Theorem 6.2 answers together. Which memory distribution do the participants have? That has the same polynomial cost whether or not their rates collide. Which participant has which weight? That becomes arbitrarily expensive as rates approach each other (Theorem 6.1) and impossible when they coincide (Theorem 3.1). A supervisor who needs the first can have it at the rate of Theorem 6.1; one who needs the second has to be told the number of cohorts and a separation, or has to obtain tagged data (Section 7).

# S11 Robustness to nonlinear and adaptive trading rules

The theorems of Sections 3–6 use the linear exponential-memory recursion (2). Real desks cap positions, trade only beyond thresholds, adapt their gains and learn online. A small probe of size $A$ moves the state by $O(A)$, so any rule whose order is a twice-differentiable function of an exponential-memory state is linear to first order and differs from it by $O(A^2)$; the theorems should then hold for small probes and fail for large ones. We check this, and locate where it fails, in simulations of five stylized rules at $a = 0.5$, $\kappa = 0.8$, total weight $q = 0.2$ and mean rate $\bar{\eta} = 0.4$, over 240 periods after a pulse probe.

The rules are: (1) *momentum with a position limit*, the cohort's order a smooth saturating (tanh) function of its memory state $y_{i,t}$, normalized to unit slope at the operating point; (2) a *threshold rule* that trades only when $|y_{i,t}| > \tau = 0.05$, in two versions, an exact dead zone and one averaged over background state variation of standard deviation 0.04; (3) an *adaptive-gain rule* in the form of recursive least squares with forgetting factor $1 - \eta_i$, whose gain falls when the regressor is large; (4) an *online neural readout*, a linear readout of a fixed two-layer tanh feature map with eight hidden units, trained by regularized stochastic gradient steps of size $\eta_i$; and (5) a *tabular value rule* that updates a 21-bin table at rate $\eta_i$, in two versions, kernel-smoothed across bins and a single-bin lookup. These are stylized: each is built so that its small-signal dynamics are an exponential filter with rate $\eta_i$, and they are not calibrated to any production system.

Two checks follow the theory. *Test A (merged measure):* one cohort with weight $q$ and position limit 1 against two cohorts with weights $q/2$ at the same rate and limits 0.55 and 1.65, so the two economies have the same merged measure; the raw response gap $\|X^{(1)}(A) - X^{(2)}(A)\|_2$ should be $O(A^2)$ (slope 2 in $\log A$) and the probe-normalised gap $\|X^{(1)}(A) - X^{(2)}(A)\|_2/A$ should be $O(A)$ (slope 1 in $\log A$) over $A \in [0.002, 0.064]$. *Test B ($\delta^{-4}$ law):* one cohort at $\bar{\eta}$ against the symmetric pair at $\bar{\eta} \pm \delta$, $\delta \in [0.015, 0.12]$, at probe size $A = 0.005$; the normalized response difference $\|\Delta h\|_2$ should scale as $\delta^2$ and the capital lower bound of Theorem S9.2 as $\delta^{-4}$. The saturation threshold $A_{\text{sat}}$ is the smallest probe at which $\|\Delta h\|_2$ at $\delta = 0.05$ departs by more than 10% from its small-probe value.

To check that these conclusions do not depend on hand-built rules, we pre-specified a second experiment in which online learners are trained from scratch inside the closed-loop market for $T_{\text{burn}} = 5{,}000$ periods of flow and noise innovations; their learned parameters are saved and reloaded into the probing simulator without rescaling. Four learning rules are used: (i) recursive least squares with forgetting factor $\lambda = 0.995$, learning its return-forecast slope; (ii) a two-layer tanh network with eight hidden units whose readout is trained by regularized online stochastic gradient descent on trading profit; (iii) a scalar policy-gradient rule trained on trading profit; and (iv) a tabular Q-learner with 21 states and 21 actions, TD(0) Bellman backups ($\gamma = 0.90$) and Boltzmann action choice. Each learner acts on an exponential-memory state $y_{i,t}$ with rate $\eta_i$: training sets its response gain and curvature, not its memory rate, so the experiment asks whether learned policies preserve the merged-measure and $\delta^{-4}$ predictions, not whether learning produces exponential memory. The specification fixed four predictions: (a) a single exponential fits each trained learner's small-probe impulse response with $R^2 \geq 0.90$; (b) two trained populations with the same merged measure differ by at most 1% in relative $L_2$ response at $A = 0.005$; (c) the capital lower bound to separate trained populations whose memory rates differ by $2\delta$ scales with log–log slope in $[-4.15, -3.85]$; and (d) we report where equivalence and the $\delta^{-4}$ law fail when learning continues during the probe.

Table S16 and Figure S4 show both where the linear reduction holds and where it breaks down. In Panel A, for small probes every smooth or dithered rule behaves like the linear model: economies with the same merged measure have raw response gap $O(A^2)$ (slopes 2.011 to 2.064) and probe-normalised gap $O(A)$ (slopes 1.011 to 1.064), while the response difference and capital lower bound scale as $\delta^2$ (slope 2.007) and $\delta^{-4}$ (slope $-4.001$), with capital at $\delta = 0.05$ within 0.1% of the linear 61.42. In Panel B, predictions (a)–(c) pass for all four trained learners: a single exponential explains $R^2 \geq 0.993$ of each small-probe impulse response, populations with the same merged measure differ by at most $10^{-5}$ in relative response at $A = 0.005$, and capital lower bound scales with slopes $-4.014$ to $-3.973$,

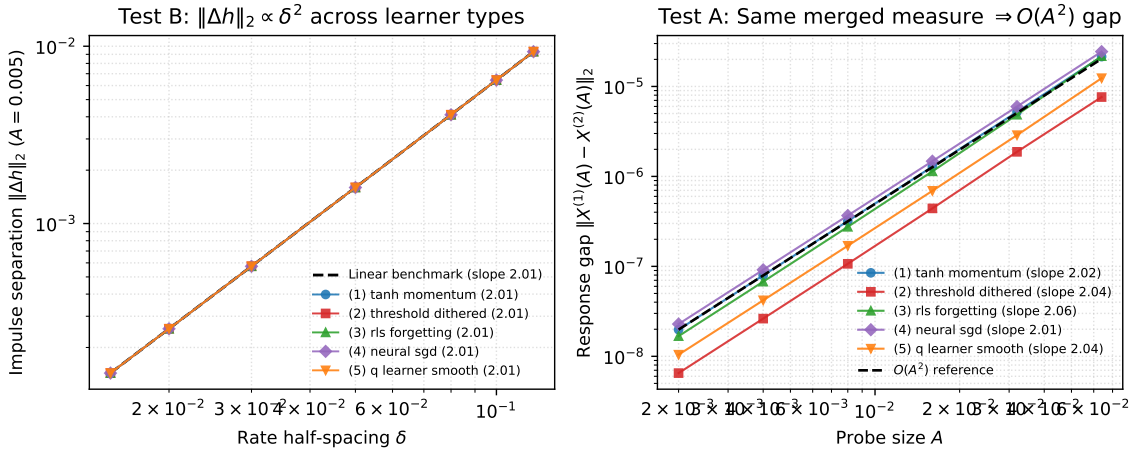



Figure S4: Left: Test B, probe-normalised response difference $\|\Delta h\|_2$ against the rate half-spacing $\delta$ at probe size $A = 0.005$, five stylized rules and the linear benchmark. Right: Test A, probe-normalised response gap $\|X^{(1)}(A) - X^{(2)}(A)\|_2/A$ between two economies with the same merged measure against the probe size $A$, with an $O(A)$ reference line (corresponding to an $O(A^2)$ raw gap).

inside the registered band. The level of capital differs across learners because training sets the response gain.

Panel C shows where this changes. When the two populations are trained on independent innovation streams, prediction (c) fails: the capital slopes are $-1.75$, $-0.97$, $-1.62$ and $-0.38$. Independently trained populations end with learned gains that differ by sampling error of order $T_{\text{burn}}^{-1/2}$, so they differ in merged weight as well as in memory rate, and a difference in weight is cheaper to detect than an $O(\delta^2)$ difference in rates. The $\delta^{-4}$ law therefore prices the separation of memory rates between populations whose feedback weights match; it is not a claim about telling apart any two trained populations. When learning continues during the probe at rate 0.35, the law survives (slopes $-4.02$ to $-3.98$) and the merged-measure gap rises only to $1.1 \times 10^{-3}$, so the breakdown we anticipated under prediction (d) did not occur in this design.

The linear theory fails in two further places. First, saturation, adaptive gains and the neural readout leave the small-signal regime once probes exceed $A_{\text{sat}} \in [0.751, 1.271]$ par units, and the dithered threshold rule leaves it already at $A_{\text{sat}} = 0.049$, the threshold itself. Second, rules with hard discontinuities are different in kind: an exact dead zone ($\tau = 0.05$) or a single-bin lookup does not respond at all to probes smaller than 0.139 and 0.149 par units ($\|\Delta h\|_2 = 0$), concealing those cohorts from small probes altogether. Neither smooth nonlinearity nor hard thresholds offers a cheaper route to separating memory rates than the linear benchmark; populations whose learned weights differ are told apart by their weights,

not their rates.

# S12 A spectral blind spot

Even without identification, could a supervisor monitor risk through eigenvalues or volatility?

**Theorem S12.1** (Spectral blindness)**.** *For any eigenvalues* $0 < \lambda_2 < \lambda_1 < 1$*, one-step price variance* $s_1 > 0$ *and stationary price variance* $V$ *large enough, there is a family of two-state markets* $J_\theta = \left(\begin{smallmatrix} \lambda_1 & \theta \\ 0 & \lambda_2 \end{smallmatrix}\right)$ *with shock covariance* $\mathrm{diag}(s_1, s_2(\theta))$ *such that all members share the eigenvalues, the one-step price variance and the stationary price variance, while the worst-case transient gain* $\sup_k \|e_1^\top J_\theta^k\|$ *grows without bound in* $\theta$*. Here* $s_1$ *is the variance of the fresh price shock given the full prior state; a forecaster who sees prices only can face additional uncertainty about the hidden state, and the family is the one with positive coupling* $\theta > 0$*.*

**In words.** Two markets can look identical by the usual risk measures, the same eigenvalues and the same price variance, and yet one of them can produce far larger losses in the short run. Monitoring eigenvalues or volatility alone therefore cannot certify that a market is safe. The gain is measured in the state coordinates of $J_\theta$; rescaling a hidden coordinate changes its numerical value while leaving the price dynamics unchanged, so translating it into market risk needs a specified shock set, units and loss.

At $\lambda = (0.8, 0.7)$, $s_1 = 1$ and $V = 5$, the gain is 1.77, 5.10, 16.95, 50.85 and 169.50 for $\theta = 1, 3, 10, 30, 100$, with stationary variance 5.000 in every case; the cumulative gain rises from 18.3 to 1,667.7. No monitor that depends only on eigenvalues and second moments of the price can be uniformly safe (Figure S5).

Two clarifications prevent over-reading. First, the cohort economy of Section 2 is itself non-normal, yet its price output does not amplify (Lemma 2.3). Large weighted gains can be manufactured by output weights alone: with $C_w = (1, 160)$, a price-and-memory functional peaks at 160.003 at $t = 0$, which is the static weight, not dynamic growth (Supplement S3.5). Second, in a network of $M$ identical cohort economies coupled by symmetric nonnegative cross-impact, the coupled operator stays symmetric in the coordinates of Lemma 2.3; with cross-impact 0.05, the spectral radius rises from 0.822 ($M = 1$) to 0.973 ($M = 10$) while the Kreiss constant stays at one, as it must for a stable symmetric operator. A stable normal contraction cannot exhibit Euclidean state-norm transient growth. This does not exclude scalar input-output overshoot relative to initial impact, which also depends on the input and output maps. The positive-cohort no-overshoot result is model-specific; the companion paper studies a depth-depletion mechanism.

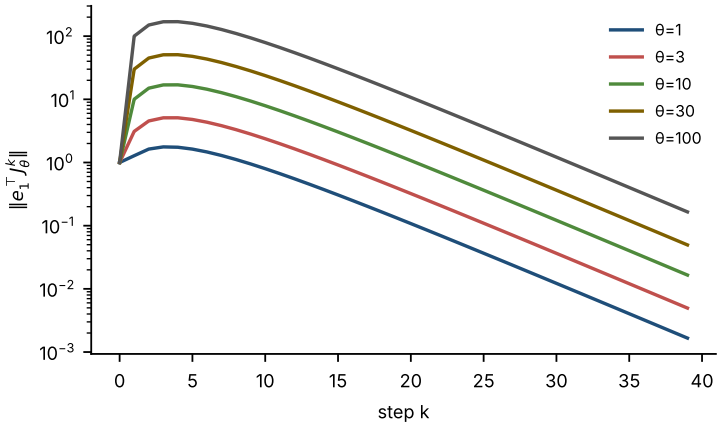



Figure S5: Price-row transient gain of the spectral-blindness family $J_\theta$: eigenvalues $(0.8, 0.7)$, one-step price variance 1 and stationary price variance 5 are the same for every $\theta$.

# S13 Empirical feedback and its measurement

The theory says what prices can reveal. The empirical work has two roles, and neither is a test of the identification bounds themselves, which concern controlled experiments no outside researcher can run. First, Supplement S19 tests the theory's policy prediction directly: that class-tagged data pin down memories that prices leave unrestricted. Second, this section asks what the reduced-form feedback indices a supervisor would compute from public data actually measure, and how much of them is feedback rather than filtering, timing or trading activity.

## S13.1 Estimand and filtering mechanics

In theory $x_t = p_t - p_t^*$ is the deviation from an unobserved fundamental. Empirical work replaces $p_t^*$ by a trailing filter. Let $p_b$ be a log price proxy on uniform bars of length $\Delta$ and $m_b = (1-\alpha)m_{b-1} + \alpha p_b$ an exponential moving average with $\alpha = 2/(W/\Delta + 1)$. The empirical state is

$$x_b^{\mathrm{EMA}} \equiv p_b - m_b = (1-\alpha)(p_b - m_{b-1}) = \frac{1-\alpha}{1-(1-\alpha)L}\Delta p_b. \tag{S11}$$

Because $x_{b+1}^{\mathrm{EMA}} = (1-\alpha)(p_{b+1} - m_b)$, a martingale price gives

$$\mathbb{E}[x_{b+1}^{\mathrm{EMA}} \mid \mathcal{F}_b] = (1-\alpha)x_b^{\mathrm{EMA}} : \tag{S12}$$

the filtered state is an AR(1) with mechanical persistence $a_0 = 1-\alpha$, which equals $29/31 \approx$

0.9355 for $W = 300$ seconds and $\Delta = 10$ seconds.

Let $f_t$ be signed flow. The empirical VAR(1)

$$\begin{pmatrix} x_{t+1}^{\mathrm{EMA}} \\ f_{t+1} \end{pmatrix} = \begin{pmatrix} J_{11} & J_{12} \\ J_{21} & J_{22} \end{pmatrix} \begin{pmatrix} x_t^{\mathrm{EMA}} \\ f_t \end{pmatrix} + e_{t+1} \tag{S13}$$

has reduced-form coefficients, not authenticated impact or learner parameters. When denominators are nonzero we use the scale-invariant indices

$$\hat{Q} = \frac{\hat{J}_{12}\hat{J}_{21}}{1 - \hat{J}_{22}}, \qquad \hat{S} = \hat{J}_{11} + \hat{Q}, \qquad \hat{L} = \frac{\hat{Q}}{1 - \hat{J}_{11}}. \tag{S14}$$

Equation (S12) implies that $\hat{J}_{11}$ must be compared with $1 - \alpha$, not with zero: persistence that a naive reading attributes to feedback can be pure filtering.

## S13.2 Data

Table S17 summarizes the Binance spot and perpetual samples used in this section.

## S13.3 Clock alignment dominates reduced-form feedback

We estimate (S13) on ten-second bars built from one-second Binance klines across a 96-date, eight-pair spot panel (the 10th and 24th of each month, 2021–2024) under three constructions (Table S18), and run the identical pipeline on a placebo random walk with each pair-day's volatility. Pairing the state with *next-bar* flow (a look-ahead) gives $\hat{Q} = 0.0081$; aligning flow with the bar in which it is observed lowers this by more than an order of magnitude, to 0.00037; and replacing the EMA state by raw returns gives an interval that includes zero. Mean $\hat{J}_{11}$ under observed flow is 0.9203, below the mechanical 0.9355, and the placebo reproduces the mechanical value (0.9351) with $\hat{Q}$ indistinguishable from zero.

## S13.4 Feedback beyond filtering mechanics

We re-estimate feedback on a separate sample and add a placebo. We use Binance BTCUSDT one-second klines on 72 sampled days in 2021–2023. The state $x_t$ is the log price deviation from its EMA (span 30 one-second bins), $f_t$ is next-second signed taker flow, and a VAR(1) gives $\hat{Q}$. Because the filter alone produces persistence, we run the identical pipeline on a placebo: a random walk with the same one-second volatility, independent of the real flow.

Table S20 gives two findings. Feedback, in the sense of a loop gain the filter cannot produce, is detectable: the real $\hat{Q}$ lies far outside the placebo band. It is also small, and

the price persistence that a naive reading would attribute to feedback is almost entirely mechanical (0.9364 against 0.9355). Because this construction pairs the state with next-second flow, part of $\hat{Q}$ reflects the timing effect of Table S18; the placebo shows only that the filter does not produce it.

## S13.5 Order splitting and the estimation of feedback

Persistent order flow makes lagged-flow regressions inconsistent. Suppose flow follows $q_t = \gamma y_t + u_t$ with memory $y_t = (1-\eta)y_{t-1} + \eta\Delta p_{t-1}$, order splitting $u_t = \rho u_{t-1} + \nu_t$, and $\Delta p_t = \kappa q_t + \lambda z_t + e_t$ with an observed exogenous price shock $z_t$. Lagged returns are correlated with $u_{t-1}$, so OLS of flow on returns is biased even when $\gamma = 0$. Quasi-differencing ($q_t - \rho q_{t-1} = \gamma(y_t - \rho y_{t-1}) + \nu_t$) or instruments that move $y$ through prices but are independent of $u$ ($z_{t-1}, z_{t-2}, z_{t-3}$) restore consistency. We compare estimators over 500 replications with $\eta = 0.3$ and $\rho = 0.8$ (Table S21).

OLS is badly biased. Cochrane–Orcutt with the true $\rho$, 2SLS and LIML have correct size and coverage; Cochrane–Orcutt with an estimated $\rho$ under-covers (Cochrane and Orcutt, 1949; Staiger and Stock, 1997). These are properties of estimators in a simulated market. On real data, a 2024 sub-panel of 192 spot units shows why the correction matters little at our bar length: with one VAR lag, the flow-equation residuals have essentially no first-order serial correlation (median $\hat{\rho} = -0.0002$), and Cochrane–Orcutt quasi-differencing moves the mean loop gain from $3.15 \times 10^{-4}$ to $3.21 \times 10^{-4}$ (post hoc). Order splitting shows up instead at longer lags, which the 30-lag control of Table S19 addresses. A real-data instrumental-variables estimate needs an instrument whose exclusion can be defended, which we do not claim to have; funding-settlement times, for example, may affect order flow directly.

Controlling for 30 ten-second bars of lagged flow lowers $\hat{Q}$ further (Table S19, column $\hat{Q}_{30}$), consistent with persistent order splitting.

## S13.6 Feedback across 30 perpetual markets

The companion paper estimates the same loop gain on one-minute bars of 30 Binance USD-M perpetuals spread across the liquidity ranking (2023–2024 training days). Measured feedback rises with depth: its Spearman correlation with log depth within 1% of the mid is +0.70 [0.41, 0.87], from about 0.012 in BTCUSDT and ETHUSDT to 0.0016 on average in the thinnest third, and the placebo pipeline gives $\hat{Q} \approx 0$ for every symbol. Feedback in this reduced-form sense is a feature of deep, algorithmically traded books.

### S13.7 Two fee events

Binance set spot trading fees on BTC pairs to zero on July 8, 2022 and reinstated them on March 22, 2023. Each event changes the cost of trading BTCUSDT while leaving the seven other pairs unchanged, and the second reverses the first. For every pair and every day within 20 days of each event we compute daily quote volume and the observed-flow $\hat{Q}$, and estimate $Y_{it} = \alpha_i + \gamma_t + \beta\,\mathrm{ZeroFee}_{it} + e_{it}$ with pair and date fixed effects and date-clustered errors, separately for each event and pooled (Table S22).

The fee change moved activity enormously: zero fees multiplied BTCUSDT volume relative to the controls by $e^{0.945} \approx 2.6$ when they began, and their removal divided it by $e^{1.599} \approx 4.9$. Measured feedback did not follow. The two point estimates have opposite signs, neither is significant, and the pooled effect is $-0.00016$ with an interval of $[-0.00087, 0.00055]$, against a sample mean of 0.00037. A strong first stage with no second-stage response says that $\hat{Q}$ is not a by-product of trading activity, and the reversal design removes the main weakness of a single treated event. With one treated asset the interval remains wide, so this is evidence that $\hat{Q}$ does not scale with activity, not a proof of invariance.

## S14 Discussion of the assumptions

**Linearity.** The model is linear in prices, flows and memory states. Linearity is what makes the merged-measure equivalence exact and the testing problems Gaussian. Nonlinear strategies (for example threshold rules or position limits) can break the equivalence class, but they do so in regions of the state space that a small probe does not visit; for small probes the linearization around the steady state is the relevant object.

**Price-first timing.** We order the price update before the cohort update. Under the alternative lagged timing the transition matrix becomes $\left(\begin{smallmatrix} a & 1 \\ \eta Q & 1-\eta \end{smallmatrix}\right)$ and its eigenvalues differ, so stability statements must be computed under the timing actually used. The identification results depend on the transfer function and survive either convention.

**Exponential memory.** Lemma 2.1 justifies exponential memory for filtering and regularized learning. Neural and reinforcement-learning policies are not exponential filters. Rules whose orders are smooth functions of an exponential-memory state are exponential filters to first order, so the results hold for small probes; Supplement S11 checks this in stylized simulations and finds where it fails, and how closely production algorithms follow such rules remains an empirical question we do not settle. Mixtures of exponentials can approximate any completely monotone kernel, so a power-law memory corresponds to a continuum of

cohorts, for which Theorem S17.2 shows that accuracy improves only logarithmically in the data.

**Algorithms that react to being probed.** The probing results assume that the cohorts keep their memories while they are probed. An algorithm that detects a probe could pause, shorten its memory or trade against the probe. Four observations limit what this does to our conclusions. First, a reaction that all cohorts share does not help separate them, and an algorithm that mimics another group's memory rate only brings the hypotheses closer, so neither lowers the costs we derive. A reaction that differs across cohorts is a new signature, but one the observer must specify and estimate, so it shifts the identification problem rather than solving it. Second, the passive route needs no probe: Theorem 6.5 shows that natural price variation faces a lower bound of the same polynomial order, so an observer who avoids detection by not trading does not escape the barrier; it pays in time instead of money. We do not show that passive data attain that order. Third, detection is itself a testing problem against the same noise: the generalized-eigenvector probe of Supplement S9 trades at most 1.24 par units in any period, and a probe sized within normal flow variation is hard to tell from noise, so the threat bites mainly for large single pulses, which our ledger shows are not optimal anyway. Fourth, what we cannot rule out is an algorithm designed to defeat surveillance, which turns identification into a game between observer and observed; our results are the benchmark against which such a game would be measured, and we leave it to future work.

**Known primitives.** The resource theorems take $\kappa$, $a$ and the noise law as known. Unknown primitives can only make separation harder, so the lower bounds remain valid. Assuming known primitives simplifies the baseline statements and strengthens the lower bounds; Corollary 6.3 shows that estimating them leaves the order of the upper bound unchanged when memory rates are bounded away from zero: the pre-probe price removes $p^*$, the impact tap gives $\kappa$, and $a$ is recovered from one further tap because it acts as an extra group with memory rate zero. In an equilibrium in which dealers set the impact $\kappa$ from the order flow they see, as in Kyle (1985), a prober's unexpected trades would also move $\kappa$; we do not model this, and expect it to raise the cost of probing rather than lower it.

**Observation contracts.** Each result names what the observer sees. A price-only observer, an observer with executed quantities, one with noisy statements, one with audited equity and one with tagged flow face different experiments. The paper's central message is that these differences, not the observer's statistical skill, determine what can be learned.

# S15 A worked example: two Kalman desks

To make the orders concrete, consider two desks that track the same trend with Kalman filters (Lemma 2.1(i)) and differ only in their signal-to-noise ratios: $\sigma_\zeta^2/\sigma_\nu^2 = 0.09$ and 0.16. Their gains are $\eta_1 = 0.2584$ and $\eta_2 = 0.3279$, so $\bar{\eta} = 0.2931$ and $\delta = 0.0348$. With equal weights, the economy is observationally the split pair of Theorem 5.1 around $\bar{\eta}$, and the response difference first appears at lag two, $d_2 = -\kappa^2 q\delta^2 = -1.55 \times 10^{-4}$ (verified).

A supervisor who wants to tell this market from a single desk with gain $\bar{\eta}$ at 5% error using a single pulse and the third observation needs $A = 2\sigma\Phi^{-1}(0.95)/(\kappa^2 q\delta^2) = 212$ units of executed quantity, and, under a 0.5% margin and 4 basis points of fees, single-pulse capital of 18,166 par units. The pulse is very inefficient: the spectral lower bound of Theorem S9.2, evaluated at this pair ($\bar{\eta} = 0.2931$, $\delta = 0.0348$), gives about 77 par units for the optimal flat probe (rescaling the $\bar{\eta} = 0.4$ constant by $\delta^{-4}$ alone would give 262, because the constant also depends on $\bar{\eta}$). Converting with the measured Roll noise (factor $(0.40/0.01)^2$), the optimal probe costs roughly $1.2 \times 10^5$ USDT and the naive pulse about $3 \times 10^7$ USDT. Two desks whose signal-to-noise ratios differ by almost a factor of two are therefore expensive, but not impossible, to tell apart by probing; three or more desks are much harder (Theorem 6.1).

# S16 Do the rates survive real market noise?

Theorems 6.1 and 6.2 are proved for observation errors that are independent and Gaussian, and Table S35 checks them with Gaussian noise. Real price changes are different: one-minute returns have heavy tails, although they are almost uncorrelated from one minute to the next. To see whether the rates depend on the Gaussian assumption, we ran a pre-specified test (opened once) in which the trader groups are synthetic, with known memory spacing, and only the noise is real. Each reset experiment returns the price response to a unit impulse plus a window of consecutive one-minute log-price changes of Binance BTCUSDT, demeaned and scaled to unit variance. The changes come from 21 days, the 6th of each month from January 2025 to September 2026, that are used nowhere else in this project; their kurtosis is 19.4, against 3 for Gaussian noise. With class tagging, the experiment also returns each group's flow response, observed through independent windows of the same noise. A control arm uses i.i.d. Gaussian noise of the same variance. For each spacing, $N^*$ is the smallest number of experiments at which every recovered weight lies within 25% of the smallest true weight in at least 75% of 200 replications, using Prony's method on prices alone or the per-group regression of Proposition 7.1 with tagging. The noise level and the spacing grids were fixed on Gaussian noise alone, before any real noise was downloaded.

Both pre-specified hypotheses hold, so the test passes (Table S23). With real noise, the price-only requirement for two groups rises from 64 experiments at $\Delta\eta = 0.5$ to 370,728 at $\Delta\eta = 0.125$, a factor of about 5,800, and the fitted slope is $-6.01$; for three groups it is $-9.81$. Over the same grid the tagged requirement stays between 256 and 724 experiments, and it falls slightly rather than rises as the spacing shrinks. At every spacing the real-noise requirement is within one step of the grid (a factor $\sqrt{2}$) of the Gaussian control. The slopes are least-squares fits of $\log N^*$ on $\log \Delta\eta$ over the stated grids (seven spacings from 0.5 to 0.125 for two groups, five from 0.6 to 0.25 for three). No uncertainty interval was pre-specified; as a post hoc description, the ordinary least-squares standard errors are 0.17 and 0.47 with real noise (0.12 and 0.31 for the Gaussian control, slopes $-6.22$ and $-9.96$), and each $N^*$ is itself read off a grid with steps of $\sqrt{2}$. For two groups, the last four price-only points lie above 4,096 and use the normal draw, so the heavy tails enter the fit only through the first three.

The test has a limit, stated in its specification. Averages of $N$ real windows are resampled exactly up to $N = 4{,}096$; beyond that they are drawn as Gaussian vectors with the real windows' covariance, as the central limit theorem suggests. Real one-minute changes are almost uncorrelated (lag-one autocorrelation between $-0.007$ and 0.009 within the windows), so beyond 4,096 the real and Gaussian arms differ little by construction. The direct evidence about heavy tails therefore comes from the points in the exact range: every tagged requirement, and the price-only requirements for two groups at the three widest spacings (64, 362 and 1,448 experiments, against 64, 362 and 1,024 with Gaussian noise). The other four price-only points for two groups, and all points for three groups, check the rate with the real noise's variance and covariance but not its tails. Averaging should make heavy tails matter less as $N$ grows, but the test does not show this.

**In words.** Where real Bitcoin price noise could be used directly, it needs about as many experiments as Gaussian noise, and over the whole grid halving the spacing between two groups still multiplies the number of experiments by about 64, while class-tagged records keep the requirement flat. At the largest sample sizes the test relies on a normal approximation to the real noise.

# S17 General memory: the minimax rate is logarithmic

Exponential memory is a modelling choice. Suppose instead that the feedback kernel is completely monotone, $k_m = \int_{[0,\rho]} r^m \, d\mu(r)$ with $\rho < 1$, where $\mu$ is any nonnegative measure of total mass $W$: a finite or infinite mixture of exponential memories with merged weight $W$.

Write $\mathcal{M}(\rho, W)$ for this class, assume the closed loop is uniformly stable, $s \equiv a+\kappa W/(1-\rho) < 1$, and measure accuracy by

$$d(\mu, \mu') = \sup\Big\{ \textstyle\int f \, d(\mu - \mu') : \;\; f \text{ 1-Lipschitz on } [0, \rho], \;\; f(0) = 0 \Big\},$$

which equals the Wasserstein-1 distance between measures of equal mass and also compares estimates whose mass is not exactly $W$. Let $E_L(|x|)$ denote the error of the best uniform approximation of $|x|$ on $[-1, 1]$ by polynomials of degree $L$, and let $c_* = \inf_{L \geq 1} L\, E_L(|x|)$. Bernstein (1914) showed that $L\, E_L(|x|)$ converges to a positive constant (0.2802; Varga and Carpenter, 1985), so $c_* > 0$; numerically $c_* = 1/4$, attained at $L = 2$ (Table S24).

**Lemma S17.1** (Moment-matched cohort pairs)**.** *For every $n \geq 2$ there are two probability measures on $[0, \rho]$, each with exactly $n$ atoms, whose moments of orders $0, \ldots, 2n-2$ coincide and whose Wasserstein-1 distance is at least $\rho\, E_{2n-2}(|x|) \geq c_* \rho/(2n-2)$.*

The atoms are the points where the best approximation of $|r - \rho/2|$ by polynomials of degree $2n-2$ attains its maximal error with alternating signs, and the weights are those of the divided difference on these points, which annihilates every polynomial of degree $2n-2$ (Supplement S1). Each measure is an $n$-atom merged measure in the class; one atom may sit at rate 0 (a cohort with $\eta = 1$), which the class allows.

**Theorem S17.2** (Minimax rate for general memory)**.** *Let $N$ independent reset experiments observe the impulse response with Gaussian noise of variance $\sigma^2$ per tap, and let $B(\rho) = (1+\sqrt{2})(1+2/\rho)$ and $C_h = \kappa^2/(1-s)^2$.*

*(i)* Lower bound. *Let $\Lambda_N = \log(N/\sigma^2) + \log\big(2C_h^2 W^2/((1-\rho)\rho)^2\big)$. Whenever $\Lambda_N > 4\log(1/\rho)$, every estimator $\hat{\mu}$ satisfies*

$$\sup_{\mu \in \mathcal{M}(\rho, W)} P_\mu\Big( d(\hat{\mu}, \mu) \geq \frac{c_*\, \rho W \log(1/\rho)}{\Lambda_N} \Big) \;\geq\; \frac{1}{4e},$$

*and the bound already holds over $n$-cohort economies, for a finite $n$ that grows with $N$.*

*(ii)* Upper bound. *With $a$ and $\kappa$ known, invert the averaged response into kernel values $\hat{k}_0, \ldots, \hat{k}_L$, with $L+1 = \lfloor \log(N/\sigma^2)/(4 \log B(\rho)) \rfloor$, and let $\hat{\mu}$ be any measure on $[0, \rho]$ minimizing $\max_{j \leq L} |\int r^j d\hat{\mu} - \hat{k}_j|$ (a linear program). For every $\beta \in (0,1)$ there is $N_0$ depending only on $(\rho, W, a, \kappa, \beta)$ such that for $N/\sigma^2 \geq N_0$,*

$$\sup_{\mu \in \mathcal{M}(\rho, W)} P_\mu\Big( d(\hat{\mu}, \mu) > \frac{3\pi \rho W \log B(\rho)}{\log(N/\sigma^2)} \Big) \;\leq\; \beta.$$

*Hence the minimax accuracy is of exact order* $1/\log(N/\sigma^2)$*, and reaching accuracy* $w$ *requires and suffices* $N/\sigma^2 = \exp(\Theta(1/w))$.

*Proof sketch.* (i) Put the measures of Lemma S17.1, scaled to mass $W$, into the model. Their difference annihilates polynomials of degree $2n-2$, so the generating function of the kernel difference is $\int(1-rz)^{-1}d(\mu_A-\mu_B) = z^{2n-1}\int r^{2n-1}(1-rz)^{-1}d(\mu_A-\mu_B)$; on $|z|=1$ it is at most $2W\rho^{2n-1}/(1-\rho)$, and Parseval bounds $\|k^A-k^B\|_2$ by the same quantity. The response difference is $H_AH_B$ times the shifted kernel difference, so $\|h^A-h^B\|_2 \le C_h\|k^A-k^B\|_2$. Choosing the smallest $n$ for which the Kullback–Leibler divergence of the $N$-sample experiments is at most one, the Bretagnolle–Huber form of Le Cam's lemma gives the bound with half the Wasserstein separation of the lemma. (ii) For a 1-Lipschitz $f$ with $f(0)=0$, Jackson's theorem gives a polynomial $q$ of degree $L$ with $\sup|f-q| \le \pi\rho/(4(L+1))$, and expanding $q$ in Chebyshev polynomials of $2r/\rho-1$ bounds the sum of its monomial coefficients by $\rho B(\rho)^{L+1}$. Hence $d(\hat\mu,\mu) \le \pi\rho(W+\delta)/(2(L+1)) + 2\delta\rho B(\rho)^{L+1}$, where $\delta$ is the largest kernel error. The kernel is recovered from the response by a series reciprocal, $K(z) = (1-az-\kappa/H(z))/(\kappa z)$, whose error is at most $2((1+s)/\kappa)^2(L+2)$ times the largest response error, so $\delta = O_P(\sigma N^{-1/2}L\sqrt{\log L})$. The choice of $L$ makes $\delta B(\rho)^{L+1} = O_P((\sigma^2/N)^{1/4}L\sqrt{\log L})$, negligible against the first term. Full proofs are in Supplement S1. □

**Machine check.** Lean verifies part (i), for $N$ reset experiments of any length, using only Lean's standard axioms. The formal proof replaces the pair of Lemma S17.1 by an elementary one: the binomial weights $W\binom{2L+1}{k}/2^{2L}$ on the rates $\rho k/(2L+1)$, odd $k$ against even $k$, which have $n = L+1$ atoms each and equal moments of orders $0,\dots,2n-2$. The 1-Lipschitz function $f(r) = (h/\pi)(1-\cos(\pi r/h))$, $h = \rho/(2L+1)$, vanishes at 0 and at even nodes and equals $2h/\pi$ at odd nodes, so the pair is at distance at least $2\rho W/(\pi(2n-1))$, more than the $c_*\rho W/(2n-2)$ of the lemma, without Bernstein's theorem. Because the raw moments agree, the kernels agree below lag $2n-1$ and differ by at most $2W\rho^{2n-1}/(1-\rho)$ in $\ell_1$ over all lags; the closed-loop $\ell_1$ bounds, the Gaussian divergence and the testing lemma then show that, with $C = 2\kappa^2W/((1-\rho)(1-s)^2)$, $\Lambda = \log(2NC^2/\sigma^2) > 0$ and $L = \lceil\Lambda/(4\log(1/\rho))\rceil$, no estimator reaches accuracy $2\rho W\log(1/\rho)/(\pi(\Lambda+6\log(1/\rho)))$ with probability at least $3/4$ under both economies. This is the order of part (i), with success probability $3/4$ in place of $1/(4e)$ and constants of its own. An estimator $\hat\mu$ enters through its integrals $\int f\,d\hat\mu$ of 1-Lipschitz functions with $f(0)=0$, which is how $d(\hat\mu,\mu)$ is defined. Lean also verifies part (ii). Mathlib has no Jackson theorem, so the formal proof builds one: on $M = n+2$ equally spaced nodes the Fejér–Korovkin kernel $K(u) = |\sum_{k\le n} a_k e^{iku}|^2/\sum_k a_k^2$, $a_k = \sin(\pi(k+1)/(n+2))$, is nonnegative, has node mean one and node mean of $K(u)(1-\cos u)$ equal to $1-\cos(\pi/(n+2))$ (from the recurrence of the sines and the orthogonality of roots of unity), so the positive

operator it defines maps an $L$-Lipschitz function on $[-1, 1]$ to a polynomial of degree at most $n$ within $L\pi/(n+2)$, by Cauchy–Schwarz on the nodes. Written in Chebyshev polynomials and rescaled to $[0, \rho]$, its coefficients sum in absolute value to at most $\rho(n+1)B(\rho)^n$. With an explicit bound for the deconvolution, $\alpha e(1+\beta')^t$ for kernel value $t$, and the sub-Gaussian union bound, any measure on $[0, \rho]$ that fits the first $L+1$ deconvolved kernel values at least as well as the truth, the linear-programming estimate among them, is within a finite-sample bound of $\mu$ with probability at least $1-\beta$; taking $L = \lfloor \log(N/\sigma^2)/(8 \log D) \rfloor$, $D = B(\rho)(1+\beta')$, gives $16\pi\rho W \log D / \log(N/\sigma^2)$ for all $N/\sigma^2$ beyond a threshold depending only on $(a, \kappa, \rho, s, W, \beta)$. The formal constants are larger than the theorem's $3\pi$ and $B(\rho)$, because the formal deconvolution bound grows geometrically in $L$; the order is the same.

**In words.** If trader memory is a general mixture of exponentials rather than a few clean ones, accuracy improves only with the logarithm of the data: to halve the error, the data must be squared. The few-groups model of the previous results is therefore the optimistic case.

Theorem S17.2 replaces a polynomial barrier with a logarithmic one. With exponential memory and $K$ known cohorts, halving the error multiplies the required number of experiments by a constant; with general memory, halving the error squares it. The lower bound holds already for finite mixtures, so knowing that the memory is a finite mixture of exponentials does not help unless the number of cohorts is also known (Theorem S10.1). The constants of the two bounds differ, by a factor that grows as $\rho \to 1$, so the dependence on how long memory lasts is not pinned down. The source of the gap is visible in the proof: as $\rho \to 1$ the moment-matched pairs separate their kernels more slowly ($\rho^{2n-1}$ decays slowly), which weakens the lower bound through $\log(1/\rho)$, while the upper bound pays $\log B(\rho)$, which stays bounded away from zero, for the conditioning of polynomial approximation on $[0, \rho]$. Logarithmic rates of this kind are familiar from deconvolution with supersmooth errors (Fan, 1991) and from moment methods for mixing distributions (Kong and Valiant, 2017; Wu and Yang, 2020); here they arise from the closed loop of a market.

Table S24 checks the construction and the estimator. The pairs of Lemma S17.1 have exactly $n$ atoms, agree in all moments below order $2n-1$ to $10^{-61}$, and remain at Wasserstein distance about $0.4/n$, while the distance between their kernels shrinks by a factor of about 0.2 per atom at $\rho = 0.9$. The linear-programming estimator applied to a continuous memory distribution has error times $\log(N/\sigma^2)$ between 0.18 and 0.28 for $N$ from $10^4$ to $10^{24}$: a $10^{20}$-fold increase in data halves the error. The exponential-memory model of the previous subsections, where the cost is polynomial in $1/\Delta\eta$, is therefore the optimistic case: relaxing the parametric form makes identification from prices much harder, not easier.

# S18 Mandated disclosure: the CFTC's trader-class reports

**The stock–flow inventory bridge.** To connect the theoretical flow recursion (2) and inventory rule (Remark 2.2) to the empirical position stocks $n_{c,t}$ reported in regulatory filings, recall from Remark 2.2 that cohort $k$'s inventory $I_{k,t} = (1-\delta_{\text{inv}})I_{k,t-1} + u_{k,t}$ under optimal trading $u_{k,t} = q_k y_{k,t} - \psi_{\text{inv},k} I_{k,t-1}$ and belief update $(1-r_kL)y_{k,t} = \eta_k x_t$ satisfies the coupled recursion

$$I_{k,t} = \rho_{\text{inv},k} I_{k,t-1} + q_k y_{k,t} + \nu_{k,t}, \qquad (1-\rho_{\text{inv},k}L)(1-r_kL)I_{k,t} = w_k x_t + (1-r_kL)\nu_{k,t}, \tag{S15}$$

where $\rho_{\text{inv},k} \equiv (1-\delta_{\text{inv}}) - \psi_{\text{inv},k} = \lambda_{\text{tc},k}(1-\delta_{\text{inv}})/(\lambda_{\text{tc},k} + \gamma_{\text{inv},k}) \in [0,1)$ is the inventory-turnover root and $r_k = 1-\eta_k \in (0,1)$ is the belief-memory root. Equation (S15) has two immediate empirical implications. First, a class position stock $n_{c,t}$ carries two autoregressive roots: the inventory-adjustment root $\rho_{\text{inv},c}$ (which approaches one when inventory holding costs $\gamma_{\text{inv},c}$ are small relative to execution costs $\lambda_{\text{tc},c}$) and the belief-memory root $r_c$. A one-lag autoregression $n_{c,t} = \alpha_c + r_c^{\text{pos}} n_{c,t-1} + w_c \,\text{ret}_t$ therefore estimates the dominant persistence of the class's position state—which prices alone cannot bound—while separating the faster belief root $r_c$ from slow turnover $\rho_{\text{inv},c}$ requires either the two-lag specification $(1-\phi_1 L - \phi_2 L^2)n_{c,t}$ or first-differenced class flows $\Delta n_{c,t}$, both of which we test below.

The cleanest test uses disclosure mandated by law. The U.S. Commodity Futures Trading Commission requires large traders in regulated futures markets to report their positions and publishes a weekly snapshot of Tuesday positions, usually on Friday, aggregated by reporting category: leveraged funds, asset managers/institutional investors and dealers/intermediaries in financial futures, and managed money, producers/merchants/processors/users and swap dealers in commodities. The categories reflect each trader's reported business, not the algorithm that traded. These are published, aggregated class positions, not class-tagged flow. We pre-specified a test before downloading any data, in eight markets that span asset classes: CME Bitcoin and Ether, the E-mini S&P 500, 10-year Treasury notes, the euro, the yen, WTI crude oil and Henry Hub natural gas, using every weekly report through October 2026 (282 to 1,039 weeks per market). For each class, $n_{c,t} = (\text{long} - \text{short})/\text{open interest}$ follows $n_{c,t} = \alpha_c + r_c n_{c,t-1} + w_c \,\text{ret}_t$, the weekly version of the cohort recursion. The price-only alternative regresses each week's return on its own twelve lags and fits a two-exponential kernel $\phi_j = b_1 r_1^{j-1} + b_2 r_2^{j-1}$ to the twelve coefficients by least squares over the grid $0 \le r_1 < r_2 \le 0.99$ in steps of 0.02. Weekly futures returns are close to serially uncorrelated, so the fitted weights are close to zero and the rates are weakly identified nuisance parameters (Andrews

and Cheng, 2012); that is part of what the wide price-only intervals show. Returns come from Binance spot (Bitcoin, Ether), the S&P 500 index, euro and yen spot rates, minus the change in the 10-year Treasury yield, and front-month crude-oil and natural-gas futures (Yahoo Finance). Only the last two are futures returns; for the other six the basis between spot and the CME contract enters the return regressor; this matters most for the sign of $w_c$ in markets where leveraged funds run basis trades. The comparison is descriptive, not a race between estimators of the same object: the class route fits the persistence of an observed position series, the price route infers latent kernel rates from returns. Intervals come from a 13-week moving-block bootstrap; near a unit root an autoregressive coefficient is estimated at rate $T^{-1}$ rather than $T^{-1/2}$, so the narrow class intervals partly reflect the near-unit-root persistence of position stocks, such percentile intervals are pointwise and can understate uncertainty, and price-only intervals that fill the admissible range are better read as a sign of weak identification than as calibrated 95% intervals. The pre-specified hypotheses were: (H1, primary) the fast class's memory rate is pinned to an interval narrower than 0.25 while both price-only intervals are wider, in at least 75% of markets; (H2) the fast class trend-follows, $w_c > 0$, in at least 75%; (H3) the fast class has the shorter memory in at least half.

The primary hypothesis holds in all eight markets (Table S25). The class reports pin the fitted persistence rate of each fast class's positions to an interval between 0.013 and 0.106 wide, while the price-only kernel leaves both rates anywhere in the admissible range; the width ratios run from 9 to 71. The ratio partly reflects how little weekly returns reveal, which is the point of the theory rather than an artefact of it. A fitted persistence of aggregated positions is not a structural learning memory: it also reflects how slowly positions turn over and how accounts enter and leave a class. In the model with inventory costs (the remark after Lemma 2.1) a position accumulates order flow, so it carries an inventory-adjustment root, close to one when inventory costs are small, on top of the belief root $r_i$, and a one-lag fit to positions mostly picks up the former. The classes differ as the cohort model allows. Leveraged funds and managed money have the less persistent positions in six of eight markets, with half-lives of 6 to 18 weeks against months to years for asset managers and producers; in Treasuries and natural gas the two persistences are indistinguishable.

The trend-following hypothesis fails. The fast class buys after rises in the euro, the yen, crude oil and natural gas, but sells after rises in Bitcoin, Ether and the S&P 500 futures. The first half matches the finding of Moskowitz et al. (2012) that speculators' positions tend to go with past returns. We did not predict this. One candidate explanation is that leveraged-fund positions in financial futures are dominated by the short futures leg of cash–futures basis trades, which a single memory per class cannot represent (Barth and Kahn, 2021). A post-hoc check is consistent with it but not decisive: weekly position changes of leveraged funds

and asset managers are strongly negatively correlated in Bitcoin, Ether, the S&P 500 and Treasuries ($-0.37$ to $-0.67$) but not in the euro or yen ($-0.04$ and $-0.05$), and holding the asset managers' change fixed removes the negative response in Bitcoin and halves it in Ether and the S&P 500. Because class positions sum to zero across all reporting categories, some negative correlation is mechanical, so this is a signature, not a test. The failure sharpens rather than weakens the case for class tagging: class labels are not strategies, and a supervisor who wants to know how a class responds to prices needs its positions, which the price process does not reveal. Negative estimated $w_c$ also lies outside the positive economy of Section 2, so the dollar bounds of Supplement S8 do not apply to these classes; the pre-specified restriction test does not use positivity.

**Post hoc robustness.** After the registered test we reran the same estimators with three changes, approved before any of them was run and reported without re-scoring the registered result: CME futures prices (BTC, ETH, ES, ZN, 6E and 6J front months) for the six markets that used spot, index or yield prices; the futures version with contract-roll weeks removed by a calendar rule fixed in advance; and 26- and 52-week bootstrap blocks. With futures prices the counts are unchanged (H1 8 of 8, H2 4 of 8, H3 6 of 8), and so is the sign of $\hat{w}_{\text{fast}}$ in every market: the fast class still sells after rises in Bitcoin, Ether and the S&P 500. The spot–futures basis in the registered returns therefore does not produce the negative signs. Longer blocks leave the counts unchanged; Bitcoin's response interval then includes zero. Removing roll weeks leaves H1 and H2 unchanged, lowers H3 to 4 of 8 and turns the Treasury response negative. H3 is lost in Bitcoin and Ether, where the rule removes 46% of the weeks (200 of 436 and 131 of 284): a report is dropped whenever any day of its week touches the five-day expiry window, so with monthly expiries the rule removes about two weeks in four, more than intended, and the fast–slow difference is no longer resolved in the shorter samples. In the equity-index, currency and Treasury markets it removes 15% of the weeks, and in crude oil and natural gas none. The primary result, that class positions pin down a persistence that prices leave open, holds in all seven variants.

**Class flows: a pre-specified test that failed.** Proposition 7.1 is about tagged flow, while the registered test uses positions. A second pre-specified test (opened once) reads the same reports as flows, the weekly change in the fast class's net position, and estimates $u_t = c + r\,u_{t-1} + w\,\text{ret}_t$ by the instrumental-variables regression of the proposition, with last week's return as the instrument for last week's flow and the same 13-week bootstrap. Its primary hypothesis, that the 95% interval for $r$ is narrower than 0.5 in at least six markets, fails. It holds in three: the S&P 500, crude oil and natural gas, with widths from 0.36 to 0.44.

The intervals are 0.57 and 0.60 wide in the euro and the yen, 1.08 in Ether, 3.4 in Bitcoin and 10.8 in Treasuries. The instrument is strong (first-stage $F \geq 10$) in seven markets, all but Treasuries ($F = 4.3$), and the flow response $w$ is nonzero in six, with the same signs as in positions: negative in Ether and the S&P 500, positive in the euro, the yen, crude oil and natural gas. Read as flows, then, the CFTC reports identify how each class's flow responds to prices, but not how persistent that flow is, at the precision the test required. In Proposition 7.1 the precision is of order $T^{-1/2}$ with a constant that grows with the noise in the reported flow ($\sigma_f^2$ in (15)). Weekly changes in aggregated positions also move when accounts enter or leave a class, which is noise for this purpose, and two markets have short samples (279 and 428 weeks); we did not test these explanations. The failure limits the claim that published class aggregates implement the tagged-flow contract of Proposition 7.1. It does not bear on the proposition itself, which is a statement about the model.

**Separating memory from turnover: a pre-specified test that failed.** A fitted persistence of positions mixes two things: in the model with inventory costs (the remark after Lemma 2.1) a class position follows a slow inventory or turnover root and, on top of it, the faster belief-memory root $r_i$. A third pre-specified test (opened once) asked whether the CFTC positions show that second root. For each class it fits $n_t = \alpha + \phi_1 n_{t-1} + \phi_2 n_{t-2} + b_0 \mathrm{ret}_t + b_1 \mathrm{ret}_{t-1}$, takes the roots $z_1 \geq z_2$ of $z^2 - \phi_1 z - \phi_2$, and requires, for the fast class in at least six markets, that the 95% interval of $z_2$ lie inside $(0, 0.9)$ and that $z_1 - z_2$ be clearly positive. It fails: the condition holds in two markets, crude oil and natural gas. The larger root is between 0.90 and 0.99 in every market, as in Table S25. The smaller root is between $-0.16$ and 0.21; it is negative in Bitcoin, Ether, the S&P 500 and Treasuries, and where it is positive its half-life is under half a week (0.22 to 0.44 weeks), far shorter than any learning memory the weekly reports could reveal. Positions therefore behave as one slow component plus short-lived noise. By the interpretation fixed before the run, this is a failure of the two-root reading of CFTC positions, and it confirms the limitation stated above: these data restrict how persistent class positions are, but they cannot separate learning memory from the turnover of positions.

**Replication on twelve new markets.** On 31 October 2026 we registered (opened once) the same test, with the code, the bootstrap and the thresholds unchanged, on twelve CFTC markets it had never been run on: gold, silver, copper, corn, soybeans, wheat, live cattle, coffee, sugar, cotton (disaggregated reports, managed money against producers), the Nasdaq-100 and the British pound (financial reports, leveraged funds against asset managers), with Yahoo Finance front-month or index prices and reports from June 2006 to 27 October 2026

(1,039 to 1,041 usable weeks). The primary hypothesis replicates in all twelve markets: the fast class's rate interval is 0.014 to 0.042 wide, while both price-only intervals are 0.85 to 0.96 wide, a width ratio of 23 to 69. Trend following (H2), which failed on the original eight markets, holds in eleven of the twelve: the managed-money or leveraged class buys after price rises everywhere except the Nasdaq-100, where it sells, as in the S&P 500. We report this as a new test on new markets; it does not undo the earlier failure, and the two results together say that the sign of the speculative class's response depends on the market, with commodities and currencies trend-following and equity indices not. H3 fails: the fast class is less persistent than the slow class in three markets (gold, silver and the Nasdaq-100); in the other nine the two classes' persistence cannot be told apart, since managed money and producers hold positions for similar times in agricultural and industrial commodities. The replication therefore extends the primary result from eight to twenty markets and leaves the limitation unchanged: class positions pin down how persistent positions are, not whether that persistence is learning memory.

**A descriptive precision conversion.** The recorded interval-width ratios are not an identified equivalent sample size: the two routes estimate different objects, price intervals hit parameter bounds, and near-unit-root positions need not obey uniform root-$T$ scaling. Table S26 reports the earlier arithmetic conversion only as a hypothetical calculation conditional on common root-$T$ scaling. It is not a lower bound on years of price data needed. A first-half/full-sample interval comparison cannot establish an asymptotic rate or repair the estimand mismatch.

**A hypothesis for the failures: offsetting strategies within a class.** The pre-specified scorecard (Table S29) reports that the directional trend-following test (H2) and the flow-persistence tests failed in several markets, notably in equity indices, where leveraged funds' positions move against recent returns rather than with them. One explanation is aggregation: the CFTC's "Leveraged Funds" category can contain cash–futures basis traders, whose futures positions may fall when prices rise, alongside momentum funds, whose positions rise. If such groups respond to the same price move with opposite signs, their responses partly offset in the reported total and the estimated persistence and price response of the class are attenuated. We have not tested this explanation, and the failures are equally consistent with weak responses, misspecification of the weekly model or noise. Two weak responses and two strong offsetting ones look alike in the aggregate; distinct memory rates can also offset at short lags and diverge at long ones. A test would compare subgroup responses to a common shock, with uncertainty intervals, against the observed aggregate, which requires

records finer than the public report. The failures therefore motivate such a test; they do not show that more granular records are necessary.

Together with Table S27, the evidence now covers two disclosure regimes, one voluntary and one mandated by law, in crypto and in regulated equity-index, rate, currency and commodity futures. In both, class-tagged positions restrict persistence rates that prices leave unrestricted; whether that persistence is learning memory or slow turnover of positions, these data do not say.

# S19 Voluntary disclosure: Binance's trader-class ratios

Binance publishes voluntarily, for one venue but every five minutes, long/short ratios for its top-trader group, by position and by account, and for all accounts. These are class-level positions, the closest public analogue to the tagged data of Proposition 7.1, but not homogeneous cohorts: the top-trader position ratio is built from position sizes, the all-account ratio from counts of long and short accounts, the top-trader group is a subset of all accounts, and its membership can change. We treat a class's reported net position as the observable counterpart of its cohort state: it obeys the same one-lag recursion class by class, so the block-diagonal information structure of Proposition 7.1 carries over. The mapping is an assumption, because the reports are positions rather than flows and returns stand in for the unobserved deviation $x_t$. For each class $c$ we estimate $n_{c,t} = r_c n_{c,t-1} + w_c \,\mathrm{ret}_t$, with $n = (R-1)/(R+1)$ from the ratio $R$. The price-only alternative regresses signed five-minute taker flow on twelve lagged five-minute returns and fits two exponentials to the kernel; it estimates the memory rates without class information. Intervals come from a day bootstrap with 200 draws.

**Reading guide.** The tests below were run in the order described, but each answers one of three questions. Every outcome is in the scorecard.

(1) *Do class ratios restrict persistence that prices leave open?* Yes. See Table S27; on the fairest, horizon-matched comparison, over 30 fresh markets, the price-only intervals are a median 7.3 times wider.

(2) *Do class-identified memories improve return forecasts?* No. Neither pre-specified out-of-sample test finds a gain.

(3) *Do class data track or respond to prices?*

- Prices and flow cannot reconstruct where the classes are positioned: the state-tracking test passes (Table S28), but its gain is persistence.

- The October 2026 feedback and flow tests failed.

We begin with the one-hour price kernel of Table S27 and end with the fairer horizon-matched comparison on 30 new markets, which reduces the width ratio to a median of 7.3. Table S27 gives three results. First, tagged class data restrict persistence rates that prices and flow leave essentially unrestricted: with the one-hour price-only kernel, the price-only intervals are 28 to 124 times wider in every one of 30 symbols on fresh days, and cover most of the admissible range. Part of this ratio is mechanical, because a series of positions identifies its own persistence directly while the price-only route must fit two exponentials to a weakly identified kernel. Restricting attention to the five symbols in which the price-only fit does identify the merged response (ACH, API3, OGN, ZEN and DASH) does not change the conclusion: the price-only rate intervals are still 0.64 to 0.92 wide and the ratios run from 27.6 to 62.5 (median 47.5). We therefore read the ratios as a measure of how little prices reveal about class memory, not as a horse race between two estimators of equal standing. The ratios also overstate the gap for a second reason: the price-only kernel spans one hour (twelve five-minute lags), while the tagged half-lives are several hours, so the price-only fit cannot see memories of that length. An exploratory check on the 30 symbols and 105 days of the follow-up test refits the price-only kernel with 48 and 96 lags (four and eight hours) and compares both routes in one unit, the half-life. With 96 lags the price-only slow half-life (median 5.6 hours) reaches the tagged scale (median 4.2 hours). The tagged intervals remain narrower in all 30 symbols, but by a median factor of 7.5 in log half-life rather than 28 to 124, and the price-only interval still reaches a bound of the fit in 25 of 30. The horizon-matched comparison was then pre-specified as a primary hypothesis of a test on 30 symbols never used in this project, before its data were downloaded, and passed: the tagged interval was narrower in 29 of 29 evaluable symbols (median factor 7.3, smallest 2.6), and the price-only interval reached a bound of the fit in 26. A pre-specified secondary of the same test qualifies what the tagged estimate measures. Day-demeaned regressions bias persistence down; corrected by a split-panel jackknife, the median tagged rate is 1.0008 per five minutes, and the corrected interval lies below one in only 7 of 87 class fits. Within a day the class ratios are close to a random walk, so the half-lives of a few hours reported here are largely produced by removing each day's mean. The evidence therefore supports a narrower claim than identification of class memory: positioning data restrict a persistence parameter that prices cannot bound. Whether that persistence reflects learning memory or slow turnover of positions, which theory's tagged contract (class-tagged flow) would distinguish, is not established by these data. Second, the classes differ where theory says the information should be: in BTCUSDT and ETHUSDT the top-trader group's positioning has a half-life of about three to five hours against about one to two days for all accounts, while in thin

symbols the classes are often indistinguishable. Third, the pre-specified test fails its 80% criterion, on the part of the prediction that concerns what prices *do* reveal: the merged response is significantly different from zero in only 5 of 30 symbols. Theorem 6.4 says the merged measure is identified; on 42 days of thin-market data it is not estimated precisely. This is a failure of the prediction as pre-specified, and we report it as one. A post hoc reading, not tested here, is that identification is an asymptotic statement and 42 days of passive thin-market data are far from that regime. Two follow-ups were pre-specified before their data were downloaded, and each ran once. On 105 untouched days in the same 30 markets, price-only intervals were again at least ten times wider than class-tagged ones in all 30 symbols (median ratio 74.9 with the one-hour kernel), but the merged response was significantly different from zero in only 9 symbols, and pinning it down was at least ten times cheaper than separating the memory rates in only 10 of 30 (23 were required): that prediction failed. On 58 further days, the advantage of tagging grew with the sample in 22 of 29 symbols, the number required, but the tagged intervals shrank more slowly than the parametric rate (20 of 29 in the predicted range, 22 required), consistent with drift in class persistence over 21 months. Complete tables for both follow-ups are in the public replication repository.

To test directly whether the memory rates identified by class-tagged positions improve out-of-sample return predictions after large order-flow shocks, we pre-specified a dedicated out-of-sample test before downloading the 22nd-of-month data for January 2025–September 2026 across 19 thin perpetuals (6,137 top-decile five-minute flow shocks on 2024 training days; 4,579 shocks on untouched 2025–2026 test days). On the training sample we estimate each symbol's two memory rates $\hat{r}_1, \hat{r}_2$ either from the class-tagged positions or from the price-only two-exponential fit, construct the filtered memory states $y_{k,t} = \hat{r}_k y_{k,t-1} + (1 - \hat{r}_k)\mathrm{ret}_t$, and forecast the 30-minute post-shock cumulative return $R_{t+1\to t+6}$ out of sample relative to an AR(1) return-and-flow baseline. The primary hypothesis fails: on the untouched test shocks, the price-only memory states yield $\Delta R^2 = -0.199$ percentage points relative to the baseline, the class-tagged memory states yield $\Delta R^2 = -1.288$ points, and the tagged-minus-price-only difference is $\Delta R^2 = -1.089$ points (95% date-block bootstrap CI $[-2.605, +0.376]$, Clark–West $t = 0.96$, $p = 0.1684$). The difference is statistically inconclusive, and there is no evidence of a gain from class data. Likewise, adding the level and one-hour change of the top-trader and all-account positions to the hourly tail-risk controls of the companion paper leaves out-of-sample $R^2$ unchanged ($-0.19$ points, CI $[-0.99, 0.83]$).

After this test failed, we pre-specified a follow-up on untouched 2nd-and-9th-of-month data for January 2025–September 2026 across the same 19 thin perpetuals (12,046 top-decile five-minute flow shocks on 24 training dates in 2024; 9,875 shocks on 42 test dates; test

opened once). It asks the narrower question that Theorem 3.1 and Proposition 7.1 answer directly: can prices and flow track the class states themselves? A two-exponential price-and-flow filter ($\alpha_{\text{fast}} = 0.50$, $\alpha_{\text{slow}} = 0.10$ on five-minute returns and signed flow, plus shock return, shock flow and symbol fixed effects) is compared with a filter that adds the current class states and their one-bar changes. Both pre-specified primaries pass by a wide margin (Table S28): adding the class states raises out-of-sample $R^2$ from $-4.8\%$ to 46.0% for the 30-minute divergence between top-trader and all-account positions, and from 4.4% to 54.8% for the top-trader state. As the next paragraph shows, the gain reflects the near-unit-root persistence of position levels, not predictability of future trading.

The gain is almost entirely persistence. In an exploratory decomposition on the same test shocks, an AR(1) on the current level alone accounts for 100.5% and 99.1% of the improvement, and when the target is the 30-minute *change* in positions the class data add $-0.26$ and $+1.41$ points. Because the class filter observes the series it forecasts, this is a weak test: it confirms that prices and flow cannot reconstruct where the classes are positioned, which Theorem 3.1 implies, but it does not show that class data reveal how their positions will move, and neither pre-specified test finds that class data improve return forecasts.

**October 2026: feedback and flow, two pre-specified tests that failed.** The state-tracking result above passes, but its gain is persistence. On 31 October 2026 we registered (opened once) two tests on a window no earlier test had used, 1–30 October 2026, for 30 liquid USD-M perpetuals fixed in advance (all 30 met the data rule), with a power check on synthetic data before registration. N2 asks whether price changes over the last half hour improve out-of-sample forecasts of the change in the top-trader position ratio beyond its own six lags, fitting on 1–20 October and evaluating on 21–30 October (about 2,600 five-minute bars per market) with a one-sided Diebold–Mariano test. It fails: $p < 0.05$ in one market (18 required), and the median out-of-sample $R^2$ gain is $-0.01$ percentage points, below the 0.8 to 1.6 points the test could detect. At five-minute frequency, then, this class position does not respond to recent prices beyond its own dynamics by a detectable amount. N3 asks whether the taker-flow imbalance, an untagged aggregate flow, pins down the combined one-hour response $\Psi = \sum_{m=1}^{12} k_m$ to past returns, with a Newey–West interval. Its primary criterion (interval excluding zero and half-width below $|\Psi|/2$) fails, holding in 5 of 30 (24 required); as a description, the interval excludes zero in 17 markets, with negative responses in 8 (among them Bitcoin and Solana) and positive in 9 (among them Dogecoin and Filecoin), and the sign is the same in both halves of the month in 21 (24 required). The one secondary prediction about rates holds in every market: the interval from ten days is 1.33 to 2.45 times as wide as the interval from thirty (median 1.69; root-$T$ shrinkage predicts 1.73). Flow data

shrink uncertainty at the rate the theory gives, but the aggregate response is small, noisy and of different signs across markets, so a month of five-minute data does not pin it down at the precision we required.

# S20 Who the results matter for

The results point in a constructive direction for several groups of readers.

- *Supervisors and policymakers* get a way to decide what data to require. If the goal is only to watch overall price risk, public prices are enough, because the combined price response is learned at the ordinary rate (Remark S25.2). Labelled records earn their cost when supervision must be targeted, for example when one algorithm must be paused while the others keep supplying liquidity, and when the reporting system covers enough markets to share its fixed cost (Proposition S27.1). In our stylised cost model this illustrates a possible economy of scale for market-wide reporting; it does not explain the legal design of regimes such as CAT. For the narrow task of attributing behaviour to classes, a persistent pseudonymous label is enough, which bears on how much personal data such a task needs but not on what enforcement needs (Section 8). Proposition 7.4 gives a stylised decision in which this matters: the size of a market-wide measure depends only on the combined response, while deciding whether halting one group is enough needs that group's own intensity and inherits the full cost of separating the groups.
- *Exchanges and trading venues*, which already see account-level flow, learn which questions about their participants public prices cannot answer, and why publishing class-level positioning, as some crypto venues already do, adds information that prices do not contain.
- *Risk managers and trading desks* get a realistic limit on what can be inferred about other traders or about crowding from market data, and a warning that common feedback measures computed from public data mostly reflect filtering and timing rather than true feedback (Supplement S13).
- *Researchers* who estimate models with several trader types get one reason why the parameters that separate the types are often imprecise, and a number that says how precise they can be before any estimation (Supplement S4).

# S21 Robustness checks, the failed tests and real noise

Rerunning the CFTC test with CME futures prices, without contract-roll weeks or with longer bootstrap blocks leaves its primary result unchanged. Read as weekly flows rather than positions, the same reports failed a second pre-specified test: they pin down how each class's flow responds to prices, but not how persistent that flow is. A third test asked whether positions carry a second, faster root that could be learning memory, as the model with inventory costs predicts; it failed too (2 of 8 markets), so these data cannot separate learning memory from the slow turnover of positions. Another pre-specified test gives synthetic groups real Bitcoin price noise instead of Gaussian noise and finds the predicted rates, a slope of $-6.01$ against $-6$ for two groups and $-9.81$ against $-10$ for three, while class-tagged requirements stay flat; at large sample sizes it uses a normal approximation to the real noise (Supplement S16). These comparisons are descriptive: a position series reveals its own persistence directly, while the price route must infer it. A replication registered on 31 October 2026 reproduces the primary CFTC result on twelve new markets (12 of 12). Two tests on fresh October 2026 Binance data failed: price changes did not forecast a class position beyond its own persistence, and aggregate taker flow did not pin down the combined response, although its uncertainty shrank at the predicted root-$T$ rate. Several predictions failed, and Table S29 lists them next to the ones that passed. The data therefore show what prices cannot pin down. They do not show that this persistence is the learning memory of the model.

# S22 Two supplementary results

Neither is needed for the main results. Markets with the same eigenvalues, the same one-step volatility and the same long-run price variance can still have very different worst-case short-run losses, so watching eigenvalues or volatility cannot replace knowing who trades (Theorem S12.1). And standard feedback measures computed from public data mostly reflect filtering and clock alignment: two fee changes that moved trading volume by factors of 2.6 and 4.9 left measured feedback unchanged (Supplement S13).

# S23 How large are the costs? A hypothetical illustration

To give a sense of scale we price the trading option in a hypothetical benchmark. The model parameters are illustrative; only the size of the measurement noise is estimated, from Bitcoin

data. An outside participant that re-learns two unknown algorithms each week with test trades would lose at least $15 million to $356 million a year per market; with three unknown algorithms it would lose more than $40 million a year per market even when their memories are far apart (Supplement S8). These are lower bounds for one stated way of learning, not estimates of what any regulator spends: regulators do not trade to learn. They are best read as the price of the information that labelled records deliver without any trading.

# S24 Execution cost and capital

A probe changes inventory as well as price. We therefore specify cash and execution timing. Price has units of currency per asset, and $\kappa$ currency per squared quantity.

**Lemma S24.1** (Fair execution)**.** *For a quote schedule refreshed after the uncontrolled state update, execution at the average price of a linear book costs, relative to the constant fundamental mark,*

$$C(v) = \tfrac{1}{2} v^\top T_h v + c\|v\|_1, \qquad (T_h)_{ij} = h_{|i-j|}. \tag{S16}$$

*In the positive stable economy $T_h$ is positive definite, so no nonzero round trip produces a deterministic gain.*

Positivity here means that the response is a positive mixture of decaying exponentials (Lemma 2.3), whose spectral density is positive. Positive, decreasing taps alone do not suffice: for $h = (1, 0.9, 0.1)$ the $3 \times 3$ matrix $T_h$ has determinant $-0.468$, so some round trip earns money. The conclusion concerns modelled execution cost only.

We verify the ledger identity (S16) against an explicit cash account to $5.7 \times 10^{-13}$ on 1,000 random probes, and the smallest eigenvalue of $T_h$ over 1,000 random stable positive economies is 0.141. A direct-only proxy $\sum \frac{1}{2}\kappa v_t^2$ omits cross-time impact: for $v = (+1, -1)$ it gives 0.80 against a true Toeplitz loss of $h_0 - h_1 = 0.3488$.

## S24.1 Capital supplied by a counterparty

**Theorem S24.2** (Sharp robust quantity cap from cash and margin)**.** *Use the positive, stable, reset economy with constant fundamental price $P_* > 0$ and no process noise. A designated dealer fills only nonnegative probes $v_t \geq 0$ at the average price $P^0_{t+1} + \kappa v_t/2$ of a linear schedule, holds inventory $I_{t+1} = I_t - v_t$ and cash $K_{t+1} = K_t + v_t(P^0_{t+1} + \kappa v_t/2) - cv_t$, starts with equity $B > 0$, and must satisfy $\mu|I_t|P_t \leq E_t$ at every time. For the robust contract*

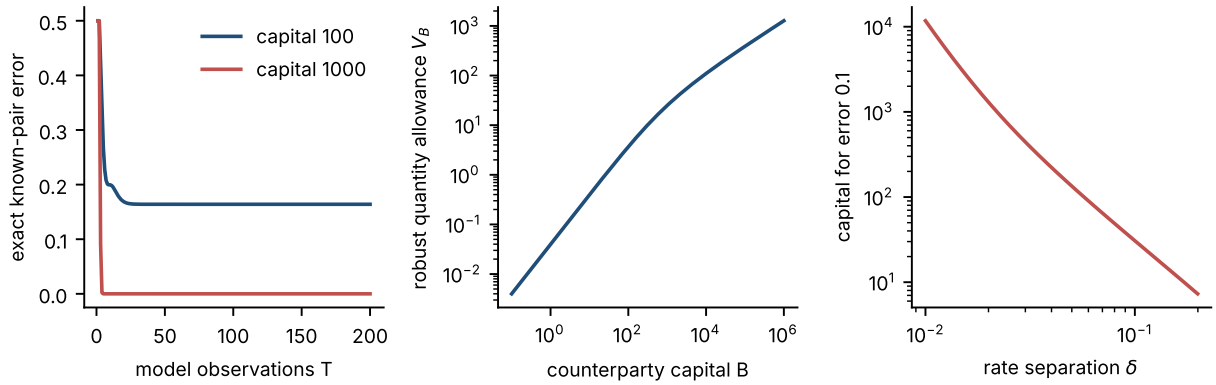



Figure S6: Known-pair experiment funded by the counterparty cash/margin contract. (a) Exact Gaussian error at fixed capital 100 and 1,000, separation $\delta = 0.05$. (b) The sharp robust quantity allowance (S17). (c) Capital for error 0.1 with 200 observations. Parameters: $a = 0.5$, $\kappa = 0.8$, $q = 0.2$, $\eta = 0.4$, observation s.d. 0.003, reference price 100, margin 0.25, cost 0.02. No process noise.

*admitting every probe with $\sum_t v_t \le V$, the largest allowance guaranteeing margin is*

$$V_B = \frac{2B}{\beta + \sqrt{\beta^2 + 4(\mu + \frac{1}{2})\kappa B}}, \qquad \beta = c + \mu P_*, \tag{S17}$$

*equivalently $(\mu + \frac{1}{2})\kappa V^2 + (c + \mu P_*)V \le B$. The cap holds across all positive cohort representations with $S < 1$, without resolving individual rates. An observer that receives only noisy prices and executed quantities has best error $\Phi[-V_B \|d_T\|_2/(2\sigma)]$ within this allowance; fixed error $\alpha < 1/2$ requires capital of order at least $\delta^{-4}$, and a single positive pulse attains that order.*

**In words.** A probe moves the price against the account that makes it, so the margin it needs grows with the square of the probe. Because the probe grows like $1/\delta^2$, the capital grows like $1/\delta^4$: halving the difference multiplies the capital needed by 16.

Even with zero initial inventory, the probe creates a quadratic adverse mark that a fixed initial price exposure would miss. Figure S6 illustrates the experiment.

**Proposition S24.3** (Exact financial disclosure changes the experiment)**.** *In this noiseless-state account, exact disclosure of the third equity mark after a single positive pulse $0 < A \le V_B$ distinguishes the known pair with zero error for every $\delta > 0$.*

**In words.** If the observer can see an account's exact equity, a single test trade settles the question, however close the two candidate markets are. The hard part of the problem is

therefore not the market itself but what the observer is allowed to see.

Exact marked equity is extra information, not a free observation assumed by the price-only theorem. Real account marks have their own precision: with reports rounded to 0.01, a guaranteed change requires a larger trade (Section 7).

### S24.2 Capital paid by the probe trader

**Lemma S24.4** (Time-domain Toeplitz coercivity: capital order for funded probes)**.** *Use the same economy and execution law. The statistical observer receives only noisy prices and executed quantities. The probe trader starts with cash $B > 0$ and zero inventory; a finite signed sequence $v$ changes inventory by $+v_t$ and cash by $K_{t+1} = K_t - v_t(P^0_{t+1} + \kappa v_t/2) - c|v_t|$, after which trading stops and the trader must remain solvent as the price returns to $P_*$. Put*

$$r = 1 - \epsilon(1 - S) < 1, \qquad m = \kappa \frac{1 - r}{1 + r} > 0, \qquad K_* = \frac{2\kappa^2 q}{\epsilon^3 (1 - S)^2}.$$

*Every solvent strategy satisfies $D_{\mathrm{KL}}(P^v_0 \| P^v_\delta) \le K_*^2 B \delta^4/(\sigma^2 m)$, so error at most $\alpha < 1/2$ requires*

$$B \;\ge\; \frac{2\sigma^2 m (1 - 2\alpha)^2}{K_*^2 \delta^4}, \tag{S18}$$

*without a quantity cap and at any horizon. Conversely, for any margin fraction $0 < \mu \le 1$, the pulse $A$ in (14) achieves error $\alpha$ and remains margin-solvent at every later mark with capital $B_A = \frac{\kappa}{2} A^2 + (c + \mu P_*) A$. Necessary and sufficient capital orders are $\delta^{-4}$.*

Lemma S24.4 is an order statement: an elementary time-domain coercivity argument for Toeplitz forms that establishes the $\delta^{-4}$ rate before the exact frequency-domain optimization of Supplement S9. Its explicit constant is very conservative, because it scales as $m/K_*^2 \propto \epsilon^7 (1 - S)^5$: at the calibrated parameters it is $4.9 \times 10^{-4}$ par units at $\delta = 0.05$ with $\epsilon = 0.35$, five orders of magnitude below any feasible design. We therefore state it as a lemma; Supplement S9 computes the sharp constant, within 5% of a feasible design at every separation.

## S25 Two unknown cohorts: pole fitting and the global two-point bound

Because the memory rates are zeros of $H$ (Theorem 3.1), extracting poles directly from the price impulse response by matrix-pencil or Prony methods targets closed-loop eigenvalues. A four-step inversion repairs this: (1) decouple $g_t = h_{t+1} - a h_t$; (2) form a Hankel matrix

of $g$ and compute a minimal realization by Kung's eigensystem realization; (3) compute its transmission zeros, which are the roots of the denominator of $G$; (4) project reconstructed flow onto the recovered memory filters to obtain the weights (Supplement S3.3). Without noise this recovers $K = 0$, $K = 1$ and well-separated two-cohort systems exactly (Supplement S3.3); with noise, per-cohort weights become unreliable once rates are close. The reason is information-theoretic, as the rest of this section shows.

## S25.1 A global two-point bound for any estimator

The Cramér–Rao benchmark of Proposition S7.1 applies only to estimators that are unbiased near the true parameter. Could biased, ridge, shrinkage or Bayesian estimators escape the $(\Delta\eta)^{-6}$ scaling? No.

**Theorem S25.1** (Global two-point bound and matching upper bound, $K = 2$)**.** *In this theorem $w_k$ denotes cohort $k$'s trading intensity $q_k$, not the merged weight $q_k\eta_k$ of Section 2, and $\Delta\eta$ is the actual spacing of the two rates. Fix total intensity $q = 0.20$, mean rate $\bar{\eta} = 0.40$, target separation $\tau = 0.01$, and $\Delta\eta \in (0, 8/11)$ (so that $\bar{\eta} - p_+\Delta\eta > 0$). Let $p_\mp = \frac{1}{2} \mp \tau/q$ and*

$$\theta_0 = (qp_-, qp_+, \bar{\eta} - p_+\Delta\eta, \bar{\eta} + p_-\Delta\eta), \qquad \theta_1 = (qp_+, qp_-, \bar{\eta} - p_-\Delta\eta, \bar{\eta} + p_+\Delta\eta). \tag{S19}$$

*(i) Label cohorts in increasing order of rate, so that weight vectors are compared rate by rate (as unordered sets the two weight vectors coincide). Both economies have spacing $\Delta\eta$, weight separation $\|w(\theta_1) - w(\theta_0)\|_\infty = 2\tau$ in this labelling, and equal rate moments $M_m = \sum_k w_k\eta_k^m$ for $m \in \{0, 1, 2\}$, while*

$$M_3(\theta_1) - M_3(\theta_0) = 4\tau p_- p_+ (\Delta\eta)^3. \tag{S20}$$

*Hence $\|h(\theta_1) - h(\theta_0)\|_2 = \Theta(\tau(\Delta\eta)^3)$.*

*(ii) For any measurable estimator (including biased, ridge, James–Stein or MAP) that reports weights in the same rate order, the supremum being over the class of two-cohort economies with total weight $q$, mean rate $\bar{\eta}$ and spacing $\Delta\eta$,*

$$\sup_\theta P_\theta(\|\hat{w} - w(\theta)\|_\infty \geq \tau) \;\geq\; 1 - \Phi\Big(\frac{\sqrt{N}\|h(\theta_1) - h(\theta_0)\|_2}{2\sigma_h}\Big), \tag{S21}$$

*so failure probability at most 0.05 requires $N \geq 4\sigma_h^2 z_{0.95}^2/\|h(\theta_1) - h(\theta_0)\|_2^2 = \Omega((\Delta\eta)^{-6})$. For $\Delta\eta \geq 2\tau q/(q + 2\tau)$ ($\approx 0.018$ at the stated values) the same bound holds without*

*any labelling convention, for the permutation-invariant loss*

$$d_\infty(\hat{\theta},\theta) = \min_{\pi\in S_2} \max_k \Big( |\hat{w}_{\pi(k)} - w_k| + |\hat{\eta}_{\pi(k)} - \eta_k| \Big),$$

*because then* $d_\infty(\theta_0,\theta_1) = \min\{2\tau(1+\Delta\eta/q),\ (1+2\tau/q)\Delta\eta\} \geq 2\tau$. *For smaller spacings the relevant permutation-invariant statement is the Wasserstein bound of Theorem 6.1.*

*(iii) Conversely, the matching upper bound* $N = O((\Delta\eta)^{-6})$ *on rich separated two-cohort classes is Theorem 6.2 with* $K = 2$. *On the separated class with weights and rates bounded away from the boundary, Toeplitz deconvolution followed by a Prony–Hankel root step has Hankel determinant* $g_1 g_3 - g_2^2 = w_1 w_2 \eta_1 \eta_2 r_1 r_2 (\Delta\eta)^2 > 0$ *for the deconvolved kernel* $g_m = \sum_i w_i \eta_i r_i^m$, *and perturbation analysis gives the same order.*

**In words.** For two unknown groups, no method can tell the two candidate markets apart reliably with fewer than a number of experiments of order $(\Delta\eta)^{-6}$, and a practical method (Prony's method, Theorem 6.2) achieves this order. Halving the spacing multiplies the data needed by 64.

We recomputed the two-point quantities exactly (Table S30): the moment gap equals $4\tau p_- p_+ (\Delta\eta)^3$ to machine precision, and the Le Cam requirement at $\Delta\eta = 0.10$ is 5,927.6 resets. The worst-case failure rates of four re-implemented estimators all lie above the lower bound (Table S30).

*Remark* S25.2 (The hard directions carry no risk). The two economies of Theorem S25.1 have the same total weight $\sum_i q_i = q$, so their summed price response $\sum_t h_t = \kappa/(1 - a - \kappa \sum_i q_i)$ is identical, 2.353 at the calibration of Table S30: the directions in which prices are least informative about the cohorts are directions in which cumulative price risk does not move. The summed response is a linear functional of the response and is estimated at the parametric rate whatever the spacing of the memories; the companion paper turns this into a certification theorem, a minimax-optimal pre-trade guard and a decision value of disclosure (Niroula et al., 2026). Identification and safety ask different questions of the same data.

## S26 Three ways to learn more, and what each costs

An observer that wants to know more, whether a supervisor, an exchange's surveillance team or a large trading desk, has three options (Section 2.1).

(a) *Wait and watch prices.* This costs time. In a stationary Gaussian market observed without error, a passive observer needs at least as many observations, in order of

magnitude, as an active one needs experiments (Theorem 6.5).

(b) *Trade against the groups* and watch how the price responds, much as one taps a wall to find a hidden beam. Every test trade moves the price against the trader, so learning this way costs money. To tell one group from two groups whose memory rates differ by $2\delta$, the quantity traded grows like $1/\delta^2$ and the capital at risk like $1/\delta^4$: halve the difference and the capital needed rises sixteen-fold (Theorems 5.1 and S24.2). If memory is a general mixture of fading memories rather than a few exponentials (with the slowest bounded away from a random walk), accuracy improves only with the logarithm of the data, so halving the error squares the data needed (Theorem S17.2).

(c) *Acquire labelled records*, such as audit-trail data or class-level position reports. This costs whatever the reporting system costs, but it removes the explosion in data: labels save a factor of order $(\Delta\eta)^{-(4K-2)}$ at fixed accuracy, where $K$ is the number of groups and $\Delta\eta$ the spacing of their memory rates (Theorem 7.2).

Proposition S27.1 says when each option is the cheapest.

# S27 When an audit trail is chosen

The dollar illustration of Supplement S8 prices disclosure; it does not say when a supervisor should buy it. Remark S25.2 shows that a supervisor who only needs to bound aggregate price risk does not need disclosure at all: the summed response is identified from public prices at the ordinary rate. Attribution matters when supervision is selective: one algorithm must be stopped or sanctioned while the others keep supplying liquidity, and the law requires attributing the conduct to it at a given error level. A supervisor without an audit trail does not run trading probes; it waits for public data to accumulate, and the cost of waiting is the harm done in the meantime.

Consider $M$ markets, a supervisory cycle of $\bar{T}$ periods, and in each market one algorithm whose conduct does harm $\ell > 0$ per period until it is attributed. With prices alone, attribution at the required error level takes $T_p(\Delta\eta) = c_p(\Delta\eta)^{-(4K-2)}$ periods; with class-tagged records it takes $T_t(\Delta\eta) = c_t(\Delta\eta)^{-2}$ periods, the orders of Theorems 6.5 and 7.2 with constants $c_p, c_t > 0$. We take the attribution times to be of exactly these orders, which is a modelling assumption: Theorem 6.5 gives only a lower bound for prices, and if price-only attribution is slower, $G$ below is larger and disclosure is chosen at least on the band the proposition describes. $T_t \propto (\Delta\eta)^{-2}$ is the rate target; for weights at fixed accuracy $T_t$ does not grow as $\Delta\eta$ shrinks, and the lower edge of the band disappears. Attribution that would take longer than the cycle does not happen within it. An audit trail costs $F$

per cycle to operate and $c$ per market to comply with, including any information leakage. The supervisor's expected loss is therefore $\mathcal{L}_{\text{price}} = M\ell \min\{\bar{T}, T_p\}$ without disclosure and $\mathcal{L}_{\text{disc}} = F + Mc + M\ell \min\{\bar{T}, T_t\}$ with it. Write $\Delta\eta_t = (c_t/\bar{T})^{1/2}$ and $\Delta\eta_p = (c_p/\bar{T})^{1/(4K-2)}$ for the spacings at which tagged and price-only attribution just fit within the cycle.

**Proposition S27.1** (When an audit trail is chosen)**.** *Let $K \geq 2$ and $\Delta\eta_t < \Delta\eta_p$, and let $G(\Delta\eta) = \min\{\bar{T}, T_p(\Delta\eta)\} - \min\{\bar{T}, T_t(\Delta\eta)\}$ be the attribution delay that disclosure saves.*

*(i) The supervisor strictly prefers disclosure if and only if $F/M + c < \ell\, G(\Delta\eta)$.*

*(ii) $G$ vanishes for $\Delta\eta \leq \Delta\eta_t$, increases on $(\Delta\eta_t, \Delta\eta_p]$ to its maximum $G^* = \bar{T} - T_t(\Delta\eta_p) > 0$, decreases wherever it is positive beyond $\Delta\eta_p$, and is negative for $\Delta\eta > (c_p/c_t)^{1/(4K-4)}$, where it returns toward zero from below. Hence the spacings at which disclosure is chosen form an interval around $\Delta\eta_p$, possibly empty: disclosure is never chosen for algorithms so similar that even tagged records cannot attribute them within the cycle, nor for algorithms so different that prices attribute them quickly.*

*(iii) The interval is non-empty if and only if the audit trail covers more than $M^* = F/(\ell G^* - c)$ markets (with $M^* = \infty$ if $\ell G^* \leq c$), and it widens as $M$ grows.*

Part (i) restates the comparison of losses. Part (ii) says that disclosure is valuable in a middle band, where algorithms are close enough that prices cannot attribute them within the cycle but far enough apart that tagged records can; the polynomial gap between the two delays, $(\Delta\eta)^{-(4K-2)}$ against $(\Delta\eta)^{-2}$, is what makes the band wide. Part (iii) is an economies-of-scale result: an audit trail with a fixed operating cost pays only when it spans enough markets. This illustrates a possible economy of scale for market-wide reporting; the cost function is stylised, not estimated from CAT, and the result neither explains CAT's legal architecture nor shows that it is optimal. A supervisor that can trade has a second option, active probing, whose cost is the frontier of Table S10; because that cost grows at the same polynomial order, it shortens price-only attribution only for well-separated algorithms and leaves the comparison for close ones unchanged. The model takes the harm $\ell$ as given and holds the algorithms' behaviour fixed across regimes; Section 9 returns to both.

# S28 Equilibrium results: proofs and illustrations

**A misjudged dealer and a targeted halt.** The next two results use the simplest linear market inside Theorem 4.5. They are summarized in Section 4.

- Dates are $t = 0, \ldots, T-1$ and the price change is $s_t = p_t - p_{t-1}$, with $p_{-1} = 0$.

- The dealer moves the price by $s_t = \lambda(y_t - \hat{f}_t)$, with $\lambda > 0$, where $\hat{f}_t = \sum_{m<t} \hat{k}_m s_{t-1-m}$ is the feedback flow it predicts from its kernel $\hat{k}$.
- The insider, with value $v$ of mean zero and variance $\Sigma_0$ independent of the noise, commits to a schedule $x = bv$, $b \in \mathbb{R}^T$. Schedules may change sign, so pump-and-dump patterns are allowed.
- The kernel error is $\Delta k = k - \hat{k}$, with $\varepsilon = \sum_{m \le T-2} |\Delta k_m|$.

Let $C$ be the lower-triangular matrix of ones and $L$ the strictly lower Toeplitz matrix with $L_{t,t-1-m} = \Delta k_m$.

**Theorem S28.1** (Rents from a misjudged aggregate response)**.** *(i) The insider's expected profit is $J(b) = \Sigma_0\big(\mathbf{1}^\top b - b^\top A b\big)$, with $A = \lambda C(I - \lambda L)^{-1}$. If $Q = \frac{1}{2}(A + A^\top) \succ 0$, the optimal schedule is $b^* = \frac{1}{2}Q^{-1}\mathbf{1}$ and the value is $V(\Delta k) = \frac{\Sigma_0}{4}\mathbf{1}^\top Q^{-1}\mathbf{1}$.*

*(ii) If $\lambda T \varepsilon \le \frac{1}{4}(1 - \lambda\varepsilon)$, then $Q \succ 0$ and*

$$|V(\Delta k) - V(0)| \le \frac{2\Sigma_0 T^2 \varepsilon}{1 - \lambda\varepsilon}.$$

*(iii) To first order, $V(\Delta k) = V(0) - \Sigma_0 \sum_m \omega_m \Delta k_m + O(\varepsilon^2)$, with $\omega_m = \frac{(T-m)(T-m-1)}{2(T+1)^2} > 0$ and $V(0) = \frac{\Sigma_0 T}{2\lambda(T+1)}$. The rent is therefore of exact order $\varepsilon$. The insider gains when the dealer overstates the feedback ($\Delta k < 0$) and loses when it understates it.*

*(iv) If the dealer estimates the aggregate kernel from $n$ observations with error $\varepsilon_n = O_p(n^{-1/2})$, the rent is $O_p(n^{-1/2})$. Separating the groups behind the same kernel still needs order $(\Delta\eta)^{-(4K-2)}$ experiments (Theorem 6.1).*

**In words.** A dealer that misjudges how strongly feedback traders respond hands the insider a rent, or charges it one, in proportion to the error. That error concerns only the aggregate response, which public data reveal at the ordinary rate. So the information that protects price formation is cheap to learn, while the information about who the groups are stays expensive. The dealer's coefficient $\lambda$ is held fixed: this is the insider's best response to a misspecified dealer, not a re-solved equilibrium.

**Theorem S28.2** (Targeted intervention in equilibrium)**.** *Take the Gauss and Radau pairs of Theorem 4.5(iv), with total weight $W$, on rates $c + ds$ with $\rho = |c| + |d| < 1$. Place them in the market above, with the same dealer rule $(\lambda, \hat{k})$, the same insider schedule and the same draws.*

*(i) The kernels agree for $m \le 2$, and $|k^G_m - k^R_m| \le 2W\binom{m}{3}|d|^3\rho^{m-3}$ for $m \ge 3$. Hence $\|k^G - k^R\|_1 \le 2W|d|^3/(1-\rho)^4$.*

*(ii) Before any intervention, the price paths differ by at most* $\lambda^2 T\|k^G - k^R\|_1\|x+z\|$ *divided by* $(1-\lambda\|k^G-\hat{k}\|_1)(1-\lambda\|k^R-\hat{k}\|_1)$*, which is* $O(|d|^3)$*. The insider's values differ by* $O(|d|^3)$*.*

*(iii) Halting the largest cohort leaves kernels* $\frac{W}{2}(c-d\sigma)^m$ *(the Gauss weights tie; we remove the cohort at* $c+d\sigma$*) and* $\frac{W}{4}(c-d)^m$*, which differ by* $\frac{W}{4}$ *at* $m=0$ *for every spacing. The post-halt responses therefore differ at order one, and so do the post-halt feedback budgets* $\sum_m k_m$*.*

**In words.** Two markets can look the same in equilibrium to within the cube of the memory spacing, yet halting their largest group leaves feedback that differs at order one. Equilibrium does not remove the case for targeted information; it sharpens it. In the simulation below, the pre-halt price gap is about $3d^3$ at all three spacings tried, while the post-halt gap stays near 0.19.

*Proof of Theorem 4.5. Part (i).* By induction, $m_{i,t} = \sum_{m<t} r_i^m s_{t-1-m}$. Multiplying by $w_i$ and summing over $i$ gives $f_t = \sum_{m<t} k_m s_{t-1-m}$.

*Part (ii).* If $k_m = k'_m$ for $m < T$, then by (i) the two aggregate flows agree on $t \le T$ for every signal. Conversely, the impulse signal $s_j = \mathbf{1}\{j=0\}$ gives $f_{m+1} = k_m$, so equal flows on $t \le T$ force $k_m = k'_m$ for $m < T$.

On an unbounded horizon, equality of all $k_m = \int r^m\, dM(r)$ is equality of all moments of two finite signed measures on $[0,1]$. Polynomials are dense in $C[0,1]$, so the measures coincide. In terms of $\eta$ this is the condition (8) of Theorem 3.1.

*Part (iii).* Fix a strategy profile and a realization. We argue by strong induction on $t \le T$. If prices and flows agree before $t$:

- the signals agree before $t$, because $S_j$ depends on prices up to $j$;
- the aggregate flows at $t$ agree, by part (i) and the equal kernels, since $f_t$ uses signals before $t$ only;
- the insider's orders at $t$ agree, by causality;
- hence the total flows $y_t$ agree, and then the prices $p_t$ agree, by causality of the dealer's rule.

The insider's profit $\sum_{t<T}(v-p_t)x_t$ and the dealer's inventory $-\sum_{t<T} y_t$ are functions of these paths.

Strategy sets are the same in the two games, because the rules are defined on the same histories. Every payoff is an expectation of the same function of $(v,z)$, so best responses coincide and so do the equilibria.

*Part (iv).* The Gauss rule has weights $\frac{1}{2}, \frac{1}{2}$ and nodes $\pm\sigma$ with $\sigma^2 = \frac{1}{3}$. The Radau rule has weights $\frac{1}{4}, \frac{3}{4}$ and nodes $-1, \frac{1}{3}$. Both integrate $1, s, s^2$ exactly, so on rates $c+ds$ the kernels agree for $m \le 2$. The largest weights are $\frac{1}{2}$ and $\frac{3}{4}$. □

**Machine check.** `Paper1EquilibriumInvariance.lean` proves the following:

- part (i): `p1_eqAggFlow_kernel`;
- both directions of part (ii) on a finite horizon: `p1_eqAggFlow_of_kernel` and `p1_kernel_of_eqAggFlow`;
- the pathwise statement of part (iii) for arbitrary causal insider, dealer and signal rules: `p1_eq_market_invariance` and `p1_eq_insider_profit_invariance`;
- part (iv): `p1_eq_group_level_differs`.

The unbounded-horizon moment statement and the passage from pathwise equality to equal equilibria are on paper.

**Cohort-specific demand shocks.** Suppose cohort $i$ also trades an idiosyncratic shock $\epsilon_{i,t}$ through its memory, so that $f_t$ gains the term $\sum_i w_i \sum_{m<t} r_i^m \epsilon_{i,t-1-m}$. The same induction shows that the equilibrium depends on the cohorts only through the kernel and the law of this aggregate shock process. For independent Gaussian shocks with variances $\sigma_i^2$, that law is determined by the spectral density $\sum_i w_i^2 \sigma_i^2 |1 - r_i e^{i\omega}|^{-2}$. This is a second merged object, and the invariance class shrinks accordingly. The paper's cohorts have no idiosyncratic shocks.

**The textbook benchmark.** Take a risk-neutral, competitive dealer that knows the kernel. It can compute $f_t$ from public prices and price only $y_t - f_t$, so predictable feedback flow then has no price impact, and the closed loop of Section 2 is absent in that equilibrium. The theorem holds in that benchmark as well. Feedback reaches prices in equilibrium when the dealer bears inventory risk, or when it does not know the kernel and must learn it. These are the cases studied in the equilibrium extension.

**Numerical illustration.** `equilibrium/e1_demo.py` runs one market. The insider submits $x_t = 0.3(v - p_{t-1})$. The dealer sets $p_t = p_{t-1} + 0.3\, y_t - 0.05\,(I_t - I_{t-1})$. The signal is $s_t = p_t - p_{t-1}$. Two cohort configurations are compared, the Gauss and Radau pairs of part (iv) on rates $0.6 + 0.2s$, with total weight 0.5, under the same $(v, z)$ draws. Over 200 draws:

- price paths coincide to machine precision on the first four dates ($t \le 3$), with largest gap $2 \times 10^{-16}$;

- from date five, where $k_3$ first differs (by $\frac{2}{9}d^3$ times the total weight), they separate, with largest gap $1.8 \times 10^{-4}$ at date five and $1.9 \times 10^{-2}$ by date twenty;
- the largest group's share of feedback positions stays at medians of 0.59 and 0.79.

The result file is `equilibrium/e1_demo_results.json`.

**Proof of Theorem S28.1.** *Part (i).* Total flow is $y = x + z + f$. The dealer subtracts $\hat{f}$, so $s = \lambda(x + z + Ls)$. The matrix $L$ is strictly lower triangular, so $I - \lambda L$ is invertible and $s = \lambda(I - \lambda L)^{-1}(x + z)$. Prices are $p = Cs$. With $x = bv$, realized profit is $\sum_t (v - p_t)x_t = v^2 \mathbf{1}^\top b - v\, b^\top C s$. Taking expectations with $E[v] = 0$ and $v$ independent of $z$ gives $J(b)$. When $Q \succ 0$, $J$ is strictly concave, and its maximizer and value are as stated.

*Part (ii).* A Toeplitz operator is bounded by the $\ell_1$ norm of its symbol, so $\|L\| \le \varepsilon$. For $\lambda\varepsilon < 1$ this gives $\|(I - \lambda L)^{-1}\| \le 1/(1 - \lambda\varepsilon)$ and $(I - \lambda L)^{-1} - I = \lambda L(I - \lambda L)^{-1}$. Since $\|C\| \le T$, we get $\|Q - Q_0\| \le \|A - A_0\| \le \lambda^2 T\varepsilon/(1 - \lambda\varepsilon)$. Moreover $Q_0 = \frac{\lambda}{2}(I + \mathbf{1}\mathbf{1}^\top) \succeq \frac{\lambda}{2} I$. Under the stated condition $\|Q - Q_0\| \le \lambda/4$, so $Q \succeq \frac{\lambda}{4} I$. Then

$$|V - V_0| = \tfrac{\Sigma_0}{4} |\mathbf{1}^\top Q^{-1}(Q_0 - Q)Q_0^{-1}\mathbf{1}| \le \tfrac{\Sigma_0}{4}\, T \cdot \tfrac{4}{\lambda} \cdot \tfrac{2}{\lambda} \cdot \tfrac{\lambda^2 T\varepsilon}{1 - \lambda\varepsilon}.$$

*Part (iii).* $Q_0^{-1}\mathbf{1} = \frac{2}{\lambda(T+1)}\mathbf{1}$, which gives $V(0)$. Since $A = A_0 + \lambda^2 CD + O(\varepsilon^2)$, the derivative of $\mathbf{1}^\top Q^{-1}\mathbf{1}$ is $-u^\top(\lambda^2 CD)u$ with $u = Q_0^{-1}\mathbf{1}$. Using $\mathbf{1}^\top C = (T, T-1, \ldots, 1)$ and $(L\mathbf{1})_t = \sum_{m<t} \Delta k_m$, we get $\mathbf{1}^\top CL\mathbf{1} = \sum_m \Delta k_m \frac{(T-m)(T-m-1)}{2}$. Multiplying by $\frac{\Sigma_0}{4} \cdot \frac{4}{\lambda^2(T+1)^2} \cdot \lambda^2$ gives $\omega_m$.

*Part (iv).* This is part (ii) with $\varepsilon = \varepsilon_n$. The rate for $\varepsilon_n$ is an assumption on the dealer's estimator, for example least squares of order flow on lagged price changes in a stationary market. It is not proved here.

**Proof of Theorem S28.2.** *Part (i).* Write $\mu_j^G, \mu_j^R$ for the $j$th moments of the two rules, which are at most one in absolute value. Expanding $(c + ds)^m$ gives

$$k_m^G - k_m^R = W \sum_{j \ge 3} \binom{m}{j} c^{m-j} d^j (\mu_j^G - \mu_j^R).$$

The terms with $j \le 2$ vanish because both rules are exact for those degrees. Since $\binom{m}{j} \le \binom{m}{3}\binom{m-3}{j-3}$, the sum is at most $2W\binom{m}{3}|d|^3\rho^{m-3}$. Finally, $\sum_{m \ge 3}\binom{m}{3}\rho^{m-3} = (1 - \rho)^{-4}$.

*Part (ii).* With residual matrices $L_G, L_R$ built from $k^G - \hat{k}$ and $k^R - \hat{k}$,

$$\begin{aligned} p^G - p^R &= \lambda C\big[(I - \lambda L_G)^{-1} - (I - \lambda L_R)^{-1}\big](x + z) \\ &= \lambda^2 C(I - \lambda L_G)^{-1}(L_G - L_R)(I - \lambda L_R)^{-1}(x + z). \end{aligned}$$

Bound each factor as in the proof of Theorem S28.1. The value comparison follows from the same perturbation argument, with base point $L_R$ instead of 0.

*Part (iii).* Removing one of the two Gauss cohorts leaves weight $W/2$. Removing the Radau cohort with weight $3/4$ leaves $W/4$. At $m = 0$ the kernel is the remaining weight.

**Proof of Theorem 4.1.** *Part (i).* Conjecture $x_t = -\kappa H_t + \sum_i \beta_i u_{i,t+1}$ and set $D = 1 - \sum_i \beta_i w_i$. The next-period expectation follows from three facts: $E_t \xi_{t+1} = 0$, $E_t H_{t+1} = \bar{a} H_t - \sum_i u_{i,t+1}$, and $E_t u_{i,t+2} = r_i u_{i,t+1} + w_i E_t x_{t+1}$. Together they give

$$D\, E_t x_{t+1} = -\kappa \bar{a} H_t + \sum_i (\kappa + \beta_i r_i) u_{i,t+1}.$$

The dealers' condition $x_t = E_t x_{t+1} - AV H_t$ must hold for all states. Matching the coefficient on $u_{i,t+1}$ gives $\beta_i = (\kappa + \beta_i r_i)/D$, that is, $\beta_i = \kappa/(D - r_i)$. The definition of $D$ then becomes $D = 1 - \kappa G(D)$, and matching on $H_t$ gives $\kappa(D - \bar{a}) = AVD$.

For the variance, $x_{t+1}$ depends on $\xi_{t+1}$ through $H_{t+1}$ and through its own effect on $u_{i,t+2}$. Its innovation $\delta$ therefore satisfies $\delta = \kappa \xi_{t+1} + \sum_i \beta_i w_i \delta$, so $\delta = \kappa \xi_{t+1}/D$ and $V = \sigma_F^2 + \kappa^2 \sigma_\xi^2 / D^2$.

Conversely, take a solution of (11). Then the conjectured law satisfies the dealers' condition in every state (`p1_inventory_foc`), and $V$ is consistent with it (`p1_inventory_innovation`). If the induced dynamics are stable, the process is stationary and the solution is an equilibrium.

*Part (ii).* With no cohorts, $\beta$ is empty and $D = 1$. Then $x_t = -\kappa H_t$ and $H_t = \bar{a} H_{t-1} - y_t$ give $x_t = \bar{a} x_{t-1} + \kappa y_t$. The quadratic $A\sigma_\xi^2 \kappa^2 - (1 - \bar{a})\kappa + A\sigma_F^2 = 0$ has discriminant $\Delta$ (`p1_inventory_benchmark`).

*Part (iii).* The system (11) involves the cohorts only through $G$. The price process is a function of the history of flows, and by Theorem 4.5(i) it depends on the cohorts only through the kernel $k_m = \int r^m \, dM$. A finite measure on $[0, 1]$ is determined by $G$ off its support, and also by its moments.

*Part (iv).* Write $\Phi(\kappa, D; M) = \Big(D + \kappa G(D) - 1,\ \kappa(D - \bar{a}) - A(\sigma_F^2 D + \kappa^2 \sigma_\xi^2 / D)\Big)$. At $M = 0$ the zero is $(\kappa_-, 1)$. There the Jacobian with respect to $(\kappa, D)$ is triangular with determinant $\sqrt{\Delta} \neq 0$: $\partial_D \Phi_1 = 1$, $\partial_\kappa \Phi_1 = G = 0$, and $\partial_\kappa \Phi_2 = (1 - \bar{a}) - 2A\kappa_- \sigma_\xi^2 = \sqrt{\Delta}$. The

implicit function theorem, applied in the norm of the statement, gives a unique zero near $(\kappa_-, 1)$ that is continuous in $M$.

Linearizing, $dD = -\kappa_- G(1)$. Differentiating the second equation and using $A(\sigma_F^2 + \kappa_-^2\sigma_\xi^2) = \kappa_-(1-\bar{a})$ gives $d\kappa = -\kappa_-(\bar{a}+2A\kappa_-\sigma_\xi^2)\, dD/\sqrt{\Delta}$, and so the stated formula. Finally, $G(1) = \sum_i w_i/(1-r_i) = \sum_i q_i$.

At $M = 0$ the transition matrix has eigenvalues $\bar{a}$ and the $r_i$, all inside the unit disk, so the dynamics remain stable for small $M$.

**Proof of Theorem 4.2.** *Part (i).* Write the implicit form of the price at $t$, $Dx_t = -\kappa H_t + \sum_i \beta_i r_i u_{i,t}$, and the price law at $t-1$, $x_{t-1} = -\kappa H_{t-1} + \sum_i \beta_i u_{i,t}$. Use $H_t = \bar{a}H_{t-1} - \sum_i u_{i,t} - \xi_t$. Then

$$Dx_t - \bar{a}x_{t-1} = \kappa\xi_t + \sum_i \Big(\kappa + \beta_i(r_i - \bar{a})\Big)u_{i,t}.$$

Since $\kappa = \beta_i(D - r_i)$, the coefficient is $\beta_i(D-\bar{a}) = \kappa(D-\bar{a})/(D-r_i)$. Divide by $D$.

The scaled flows inherit the cohort recursion, because the scaling is constant (`p1_eq_scaled_cohort`). The states $(H_t, u_{\cdot,t+1})$ and $(x_t, \tilde{u}_{\cdot,t+1})$ are related by an invertible linear map, since $\kappa \neq 0$ and every scale factor is nonzero. So the two dynamics are conjugate and have the same eigenvalues.

*Part (ii).* The effective weights are nonzero because $D \neq \bar{a}$. Apply Theorem 6.4(i) to the closed loop of part (i).

*Part (iii).* Since $G(D) = \sum_i w_i/(D-r_i) = \sum_i \tilde{w}_i/(D-\bar{a})$, the equation $D = 1 - \kappa G(D)$ reads $D = 1 - \kappa^* D\,\tilde{M}([0,1])/(D(1-a^*))$. Solve for $D$; then $\bar{a} = a^* D$ and $\kappa = \kappa^* D$. Conversely, $(\bar{a}, \kappa, D, M)$ determine the effective objects. So two equilibria give the same price response if and only if their effective objects agree, and then their merged measures agree. This is consistent with Theorem 4.5.

*Part (iv).*

- *$D$ exceeds every rate.* The hypothesis of Theorem 6.1 is $a^* + \kappa^* W/(1 - r_c - \Delta\eta) < 1$. It gives $\kappa^* W/(1-a^*) < 1 - r_c - \Delta\eta$, so $D > r_c + \Delta\eta$, which is at least every rate.
- *Each economy is an equilibrium.* With the stated choices, $D + \kappa G(D) = 1$ holds (`p1_eq_realize_D`), and so does $\kappa(D-\bar{a}) = AVD$ with $V = \sigma_F^2 + \kappa^{*2}\sigma_\xi^2$ (`p1_eq_realize_dealer`). By part (i) the equilibrium price response is the given closed loop, which is stable. So each economy is an equilibrium by Theorem 4.1(i).
- *The two economies share everything but the cohorts.* They have the same total effective weight $W$, so they share $D$, $\kappa$ and the primitives, and their price responses are those of Theorem 6.1.

- *Upper bound.* On a class where the effective parameters are known or estimated at the parametric rate (Corollary 6.3), Theorem 6.2 recovers $\tilde{w}$. Then $D$ follows from part (iii), and $w_i = \tilde{w}_i(D - r_i)/(D - \bar{a})$ is Lipschitz on the class.

**Proof of Theorem 4.4.** *Part (i).*

- *$D$ lies in $(m, 1)$.* Let $\kappa > 0$ and $D > 0$. The dealer equation $\kappa(D - \bar{a}) = AVD > 0$ gives $D > \bar{a}$. The cohort equation is $(D - 1)(D - r) + \kappa w = 0$. If $D < r$, it forces $D > 1$, which is impossible; and $D = r$ is excluded. So $D \in (r, 1)$, and $\kappa = \kappa_w(D)$.
- *The equilibria are the solutions of $\Psi_w(D) = A$.* Multiplying the dealer equation by $D$ gives $\kappa(D - \bar{a})D = A(\sigma_F^2 D^2 + \kappa^2 \sigma_\xi^2)$, which is $\Psi_w(D) = A$. Conversely, every such $D$ solves the system.
- *Stability.* By Theorem 4.2(i) the dynamics are those of the closed loop with $c = \kappa^* \tilde{w} = (1-D)(D-\bar{a})/D \in (0, 1)$. Its characteristic polynomial is $p(\lambda) = \lambda^2 - (a^* + r + c)\lambda + a^* r$. The roots are real and nonnegative, because the discriminant is at least $(a^* - r)^2$ and the coefficients alternate in sign. Next, $p(1) = (1 - a^*)(1 - r) - c = (D - \bar{a})(D - r)/D > 0$, so 1 lies outside the interval between the roots. Their product $a^* r$ is below 1, so both roots are below 1.
- *Existence and count.* $\Psi_w$ is continuous and positive on $(m, 1)$ and tends to 0 at both ends: as $D \to 1$, $\kappa_w \to 0$; as $D \to m$, either $D - \bar{a} \to 0$ or $\kappa_w \to 0$. So its maximum $A^*$ is attained. Every $A \in (0, A^*)$ is taken at least twice, once on each side of a maximizer, and no $A > A^*$ is taken.

*Part (ii).*

- *A parametrization by $\kappa$.* For fixed $\kappa \in (0, (1 - r)^2/(4w)]$, the solutions of $\kappa_w(D) = \kappa$ are $D_\pm(\kappa, w) = \left((1 + r) \pm \sqrt{(1 - r)^2 - 4\kappa w}\right)/2$.
- *The larger root gives the maximum.* $\tilde{\Psi}(\kappa, D) = \kappa(D - \bar{a})D/(\sigma_F^2 D^2 + \kappa^2 \sigma_\xi^2)$ is strictly increasing in $D$ for $D > \bar{a}$. The numerator of its $D$-derivative is $\bar{a}\sigma_F^2 D^2 + (2D - \bar{a})\kappa^2 \sigma_\xi^2 > 0$. So $A^*(w) = \max_\kappa \tilde{\Psi}(\kappa, D_+(\kappa, w))$.
- *Strict decrease.* Let $w_1 < w_2$, and let $\kappa^\circ$ attain $A^*(w_2)$. Then $\kappa^\circ$ is feasible at $w_1$, and $D_+(\kappa^\circ, w_1) > D_+(\kappa^\circ, w_2)$. So $A^*(w_1) \geq \tilde{\Psi}(\kappa^\circ, D_+(\kappa^\circ, w_1)) > A^*(w_2)$.
- *The limit as $w \downarrow 0$.* The bound $A^* < A_0^*$ of part (iii) is one half. For the other, fix $\kappa_0 = \sigma_F/\sigma_\xi$. As $w \downarrow 0$, $D_+(\kappa_0, w) \to 1$, and $\tilde{\Psi}(\kappa_0, D_+) \to (1 - \bar{a})/(2\sigma_F \sigma_\xi)$.

*Part (iii).*

- *The bound $A_0^*$.* By the arithmetic–geometric mean inequality, $\sigma_F^2 D^2 + \kappa^2 \sigma_\xi^2 \geq 2\sigma_F \sigma_\xi \kappa D$. So $\Psi_w(D) \leq (D - \bar{a})/(2\sigma_F \sigma_\xi) < A_0^*$. Machine-checked in squared form as $4A^2 \sigma_F^2 \sigma_\xi^2 \leq$

$(D-\bar{a})^2$ (`p1_fragility_benchmark_bound`).

- *The bound* $\Phi^*/(\sigma_F^2 w)$. Dropping $\kappa^2\sigma_\xi^2 > 0$ from the denominator gives $\Psi_w(D) < (1-D)(D-r)(D-\bar{a})/(\sigma_F^2 wD) \le \Phi^*/(\sigma_F^2 w)$. Machine-checked as $A\sigma_F^2 wD \le (1-D)(D-r)(D-\bar{a})$ (`p1_fragility_feedback_bound`).
- *The bound on* $\Phi^*$. It uses $(1-D)(D-r) \le (1-r)^2/4$ and $(D-\bar{a})/D \le 1-\bar{a}$.
- *The limit as* $w \to \infty$. We have $\kappa_w = O(1/w)$ uniformly on $[m,1]$. So $w\Psi_w(D) \to (1-D)(D-r)(D-\bar{a})/(\sigma_F^2 D)$ uniformly, and the maxima converge.

*Part (iv).*

- $\kappa$ *is determined by* $D$. Let $D > \max_i r_i$. All weights are positive, so $G(D) > 0$. Then $D = 1-\kappa G(D)$ with $\kappa > 0$ gives $D < 1$ and $\kappa = (1-D)/G(D)$. The dealer equation then reads as in part (i), and $D > \bar{a}$ as there.
- *Comparison with* $G(1)$. For $D \in (\max_i r_i, 1)$, each $D - r_i < 1 - r_i$, so $G(D) > G(1)$.
- *The bound.* The dealer equation gives $A\sigma_F^2 D < \kappa(D-\bar{a}) < (1-D)(D-\bar{a})/G(1)$. Moreover

$$(1-D)(D-\bar{a}) = D(1-\sqrt{\bar{a}})^2 - (D-\sqrt{\bar{a}})^2 \le D(1-\sqrt{\bar{a}})^2.$$

  Machine-checked as `p1_fragility_K_bound`. The bound $A_0^*$ follows as in part (iii).

- *Stability.* By Theorem 4.2(i), the dynamics are those of the closed loop with $(a^*, \kappa^*, \tilde{w})$. Here $a^* \ge 0$, $\kappa^* > 0$, $\tilde{w}_i > 0$ and $r_i \ge 0$, so its transition matrix in the state $(x_t, y_{1,t}, \ldots, y_{K,t})$ is nonnegative.
  - *The spectral radius is the root above the rates.* By Perron–Frobenius, the spectral radius $\rho$ is an eigenvalue. Every eigenvalue $z \notin \{0, r_i\}$ solves $F(z) = 1$, where $F(z) = a^*/z + \kappa^* \sum_i \tilde{w}_i/(z-r_i)$. No $r_i$ is an eigenvalue, because $\tilde{w}_i \ne 0$ (Theorem 6.4(i)). $F$ decreases strictly from $+\infty$ to 0 on $(\max_i r_i, \infty)$, so it has exactly one root $z^*$ there, and $\rho \ge z^*$. Being real and at least $z^*$, $\rho$ equals $z^*$.
  - *The criterion.* Hence $\rho < 1$ if and only if $F(1) < 1$, that is, $a^* + \kappa^* \sum_i \tilde{w}_i/(1-r_i) < 1$. With $g_i = w_i/(D-r_i)$ this reads $\kappa \sum_i g_i/(1-r_i) < 1$ (`p1_eq_effective_stability_iff`).
  - *It holds.* Each $1 - r_i > 1 - D > 0$ and $\kappa \sum_i g_i = 1 - D$, so $\kappa \sum_i g_i/(1-r_i) < 1$ (`p1_eq_stability_K`).

**Proof of Corollary 4.3.** *Part (i).*

- The cohort recursion is part of the model.

- By Theorem 4.2(i), the price is the closed loop with $a^* \geq 0$, $\kappa^* > 0$ and $\tilde{w}_i > 0$, driven by the i.i.d. flow $\xi$.
- It is stable by Theorem 4.4(iv).
- These are the hypotheses of Proposition 7.1. Its regression involves only cohort $i$'s own flow and the price, so it delivers the true $(r_i, w_i)$.

*Part (ii).*

- *Price-only side.* By Theorem 4.2, the price-only experiment is the reduced form at the effective parameters. The realization of Theorem 4.2(iv) puts the Gauss and Gauss–Radau pair inside the equilibrium class, so the price-only side of Theorem 7.2 holds with the effective weights.
- *Tagged side.* Each cohort's flow response to a unit impulse is $f_{i,t} = w_i \sum_{s \leq t} r_i^{t-s} h_s$, where $h$ is the equilibrium response and $h_0 = \kappa^* > 0$. That is the tagged record of Theorem 7.2, and its argument uses only that $h$ is bounded with $h_0 > 0$.
- On the stated classes the true weights are a Lipschitz function of the effective ones, and conversely, so the orders are unchanged.

**Numerical checks of Theorems 4.2 and 4.4.** The script `equilibrium/e4_reduced_form.py` runs three checks.

- *Reduced form.* For four solved equilibria, the equilibrium price path and the reduced-form path with $(a^*, \kappa^*, \tilde{w})$ agree to $4.4 \times 10^{-16}$ over 5,000 simulated periods. The four are: one cohort with positive feedback, one with negative feedback, two cohorts, and the Gauss pair. Their impulse responses agree to $10^{-16}$. For one cohort of weight 0.02 at $r = 0.7$, for example, $a^* = 0.629$, $\kappa^* = 0.621$ and $\tilde{w} = 0.0279$.
- *Realization.* Take the Gauss and Gauss–Radau pair with $K = 2$, $a^* = 0.5$, $\kappa^* = 0.5$, $W = 0.2$, $r_c = 0.6$ and $\Delta\eta = 0.1$, so that the stability index of Theorem 6.1 is 0.833. Both economies are equilibria with $D = 0.8$, $\bar{a} = 0.4$, $\kappa = 0.4$ and $A = 0.188$, and both equilibrium equations hold to machine precision.
  - Their true cohort weights are $(0.064, 0.036)$ and $(0.0375, 0.0625)$.
  - Their equilibrium impulse responses differ in $\ell_2$ by $1.22\,d^3$, $1.25\,d^3$ and $1.29\,d^3$ at $d = 0.1$, 0.05 and 0.025. That is the order $(\Delta\eta)^{2K-1}$.
- *Fragility.* The maximum of $\Psi_w$ reproduces the bisection thresholds of the solver to $10^{-12}$ ($A^* = 0.2547$, 0.1297, 0.0677, 0.0342 at $w = 0.02$, 0.05, 0.1, 0.2).
  - $A^* = 0.3967$ at $w = 5 \times 10^{-4}$, against the limit 0.4.

– $w\,A^* = 0.006875$ at $w = 1$ and 0.0068764 at $w = 10$, against $\Phi^* = 0.0068764$.

*Several cohorts.* `equilibrium/e4_fragility_K.py` scans $D$ for three populations at weights 0.005 to 1 per cohort. The populations are two cohorts at rates 0.5 and 0.85, three at 0.3, 0.6 and 0.8, and the Gauss pair.

- The exact threshold $A^*$ lies below the part (iv) bound in all 15 cases, at 28% to 86% of it. For example, for two cohorts with $G(1) = 1.73$, $A^* = 0.0084$ against a bound of 0.0293.
- Every equilibrium on the threshold curve is stable. At the threshold the spectral radius is between 0.79 and 0.93; it approaches 1 only at the end of the curve where $D \to \max_i r_i$ and $A \to 0$.

The result files are `equilibrium/e4_reduced_form_results.json` and `equilibrium/e4_fragility_K_results.json`.

**Numerical solution of Theorem 4.1.** The script `equilibrium/e4_inventory.py` solves the system (11) at $A = 0.2$, $\sigma_F^2 = 1$, $\sigma_\xi^2 = 0.25$ and $\bar{a} = 0.6$.

*How the solver finds equilibria.*

- For one cohort with $w > 0$, every equilibrium with $\kappa > 0$ and $D > 0$ has $D \in (\max(\bar{a}, r), 1)$ and $\kappa = (1 - D)(D - r)/w$. A scan over $D$ therefore finds all of them.
- For several cohorts, the solver follows the branch of $D(\kappa)$ that is continuous with $D = 1$.
- Each solution is checked by simulating $2 \times 10^5$ periods of the closed loop. The check computes the dealers' first-order residual and the innovation variance of the price.

*Benchmark.* Here $\Delta = 0.12$, and the two equilibria are $\kappa_- = 0.536$ and $\kappa_+ = 7.46$, as in part (ii).

*Results.*

| Cohorts (weights at memories) | $\kappa$ | $D$ | $\kappa/D$ | spectral radius |
|---|---|---|---|---|
| none | 0.5359 | 1 | 0.5359 | 0.600 |
| 0.02 at 0.7 | 0.5916 | 0.9533 | 0.6206 | 0.787 |
| −0.2 at 0.7 | 0.4212 | 1.1767 | 0.3579 | 0.597 |
| $0.005, 0.005$ at $0.5, 0.85$ | 0.569147 | 0.970292 | 0.586573 | 0.881 |
| $0.0025, 0.0025, 0.005$ at $0.5, 0.5, 0.85$ | 0.569147 | 0.970292 | 0.586573 | 0.881 |
| Gauss pair, $0.01, 0.01$ | 0.5759 | 0.9649 | 0.5969 | 0.766 |
| Radau pair, $0.005, 0.015$ | 0.5743 | 0.9662 | 0.5945 | 0.738 |
| 0.2 at 0.7; or $0.1, 0.1$ at $0.5, 0.85$ | no equilibrium | | | |

- *The simulation checks hold in every case.* The relative first-order residual is below $4 \times 10^{-16}$. The simulated innovation variance matches $\kappa^2\sigma_\xi^2/D^2$ to within 0.3%.

- *Invariance (part (iii)).* The split configuration has the same merged measure as the two-cohort one, and its coefficients agree to all printed digits.
- *Gauss and Radau pairs.* They agree on the first three kernel moments, and their impacts differ by 0.0016.
- *Negative feedback* lowers impact.

*First-order formula of part (iv)*, for one cohort at $r = 0.7$:

- at $w = 0.005$, $\kappa = 0.5457$ against the first-order value 0.5449, and $D = 0.9906$ against 0.9911;
- at $w = 0.01$, $\kappa = 0.5574$ against 0.5540.

Doubling $w$ multiplies the error by about 4.5, as a second-order remainder should.

*Existence region.* For one cohort at $r = 0.7$, an equilibrium exists for dealer risk aversion below a threshold $A^*$. This is computed by bisection, with the full scan over $D$ at each step:

| $w$ | 0 | 0.02 | 0.05 | 0.1 | 0.2 |
|---|---|---|---|---|---|
| $G(1) = w/(1-r)$ | 0 | 0.067 | 0.167 | 0.333 | 0.667 |
| $A^*$ | 0.400 | 0.255 | 0.130 | 0.068 | 0.034 |

- Without feedback, $A^* = (1-\bar{a})/(2\sigma_F\sigma_\xi) = 0.4$, which is the closed form of part (ii).
- For larger $G(1)$, $A^*$ falls roughly in proportion to $1/G(1)$.
- At the threshold the two equilibria merge and disappear.

The results file is `equilibrium/e4_inventory_results.json`.

**Numerical checks.**

- `equilibrium/e2_rents.py`, with $T = 10$, $\lambda = 0.5$ and $\Delta k_m = \pm 0.4\,\epsilon\,(0.7)^m$:
  - the exact rent is $-0.322\epsilon$ when the dealer understates feedback and $+0.322\epsilon$ when it overstates it, as $\epsilon \to 0$;
  - this matches the first-order formula $\mp\Sigma_0 \sum_m \omega_m |\Delta k_m|$ to three digits at $\epsilon = 0.0125$, and to within 6% at $\epsilon = 0.4$.
- `equilibrium/e3_halt.py`, with the market of `e1_demo.py`, $c = 0.6$, $W = 0.5$, 200 draws and 40 dates:
  - the largest pre-halt price gap is $3.8 \times 10^{-4}$, $2.9 \times 10^{-3}$ and $2.3 \times 10^{-2}$ for $d = 0.05$, 0.1 and 0.2, a ratio to $d^3$ of 3.0, 2.9 and 2.8;
  - the largest post-halt gap is 0.196, 0.190 and 0.177.

The result files are `e2_rents_results.json` and `e3_halt_results.json`. Theorems S28.1 and S28.2 are proved on paper and are not machine-checked.

# S29 The supervisory decision in detail

This section completes Section 7.2. It was moved from the main text to keep that text to journal length; nothing was removed.

**Proof of Proposition 7.4.**

*Proof of Proposition 7.4.* Part (i) follows by scaling the intensities. For (ii), expand $(1-r_c-\Delta\eta s)^{-1}$: the first three moments agree, so aggregate budgets differ by order $(\Delta\eta)^3$. Each intensity converges to $W\omega_i/(1-r_c)$, giving the stated dominant-weight gap. This exceeds the aggregate discrepancy at sufficiently small spacing. A ceiling between the two post-halt budgets reduces the action problem to testing two Gaussian mean vectors. Their common-covariance testing error is $\Phi(-\sqrt{N}\|h^G-h^R\|_2/(2\sigma))$; the response bound in Theorem 6.1 gives the lower bound. Part (iii) follows from continuity of $w/(1-r)$ on the declared compact parameter set and the fixed margin. □

**General-$K$ scope.** The same reduction gives exponent $4K-2$ only for a pair whose price responses have that moment-matching order *and* whose dominant intensities have a separately established fixed gap. Moment matching alone does not supply the latter. In particular, the symmetric odd/even binomial pair has equal maximal normalized weights. The $K=3$ rows below are numerical examples, not a construction proving this condition for every $K$.

**Explicit robustness condition.** Smooth best responses alone do not preserve separation: the constant smooth map $\Phi(M)=0$ sends both post-halt budgets to $a$. Instead assume that, over the stated action and holding horizon, $|\widetilde{R}_j - R_j| \le \varepsilon$ for $j \in \{G,R\}$, with $R_R < R_G$. Then

$$R_R+\varepsilon \le \bar{s} < R_G-\varepsilon \quad \Longrightarrow \quad \widetilde{R}_R \le \bar{s} < \widetilde{R}_G. \tag{S22}$$

Indeed, $\widetilde{R}_R \le R_R+\varepsilon$ and $\widetilde{R}_G \ge R_G-\varepsilon$. A nonempty robust window exists if $2\varepsilon < R_G-R_R$. This is a maintained response bound, not an equilibrium result. The same allowance must cover the alternative uniform action. At the $K=2$, $\Delta\eta=0.10$ calibration, $R_R=0.532$, $R_G=0.569909$, and $\bar{s}=0.55$ has margin 0.018; an allowance 0.005 fits within it.

**A complete conditional cost-and-error model.** Let $j$ denote the economy, $\pi_j$ its declared prior, $H$ the holding horizon, $b$ the surplus per active class per period, $C_h$ the implementation cost of halting and $L$ the penalty per period for violating the ceiling. These are

assumed accounting primitives, not estimated social welfare. An envelope $S_{\max}$ supports the robust uniform action

$$\theta_\varepsilon = \min\{1, \max\{0, (\bar{s} - \varepsilon - a)/(S_{\max} - a)\}\}, \qquad L_T = HbK(1 - \theta_\varepsilon).$$

This action is feasible only if its worst-case budget meets the ceiling (in particular $a + \varepsilon \le \bar{s}$). For a halt,

$$L_{h,j} = Hb + C_h + HL\,\mathbf{1}\{R_j + d_j > \bar{s}\}, \quad \mathcal{L} = F + Nc + \sum_j \pi_j\{p_j L_{h,j} + (1 - p_j)L_T\}.$$

Here $F + Nc$ is the information bill, $p_j$ the probability of halting, and $|d_j| \le \varepsilon$ the action response. Require $\Pr_j(\text{halt and violate}) \le \alpha$ separately in each world. Compare this loss with the best feasible no-purchase lottery over the *same two actions*, and allow refusing to buy information. Neither arbitrary alternative policies nor an optimum over sample size are claimed. Active probing additionally incurs its execution loss and capital requirement; these are not automatically the per-record fee $c$ and must not be counted twice.

**Existing runs and the remaining synthetic check.** The archived `R_P1_targeted_intervention.json` supplies six response/decision rows. Recomputing them independently from the stated state recursion agrees to $2.1 \times 10^{-17}$ in the response discrepancy. Those rows supply $S_j, R_j$ and Gaussian testing distances; they contain no measured disclosure fees, intervention payoffs or strategic response bounds. The existing GPU and historical diagnostics cannot fill these quantities by relabelling their outputs. The new CPU study `studies/supervisory_release_20261105/run_study.py` therefore evaluates only this missing accounting layer, with exact Gaussian probabilities rather than a new large simulation.

For this check alone, assume independent scalar disclosures $Z_n \sim N(R_j, \sigma_t^2)$ and halt when $\bar{Z} + z_{1-\alpha}\sigma_t/\sqrt{N} + \varepsilon \le \bar{s}$. This is an illustrative measurement contract, *not* a claim that class positions or tagged flow directly observe post-halt budgets. If a world is unsafe under the response allowance, this gate falsely halts with probability at most $\alpha$ under the stated unbiased Gaussian law. The price comparator observes the full 400-tap Gaussian reset record of the same known pair, with a likelihood-ratio rule constrained by the same per-world error budget. Both information channels have the same assumed monetary fee here; neither represents free passive observation.

With $K = 2$, $N = 32$, $\sigma_t = 0.02$, $\varepsilon = 0.005$, $\alpha = 0.05$, $\pi_R = 1/2$, $H = b = 1$, $C_h = 0.02$, $L = 10$, $F = 0.05$, and $c = 0.0005$, the uniform loss is 1.449219, the information bill is 0.066,

and disclosure loss is 1.305132. Its conditional net advantage is 0.144087 surplus units. Halt probabilities are approximately zero in G and 0.978928 in R. Buying the price record yields loss 1.515219 at this sample size; its optimal acquisition decision is to decline and retain the no-data action. An informed oracle before information fees has loss 1.234609.

Across 2,304 deterministic configurations varying fees, noise, horizon, prior, sample size and allowance, paid disclosure improves on the no-data benchmark in 1,207, worsens it in 1,079, and ties in 18. These counts have no population or statistical-significance interpretation. Negative cases remain in the output. A downward measurement bias of 0.03 produces false-halt probability 0.418776; a future response of 0.025 despite an allowance of 0.005 makes halting unsafe in R with probability 0.978928. The latter also invalidates the uniform-action guarantee, so payoff comparisons under that misspecification are not presented as a fair safety comparison. The experiment establishes conditional accounting and failure modes, not real-market gains, a validated equilibrium response, or the general-$K$ theorem.

**Proposition S29.1** (A robust scalar disclosure gate)**.** *Let $Z_1, \dots, Z_N$ be independent $N(R, v)$ readings of the fixed post-halt feedback budget, $N > 0$, and let the realized action response $d$ satisfy $d \le \varepsilon$. For $t > 0$, halt only if $\bar{Z} + t + \varepsilon \le \bar{s}$. Then the probability of halting while the realized budget exceeds the ceiling is at most*

$$\Pr\{\bar{Z} + t + \varepsilon \le \bar{s},\ R + d > \bar{s}\} \le \frac{v}{Nt^2}.$$

*Consequently $v \le \alpha N t^2$ gives a per-world error bound $\alpha$.*

*Proof.* On an unsafe-halt event, $\bar{Z} + t + \varepsilon \le \bar{s} < R + d \le R + \varepsilon$, so $t \le |\bar{Z} - R|$. Chebyshev's inequality for the actual Gaussian sample mean gives the bound. Lean checks the product Gaussian law, the sample statistic, this event inclusion and the probability inequality in `p1_supervisory_gaussian_unsafe_halt`; it is not merely a check of a scalar named "probability bound." The response allowance and the interpretation of $Z$ remain explicit assumptions. □

This conservative gate uses $t = \sigma_t / \sqrt{N\alpha}$. It differs from the sharper Gaussian-quantile gate above, whose exact normal probabilities are numerically evaluated but whose quantile identity is not formalized here. Under the same calibration with $N = 128$, the conservative gate has radius 0.007906, information bill 0.114, loss 1.349034 and conditional net advantage 0.100185. Its halt probabilities are approximately zero in G and 0.998023 in R. Across the same 2,304 sensitivity configurations it improves on the fixed-menu no-data benchmark in 781, worsens it in 1,499 and ties in 24. Both gates' rows remain in the result file: 4,608

configurations in total. The welfare and error interpretation remains conditional on the declared observation and response contracts.

**Operational bridge from tagged flows to a halt decision.** The scalar-reading contract above can be replaced, within the reset model, by five observed channels: two first class-flow responses, two second responses, and the first price response. For a unit pulse write $f_{i,0} = \kappa w_i$, $f_{i,1} = w_i(\kappa r_i + h_1)$, and $h_1 = \kappa a + \kappa^2 \sum_i w_i$. Thus

$$w_i = f_{i,0}/\kappa, \qquad r_i = f_{i,1}/f_{i,0} - h_1/\kappa, \qquad q_i = w_i/(1 - r_i).$$

Assume known impact, verified stable class membership, $0 \le r_i \le 0.95$, independent unbiased Gaussian reset readings with per-channel variance $\sigma_f^2$, and $N$ independent records. A simultaneous radius $e = \sigma_f\sqrt{5/(N\alpha)}$ covers the five sample means with probability at least $1 - \alpha$ by Chebyshev and a union bound. Interval division, only when the first-flow lower endpoint is positive, gives bounds on $w_i, r_i, q_i$; intersect rate bounds with the declared class. Halt only if the remaining upper budget plus the response allowance meets the ceiling. Otherwise retain the same robust uniform fallback. On simultaneous coverage, every selected halt is safe. This supplies a conditional observation-to-action argument without feeding the unknown $R$, $w$ or $r$ to the rule.

The separate Lean module `Paper1FlowDecision.lean` covers exact flow inversion, ratio and clipped intervals, the intensity interval, selected-halt inequality, actual product-Gaussian joint mean bound and a probability-composition theorem with an explicit rule-soundness premise. Their deterministic composition, clipping implementation and executable simulation still require separate correspondence checks; no claim of a fully formalized executable decision algorithm is made. The pinned build receipt records the checked declarations.

The exploratory script `studies/paper1_uplift_20261105/flow_decision.py` runs 20,000 independent replications per world in each of 60 configurations: $N \in \{32, 128, 512, 2048, 8192\}$, $\sigma_f \in \{0.0005, 0.002, 0.005\}$, $H \in \{1, 10\}$ and $F \in \{0.05, 0.25\}$. Other quantities use the two-group calibration above, a per-channel-vector record fee 0.0005, and the same class-halt or throttle action menu. Fourteen configurations have positive estimated net information value; the others are retained. The largest observed gain is exploratory, selected from the grid: at $N = 128$, $\sigma_f = 0.0005$, $H = 10$ and $F = 0.05$, no-data loss is 14.492188, information-inclusive loss 12.370094 and net gain 2.122094 surplus units. A simultaneous grid-adjusted Hoeffding interval is reported in the result file; zero observed unsafe halts is not a proof of zero probability. Costs, horizons and response bounds are synthetic, and no real-market gain or calibrated welfare conclusion follows. Six additional fixed stress

cases retain both smaller biases with no observed unsafe halt and failures: a downward second-flow bias of 0.010 produces unsafe halts in G in all 20,000 replications, while a response 0.030 exceeding the allowance 0.005 produces unsafe halts in R in all replications. Swapped second-flow labels lead to abstention in both worlds. These are synthetic contract falsifications, not guarantees under misspecification.

**Observation assumptions in market data.** The required class records are signed executed quantities with stable verified labels, synchronized timing, known probe history and a justified error model. Public position ratios and weekly position stocks do not satisfy this contract. Binance's top-position ratio concerns the top 20 percent of users by margin balance; the top-account ratio counts accounts once. Their populations can change and overlap with the all-account population, so those series are not disjoint cohorts or algorithm labels. CFTC reports describe Tuesday positions generally published on Friday, and business-purpose categories can contain different trading motives. Provider definitions are available at https://www.cftc.gov/MarketReports/CommitmentsofTraders/index.htm and https://developers.binance.info/en/docs/catalog/core-trading-derivatives-trading-usd-s-m-futures/api/rest-api/market-data. Neither source identifies a learning update rule. Their fitted persistence and sample-size ratios remain descriptive diagnostics. Converting stock persistence to flow response also requires holdings, valuation, transfers, liquidations, reporting delay and changing-membership adjustments; a stock–flow identity alone supplies no causal identification. Real-market use requires verified event-level class records or a justified observation model, external excitation or valid instruments, and a separately tested post-action response allowance.

**Proposition S29.2** (No-data action and information purchase)**.** *For the fixed halt/throttle menu, let* $\overline{L}_h = \sum_j \pi_j L_{h,j}$ *and let* $p_{\max} = \alpha$ *if a halt is unsafe in at least one world, otherwise* $p_{\max} = 1$*. Assuming the fallback meets the ceiling, the best constant no-data halt probability is* $p = 0$ *when* $\overline{L}_h \geq L_T$*, and* $p = p_{\max}$ *when* $\overline{L}_h < L_T$*. A specified information rule with expected action loss* $L_I$ *is strictly preferred if and only if*

$$F + Nc < \mathcal{L}_0 - L_I.$$

*Allowing refusal to purchase gives loss* $\min\{\mathcal{L}_0, F + Nc + L_I\} \leq \mathcal{L}_0$.

*Proof.* The no-data loss is affine in its feasible halt probability: $L_T + p(\overline{L}_h - L_T)$. Its slope determines the minimizing endpoint. Subtracting the information rule's total loss gives the break-even inequality, and the minimum inequality proves the optional-purchase statement.

These algebraic decisions are machine checked in `Paper1SupervisoryDecision.lean`. They do not establish an optimum over other actions, sample sizes, or information contracts. □

## S30 How to read the paper, and the main symbols

Every main result is stated formally and then explained in a short paragraph headed "In words", which says what it means for a trader or a supervisor; proofs are in the online supplement. Supplement S15 works through two desks that use Kalman filters, in numbers, and Table S8 maps each question to its answer. Readers who want the finance first can read this introduction, Section 2.1, the worked example and Section 7, and take the statistical results as given. Table S31 explains the main symbols.

## S31 Reproducibility and formal verification

Every number in the paper is produced by a script or notebook cell that writes a result file from public data or from a closed form; Supplement S34 maps each table, figure and number to its source. "Registered" or "pre-specified" in this paper means that the authors wrote the specification and committed it to the public repository before opening the test data; this is the authors' declaration, not independent preregistration. For the October 2026 follow-up tests (Supplement S11 and Supplement S19) each specification reached the public repository in the same push as its results, so the order rests on the commit history alone. For the real-noise test, the class-flow test and the CFTC robustness checks (Supplement S16 and S18), specifications and code were pushed in a separate commit (`84af69f`, 03:43 UTC on 31 October 2026) before the first of them ran (03:56 UTC). The proofs have been checked by the authors and against numerical computations, and they have not yet been independently reviewed; the numerical checks reproduce results with the same code unless stated otherwise. Lean 4 with Mathlib machine-checks, using only Lean's standard axioms, three of the main results: Theorem 3.1(i) for the model itself (for known $a$ and $\kappa \neq 0$, two economies have the same price response if and only if they have the same merged measure, $2K+1$ noise-free periods already determine that measure, and with distinct nonzero rates the transfer function has exactly $1+K$ poles; the classical step from pole count to minimal state dimension is not formalized), and the lower bound of Theorem 6.1 on a finite observation horizon, from the moment gap through the closed loop and the Gaussian divergence to the testing step. The check holds for every $K$ with no added assumption: in place of the Gauss and Gauss–Radau rules it uses another moment-matched pair, binomial weights on $2K$ equally spaced nodes, which gives the same order; its constant depends on the number of observed periods per

experiment. Lean checks the matching upper bound of Theorem 6.2 for every $K$: with probability at least $1-\beta$ the Prony estimate exists, is unique and recovers every merged weight to within $C\sqrt{\log(4K/\beta)}\,\epsilon_N$, with constants that do not depend on the spacing; the formal version allows sub-Gaussian noise, which includes the Gaussian case, and assumes memory rates of modulus at most one, and a separate formal lemma shows that Prony's method computes exactly the estimate that the theorem bounds. Lean also checks the reset-experiment form of the tagged-flow result: class-tagged records recover every group's weight with an error bound that does not depend on the memory rates. Six further results are checked in Lean as well: Theorem 3.1(ii), the $\Omega(\delta^{-4})$ separation bound for power-constrained probes (stated for tests whose two error probabilities are at most 1/4); the spectral-blindness theorem (Theorem S12.1); the zero structure, minimum phase, and spectral-factorization rate identification of $H$ in Theorem 6.4; a finite-sample tagged rate-recovery bound giving the $(\Delta\eta)^{-2}$ upper bound of Theorem 7.2(ii); the uniform nonsingularity of the instrument matrix in Proposition 7.1; and the sample bounds of Theorem 6.5(i) and (iii) (every $K$) for passive data, for a market that has run any number of periods from rest before observation starts (built on a new formal lemma: the Kullback–Leibler divergence of a unit-triangular linear image of i.i.d. Gaussians is half the squared Frobenius norm of its off-diagonal part). From the project's earlier formalization, Lean also covers the microfoundations of exponential memory (Lemma 2.1, in the steady state), the positive geometry of the price response (Lemma 2.3), fair execution (Lemma S24.1), the robust quantity cap (Theorem S24.2) and exact disclosure (Proposition S24.3). Lean checks the lower-bound halves of the two probing-cost results end to end: for Theorem 5.1(i), any probe with total executed quantity $V$ gives divergence at most $K_*^2V^2\delta^4/(2\sigma^2)$, so a test whose two error probabilities are at most 1/4 needs $V^2 \geq \sigma^2/(2K_*^2\delta^4)$; for Lemma S24.4, a trader who starts with cash $B$ and is solvent at the fundamental mark after the probe gives divergence at most $K_*^2B\delta^4/(\sigma^2 m)$, so the same test needs $B \geq \sigma^2 m/(4K_*^2\delta^4)$, where the formal $m>0$ has the paper's form $\kappa(1-r)/(1+r)$ with some $r<1$ rather than the explicit $r$ of the lemma. Not checked: the exact error formula of Theorem 5.1(ii), the converse that a single pulse stays margin-solvent, and, for Theorem S3.1 and Proposition S3.2, the path divergence of the autoregression and the variance formula (Lean checks their decision and algebraic steps; Mathlib has no Pinsker inequality). Lean checks the supervisory separation, Proposition 7.4, for two groups, and its market-wide part for every $K$. Lean also checks both parts of the general-memory theorem, Theorem S17.2, so that its minimax accuracy is machine-checked to be of order $1/\log(N/\sigma^2)$. For the lower bound it uses a moment-matched pair that needs no approximation theory (the same binomial pair, placed on $[0,\rho]$, separated by a scaled cosine): no estimator reaches accuracy $2\rho W\log(1/\rho)/(\pi(\Lambda+6\log(1/\rho)))$ with probability 3/4 under both economies, where

$\Lambda = \log(2NC^2/\sigma^2)$ and $C = 2\kappa^2 W/((1-\rho)(1-s)^2)$. For the upper bound, since Mathlib has no Jackson theorem, the proof builds one with explicit constants from a discrete Fejér–Korovkin operator, and then shows that, for $N/\sigma^2$ beyond a threshold depending only on the class, the moment estimator is within $16\pi\rho W \log D/\log(N/\sigma^2)$ with probability at least $1-\beta$, with an explicit $D$; the constants are larger than those of the theorem. Lean also machine-checks the exact two-root stock–flow inventory bridge linking Remark 2.1 to position stocks and flows and the Lipschitz stability of the closed-loop deconvolution under plug-in estimation of $(a, \kappa, p^*)$ in Corollary 6.3. The divergence-rate form of the passive-data theorem and the matching lower bound on its divergence, and the ergodic and central limit steps of Proposition 7.1 are not formally verified; Supplement S3.7 reports numerical stress tests of them. The results whose proofs most need an independent check are Theorems 6.5, S9.2 and S9.4. Numerical constants such as the capital bounds are computed at a declared parameter set and are not separate theorems.

**Formal verification log.** The paragraphs below record what was machine-checked in each revision; they were moved here from Section 7.2.

**Formal verification scope, 5 November 2026.** This revision builds the pinned Lean 4.34.1/Mathlib project and audits its typed declarations using only standard foundations. The exact two-point Gauss–Radau witness in Table S36 is now covered by `Paper1GaussRadau.lean`: positive normalized weights, moments of orders zero through two, the third-moment gap, a dominant-intensity gap of at least $W/(8u)$ for $0 \le \Delta\eta \le u/3$, the aggregate-budget gap and the actual finite-Gaussian-record testing bound. Thus this example no longer relies solely on the other matched pair used by the earlier supervisory theorem. The finite-horizon bound has constants depending on the declared record length and response envelope. Its proof does not supply an all-$K$ quadrature construction. `Paper1SupervisoryDecision.lean` additionally checks the robust threshold/fallback, the actual unsafe-halt event for Proposition S29.1, the fixed-menu no-data optimum and the acquisition algebra of Proposition S29.2. The new receipt binds the build to hashes of the active Paper 1 sources. Declaration counts include supporting results across the project and are not counts of fully formalized manuscript statements. Independent human review, novelty and empirical market assumptions are not certified by Lean.

**Formal verification additions, 6 November 2026.** Pinsker's inequality, which Mathlib lacks, is now proved in Lean in the form the paper uses (`Paper1Pinsker.lean`): $2(P(A) - Q(A))^2 \le D_{\mathrm{KL}}(P\|Q)$ for every measurable event $A$. With it the testing step holds at every

error level, and Lean checks the following in the paper's own form (`Paper1GeneralLevel.lean`): Theorems 6.1 and S25.1 on a finite horizon, where any test with error probabilities $\alpha$ and $\beta$ needs $4\sigma^2(1-\alpha-\beta)^2 \le NT(\varepsilon C_T)^2$; Theorem 5.1(i) as stated, a worst error of at least $\frac{1}{2}(1 - K_*V\delta^2/(2\sigma))$ and hence $V \ge 2\sigma(1-2\alpha)/(K_*\delta^2)$; and the capital bound (S18) of Lemma S24.4, with a formal margin constant $m > 0$ of the paper's form. Theorem S3.1 is machine-checked end to end for rules given by measurable sets of paths (`Paper1Abstain.lean`): the autoregressive path laws are unit-triangular images of i.i.d. Gaussians, the divergence between two of them is at most $(s_1 - s_0)^2 M\, T(T-1)/4$ when $s_0^{2k} \le M$, and both the abstention bound and the binary-rule bound follow as stated; randomized rules are not covered. Corollary 6.3(i)–(ii) is machine-checked (`Paper1UnknownPrimitives.lean`): if two economies with unknown and possibly different $(a, \kappa)$ have the same price response and nonzero memory rates, they have the same $\kappa$, the same $a$ and the same merged measure; part (iii) is proved on paper.

**Formal verification additions, 7 November 2026.** Proposition S7.1(i) is machine-checked on the raw records (`Paper1GaussianHCR.lean`, `Paper1GaussianCR.lean`, `Paper1CRActual.lean`, `Paper1CRRaw.lean`). The Gaussian likelihood ratio and its second moment are derived from the densities. The Hammersley–Chapman–Robbins inequality then holds for every square-integrable statistic. A limit along a line gives $\mathrm{Var}_\theta(\hat{w}_k) \ge \sigma_h^2[F_{\mathrm{unit}}^{-1}]_{kk}$ for every statistic of all $N$ raw reset records that is unbiased near $\theta$. Here $J$ is the true Jacobian of the actual closed-loop response; Lean proves that this response is differentiable in the cohort sizes and rates. Part (iii) is machine-checked for both weight coordinates on the stated family (`Paper1FisherBounds.lean`, `Paper1CRTaps.lean`, `Paper1CRS71iii.lean`, `Paper1CRS71iiiEta.lean`, `Paper1CRS71iiiW2.lean`). The taps are those of the actual closed-loop recursion and $J$ is their true Jacobian. The lower bound uses the direction $v = (1, -1, \Delta\eta/(2w_1), \Delta\eta/(2w_2))$, along which every kernel moment moves by $(\Delta\eta)^3 Q_m$ with $|Q_m| \le m^3$, so $\|Jv\|^2 \le B_T(\Delta\eta)^6$. The upper bound uses an explicit combination of $h_1, \dots, h_4$ that recovers $w_k$ to first order, built from the inverse row of the confluent Vandermonde Jacobian. Switching between $r_i$ and $\eta_i = 1 - r_i$ leaves the weight entries unchanged. Theorem 6.4(ii) is machine-checked for every $K$, starting from equal spectra at every frequency (`Paper1SpectralID.lean`, `Paper1SpectralH.lean`). Two stable economies satisfying (i), with possibly different numbers of cohorts, anchors, impacts and innovation variances, have equal spectra only if they have equal $\sigma_\xi^2\kappa^2$, equal rate polynomials $\prod_i(z - r_i)$ (hence the same rates) and equal denominators of $H$. With $\kappa$ known they also have the same anchor and the same merged weights. Stability enters as the condition that the poles of $H$ lie in the open unit disk. This replaces the earlier two-cohort result, which

started from equal numerator polynomials rather than equal spectra. Proposition S3.2 is machine-checked end to end for the explicit block model (`Paper1PassiveVariance.lean`). This covers the block mean and second moment from the innovation and noise moments, Var $\leq R$ when $\mathbb{E}w^4 \leq 3s_w^4$, Chebyshev for $n$ independent blocks, and both midpoint errors at most $\alpha$ once $n \geq 4R/(\alpha\Delta^2)$. The converse in Lemma S24.4 is machine-checked (`Paper1CapitalConverse.lean`): the error probabilities of the single-pulse test on the third observation, and margin solvency of the funded pulse trader at every later mark. Parts (ii) of Propositions S5.1 and S5.2 are machine-checked (`Paper1KalmanTied.lean`). In Proposition S5.2, a bound on the second derivative of each tap function $w(\eta)(1-\eta)^m$ is a stated smoothness hypothesis, and part (iii), which uses Brouwer's theorem, is not machine-checked. Still written-only: the stationary IV central limit theorem of Proposition 7.1, and the divergence-rate form and the $\Theta$ bounds of Theorem 6.5.

## S32 Further tables and figures for the main text

These tables and figures belong to the main text; they are collected here to keep it to journal length.

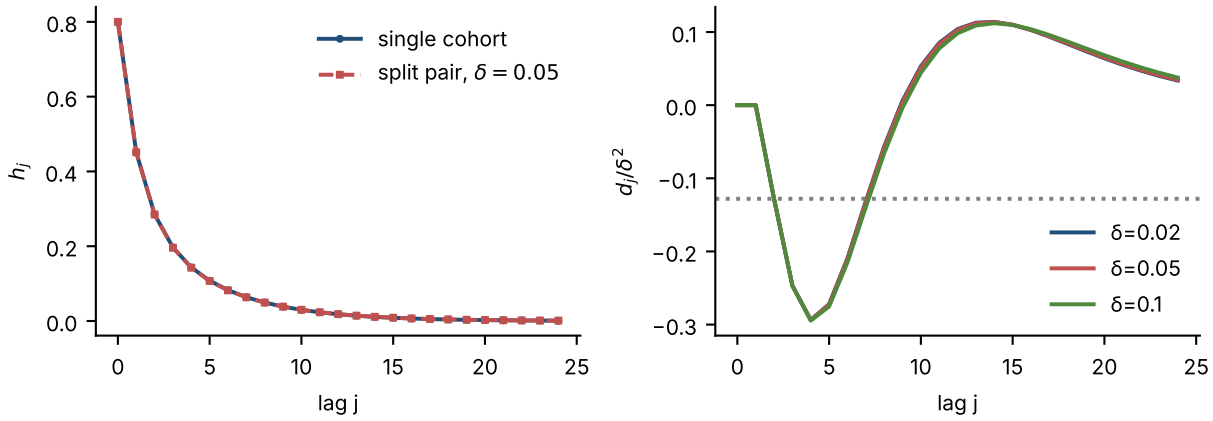



Figure S7: Left: impulse responses of the single cohort and the split pair ($\delta = 0.05$) are nearly indistinguishable. Right: the difference scaled by $\delta^2$ collapses onto one curve, with first nonzero value $-\kappa^2 q$ at lag 2 (dotted). Calibration $a = 0.5$, $\kappa = 0.8$, $q = 0.2$, $\eta = 0.4$.

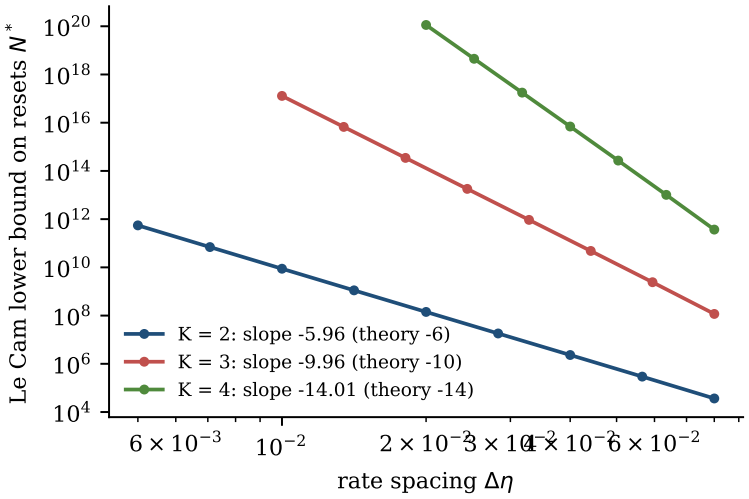



Figure S8: Two-point lower bound on the number of reset experiments needed to separate $K$-cohort economies, as a function of the rate spacing.

Table S8: Map of results

| Question | Answer | Where |
|---|---|---|
| What do prices identify? | The merged rate–weight measure only | Thm 3.1 |
| Quantity to separate a known pair | $\Theta(\delta^{-2})$, exact error formula | Thm 5.1 |
| Capital to separate a known pair | $\Theta(\delta^{-4})$, counterparty or own funds | Thm S24.2, Lem. S24.4 |
| Constant at the calibration | $[61.41, 64.48]$ par units; $\approx 10^5$ USDT | Thms S9.2–S9.4, Table S14 |
| Does adaptivity help? | At most by a factor $\approx 0.49$ in expected loss | Thm S9.3 |
| Unknown mixtures | $(\Delta\eta)^{-(4K-2)}$, tight on rich classes for every $K$ | Thms 6.1, 6.2 |
| General memory | Accuracy $\asymp 1/\log N$, lower and upper bounds | Lem. S17.1, Thm S17.2 |
| Passive data | Identifies $M$; at least the same polynomial cost for every $K$ | Thms 6.4, 6.5, Prop. S3.2 |
| Nonlinear and adaptive rules? | Settled online learners obey $\delta^{-4}$; fast adaptation or thresholds break it | Table S16 |
| What removes the barrier? | Tagged flow, audited equity | Props S24.3, 7.1 |
| Does attribution change a decision? | Yes in a stylised case: a market-wide limit needs only $S$; whether a targeted halt suffices needs each $q_i$ | Prop. 7.4 |
| What is tagging worth? | Factor $(\Delta\eta)^{-(4K-2)}$ (fixed accuracy) or $(\Delta\eta)^{-(4K-4)}$ (rates); $\approx 333\times$ for BTCUSDT (illustrative) | Thm 7.2 |
| Hypothetical probing loss? | At least \$15M–\$356M a year per market to re-learn two unknown groups weekly at $\Delta\eta = 0.10$; illustrative, not a social cost | Table S10 |
| Do class data restrict class persistence? | Yes: CFTC 8 of 8; Binance 30 of 30 and 29 of 29 new markets (registered); whether it is memory is open | Tables S25, S27 |
| Do the rates survive real price noise? | Yes where it was used directly; large samples use a normal approximation | Table S23 |
| Can eigenvalues certify safety? | No | Thm S12.1 |
| What do feedback indices measure? | Mostly filtering and timing; small real feedback | Tables S18, S20 |
| Does feedback track activity? | No: fees move volume 2.6–4.9×, not $\hat{Q}$ | Table S22 |

Table S9: Cramér–Rao sample-size benchmark $N_{\mathrm{CR}}$ of (S1) for multi-cohort weight recovery ($T = 120$ taps)

| Scenario | $K$ | $N_{\mathrm{CR}}$ probes | $\sigma_{\mathrm{crit}}$ | ML success ($f = 1$) | RMSE/CRB |
|---|---|---|---|---|---|
| $\eta = 0.30$ | 1 | 0.1 | $6.05 \times 10^{-3}$ | 94% | 1.02 |
| $\Delta\eta = 0.30$ | 2 | 47.8 | $2.31 \times 10^{-4}$ | 94% | 1.01 |
| $\Delta\eta = 0.20$ | 2 | 597 | $6.55 \times 10^{-5}$ | 96% | 0.98 |
| $\Delta\eta = 0.10$ ($\delta = 0.05$ pair) | 2 | 40,374 | $7.96 \times 10^{-6}$ | 93% | 1.03 |
| $\Delta\eta = 0.05$ | 2 | 2,620,320 | $9.88 \times 10^{-7}$ | 95% | 1.03 |
| spacing 0.10 | 3 | 17,349,791 | $3.84 \times 10^{-7}$ | 98% | 1.22 |
| spacing 0.20 | 3 | 27,038 | $9.73 \times 10^{-6}$ | 98% | 0.88 |

Tolerance $\pm 0.01$ at 95%. $\sigma_{\mathrm{crit}}$ is the single-probe noise at which one probe suffices; operational single-probe noise is $1.6 \times 10^{-3}$. ML columns: least squares over all weights and rates from six generic starts (never the true values), 200 replications per row at $\sigma_{\mathrm{crit}}$; success means every weight within $\pm 0.01$. Fitted log–log slope of $N_{\mathrm{CR}}$ in $\Delta\eta$: $-6.04$. $N_{\mathrm{CR}}$ is a benchmark for an efficient, approximately normal estimator, not a lower bound for every procedure (Proposition S7.1(ii)).

Table S10: Price-only round-trip probing loss per market and annualised weekly-renewal cost (hypothetical outside prober)

| | Loss / probe (USDT) | | Annual (52×, USDT) | |
|---|---|---|---|---|
| Economy ($\Delta\eta$) | Roll | 1-sec | Roll | 1-sec |
| $K = 2$, known (0.20) | 6,200 | 149,901 | 322,400 | 7,794,852 |
| $K = 2$, known (0.10) | 99,192 | 2,398,422 | 5,157,984 | 124,717,944 |
| $K = 2$, known (0.04) | 3,874,696 | 93,688,369 | 201,484,192 | 4,871,795,188 |
| $K = 2$, Radau (0.20) | 4,292 | 103,788 | 223,184 | 5,396,976 |
| $K = 2$, Radau (0.10) | 283,488 | 6,854,612 | 14,741,376 | 356,439,824 |
| $K = 3$, Radau (0.20) | 804,403 | 19,450,107 | 41,828,956 | 1,011,405,564 |
| $K = 3$, Radau (0.10) | 1,137,325,880 | 27,500,017,407 | 59,140,945,760 | 1,430,000,905,164 |
| $K = 4$, Radau (0.20) | 156,632,086 | 3,787,291,895 | 8,144,868,472 | 196,939,178,540 |

Separation variable $\Delta\eta$ throughout ($\Delta\eta = 2\delta$ for the symmetric known pair). Gauss–Radau rows use the exact $K$-cohort quadrature pair of Theorem 6.1 and Table S33 (moment gaps $\Delta_2 = 0.222222$, $\Delta_3 = 0.053333$, $\Delta_4 = 0.013061$ at $W = 0.08$, $r_c = 0.60$). Loss per probe is the spectral round-trip trading loss bound of Theorem S9.2 at the median BTCUSDT Roll noise (0.40 USDT) and median one-second price-change standard deviation (1.98 USDT); annual loss multiplies by 52 weekly renewals. Probing cost is a lower bound on the cost of learning from prices by trading, not on passive learning; it is a private transfer to other traders, not a social or resource cost, and it is not comparable with audit-trail budgets.

Table S11: Account-level initial capital at $\delta = 0.05$ (par units)

| Design | $C_0$ | Error | End inv. | Peak margin | Max $\lvert v_t \rvert$ | Trades |
|---|---|---|---|---|---|---|
| 12 buys, $T = 20$ apart, never unwound | 229.13 | 0.0578 | 75.00 | 39.38 | 6.25 | 12 |
| 6 buy/sell round trips | 189.45 | 0.0308 | 0.00 | 3.28 | 6.25 | 12 |
| 5 buy/sell round trips | 157.89 | 0.0442 | 0.00 | 3.28 | 6.25 | 10 |
| 12 buys, each unwound after 20 steps | 378.82 | 0.0040 | 0.00 | 3.28 | 6.25 | 24 |
| Generalized-eigenvector probe ($L = 120$) | 65.06 | 0.0500 | 0.00 | 2.74 | 1.24 | 120 |
| Generalized-eigenvector probe ($L = 160$) | 64.86 | 0.0500 | 0.00 | 2.17 | 1.09 | 160 |
| Fee-refined probe ($L = 240$) | 64.48 | 0.0500 | 0.00 | — | 1.30 | 240 |

The pulse designs place a pulse every 40 steps and close it 20 steps later; the fee-refined probe is re-optimized by sequential quadratic programming over $L = 240$. The numerically evaluated lower bound of Theorem S9.2 is 61.41; the explicit constant of Lemma S24.4 is $4.9 \times 10^{-4}$.

Table S12: Capital lower bounds and feasible upper bounds across separations (par units)

| $\delta$ | $\omega^*$ | $C^*_{\mathrm{det,quad}}$ | $\delta^4 C^*_{\mathrm{det,quad}}$ | Certified $C_{\mathrm{det}}$ | Feasible UB | Adaptive LB |
|---|---|---|---|---|---|---|
| 0.02 | 0.2546 | 2397.85 | $3.84 \times 10^{-4}$ | 2399.58 | 2428.25 | 1176.04 |
| 0.03 | 0.2541 | 473.19 | $3.83 \times 10^{-4}$ | 473.97 | 484.50 | 232.51 |
| 0.05 | 0.2526 | 61.14 | $3.82 \times 10^{-4}$ | 61.41 | 64.48 | 30.22 |
| 0.08 | 0.2488 | 9.26 | $3.79 \times 10^{-4}$ | 9.37 | 10.38 | 4.64 |
| 0.10 | 0.2453 | 3.77 | $3.77 \times 10^{-4}$ | 3.83 | 4.44 | 1.91 |
| 0.15 | 0.2331 | 0.73 | $3.67 \times 10^{-4}$ | 0.76 | 1.01 | 0.38 |

Calibration $a = 0.5$, $\kappa = 0.8$, $q = 0.2$, $\eta = 0.4$, $\sigma = 0.01$, $\alpha = 0.05$. the numerical-bound column adds the fee lower bound to the fee-free infimum; the feasible column is the best flat probe found over $L \le 240$; the adaptive column adds the fee lower bound to the adaptive bound.

Table S13: Finite-horizon capital at $\delta = 0.05$ (par units)

| Horizon $L$ | Quadratic part $c_{\mathrm{req}}/\lambda_{L-1}$ | Full bound $C_{\mathrm{RH}}(L)$ | Best feasible probe | Gap to $C_{\mathrm{det}} = 61.41$ |
|---|---|---|---|---|
| 80 | 62.55 | 64.72 | 65.02 | 5.9% |
| 120 | 61.84 | 63.39 | 64.59 | 5.2% |
| 160 | 61.54 | 62.46 | 64.50 | 5.0% |
| 240 | 61.32 | 61.76 | 64.48 | 5.0% |
| 320 | 61.25 | — | $\le 64.48$ | $\le 5.0\%$ |
| 480 | 61.19 | — | $\le 64.48$ | $\le 5.0\%$ |

Quadratic bound $c_{\mathrm{req}}/\lambda_{L-1}$ from the generalized eigenproblem of Theorem S9.4 on flat probes (equivalently $c_{\mathrm{req}}/(2\lambda'_L)$ with $\lambda'$ computed against $T_{h_0}$ rather than $\frac{1}{2}T_{h_0}$); it converges to the infinite-horizon value 61.14. The full bound minimizes (S10) over 3,600 values of $\theta$; at $L = 80$ and 120 a direct numerical minimization of loss plus fee under the separation constraint gives 64.76 and 64.43, above the bound. Feasible probes: top generalized eigenvector refined by sequential quadratic programming on the full ledger; at $L = 320$ and 480 the $L = 240$ probe padded with zeros is feasible at the same cost, so the best feasible value can only be lower.

Table S14: Capital lower bound under alternative observation-noise calibrations (USDT)

| Noise calibration | $\delta = 0.02$ | $\delta = 0.05$ | $\delta = 0.10$ |
|---|---|---|---|
| Roll estimator, median (0.40 USDT) | 3,874,696 | 99,192 | 6,200 |
| One-second price-change s.d., median (1.98 USDT) | 93,688,369 | 2,398,422 | 149,901 |
| Earlier assumption (4 USDT, upper scenario) | 383,812,500 | 9,825,600 | 614,100 |

$C(\sigma, \delta) = 61.41 \times (\sigma/0.01)^2 \times (0.05/\delta)^4$, with $\sigma$ the unrounded median. Noise estimated on 11 days of checksum-verified Binance BTCUSDT one-second klines.

Table S15: Daily observation-noise estimates, Binance BTCUSDT one-second closes

| Date | Roll $\hat{\sigma}$ (USDT) | One-second change s.d. (USDT) | Mean price (USDT) |
|---|---|---|---|
| 2023-01-10 | 0.152 | 0.679 | 17,288 |
| 2023-01-11 | 0.092 | 0.891 | 17,456 |
| 2023-01-12 | 0.232 | 2.292 | 18,343 |
| 2023-01-13 | 0.348 | 1.762 | 19,075 |
| 2023-01-14 | 0.773 | 3.511 | 20,818 |
| 2023-01-15 | 0.482 | 1.790 | 20,778 |
| 2023-01-16 | 0.629 | 2.357 | 21,036 |
| 2023-01-17 | 0.457 | 1.976 | 21,194 |
| 2023-01-18 | 0.695 | 2.541 | 21,122 |
| 2023-01-19 | 0.402 | 1.316 | 20,850 |
| 2023-01-20 | 0.316 | 2.038 | 21,306 |
| Median | 0.402 | 1.976 | |

The Roll estimator measures the negative first-order autocovariance of price changes, which reflects bid-ask bounce and recording noise rather than price volatility.

Table S16: Merged-measure equivalence and the $\delta^{-4}$ law under stylized nonlinear rules and closed-loop trained online learners

| Rule | Test A raw ($O(A^2)$) | Test A norm. ($O(A)$) | Test B slope ($\|\Delta h\|_2$) | Test B slope (capital) | Capital at $\delta = 0.05$ | $A_{\text{sat}}$ / $R^2_{\text{exp}}$ |
|---|---|---|---|---|---|---|
| *Panel A. Stylized nonlinear and adaptive rules (fixed operating point)* | | | | | | |
| Linear benchmark | — | — | 2.007 | −4.001 | 61.42 | — |
| 1. Momentum (tanh limit) | 2.015 | 1.015 | 2.007 | −4.001 | 61.47 | 0.834 |
| 2. Threshold rule, averaged | 2.043 | 1.043 | 2.007 | −4.001 | 61.42 | 0.049 |
| 3. Adaptive gain (RLS) | 2.064 | 1.064 | 2.007 | −4.001 | 61.46 | 0.751 |
| 4. Online neural readout | 2.011 | 1.011 | 2.007 | −4.001 | 61.48 | 0.751 |
| 5. Tabular value, smoothed | 2.039 | 1.039 | 2.007 | −4.001 | 61.44 | 1.271 |
| *Panel B. Learners trained in the closed loop, saved and reloaded (frozen during the probe; shared training stream)* | | | | | | |
| 6. Recursive least squares | 3.000 | 2.000 | 2.008 | −4.014 | 54.58 | $R^2 = 0.993$ |
| 7. Two-layer network (SGD) | 1.878 | 0.878 | 2.007 | −4.012 | 65.19 | $R^2 = 0.994$ |
| 8. Policy gradient on profit | 3.000 | 2.000 | 2.004 | −4.006 | 58.05 | $R^2 = 0.995$ |
| 9. Tabular Q-learner (TD(0)) | 2.108 | 1.108 | 1.994 | −3.973 | 186,821 | $R^2 = 1.000$ |
| *Panel C. Where the trained-learner results change (range over the four learners)* | | | | | | |
| Independent training streams | rel. gap $\leq 10^{-5}$ | | 0.13 to 0.92 | −1.75 to −0.38 | — | (c) fails |
| Learning during the probe | rel. gap $\leq 1.1 \times 10^{-3}$ | | 1.99 to 2.01 | −4.02 to −3.98 | — | (c) holds |

Exact dead zone ($\tau = 0.05$): $\|\Delta h\|_2 = 0$ at $A = 0.005$; the probe moves the state past $\tau$ only for $A > 0.139$.
Single-bin value lookup: $\|\Delta h\|_2 = 0$ at $A = 0.005$; the probe leaves the central bin only for $A > 0.149$.

Panel A: deterministic simulations around the operating point; Test A reports the raw response-gap slope in $\log A$ ($O(A^2)$) and the probe-normalised gap slope $\|X^{(1)}(A) - X^{(2)}(A)\|_2/A$ ($O(A)$, plotted in Figure S4). Panel B: learners trained for 5,000 periods, parameters saved and reloaded without rescaling and frozen during the probe; the two compared populations are trained on the same innovation stream; relative equivalence gaps at $A = 0.005$ are at most $10^{-5}$. A raw Test A slope above 2 means the gap is of higher order than $A^2$. Panel C: the same learners trained on independent innovation streams (first row) or updated at learning rate 0.35 during the probe (second row). Capital in par units, lower bound of Theorem S9.2 (61.42 for the linear model); the Q-learner settles on a small response gain (0.022), so separating it costs far more capital.

Table S17: Samples

| Sample | Coverage | Size | Source |
|---|---|---|---|
| Binance spot panel, 8 pairs | 96 dates, 2021–2024 | 66.3M one-second bars | Binance archive |
| Binance spot fee-event windows, 8 pairs | ±20 days around 2022-07-08 and 2023-03-22 | 640 pair-days | Binance archive |
| Binance USD-M perpetuals, 30 symbols | 90 dates, 2023–2026 | 2,565 symbol-days | Binance archive |
| Binance BTCUSDT one-second panel | 72 dates, 2021–2023 | 6.2M one-second bars | Binance archive |
| Binance BTCUSDT noise-calibration days | January 10–20, 2023 | 11 days | Binance archive |

Binance files are verified against the published SHA-256 checksums (every file used). Sampled dates are the 10th and 24th of each month.

Table S18: Multi-year Binance spot panel: reduced-form indices under three timing constructions

| Construction | $\hat{Q}$ | Date interval | $\hat{S}$ | Date interval | $\hat{Q} > 0$ |
|---|---|---|---|---|---|
| Forward-paired flow (look-ahead) | 0.0081424 | [0.0072965, 0.0089882] | 0.92074 | [0.91665, 0.92482] | 87.4% |
| Observed-flow EMA | 0.0003733 | [0.0002351, 0.0005116] | 0.92072 | [0.91640, 0.92504] | 62.4% |
| Observed, 30 flow lags | 0.0002580 | [0.0001415, 0.0003745] | 0.91738 | [0.91167, 0.92308] | 59.1% |
| Observed raw return | −0.0001555 | [−0.0003732, 0.0000622] | −0.04051 | [−0.05078, −0.03025] | 49.2% |
| Placebo random walk | 0.0000016 | [−0.0000020, 0.0000052] | 0.93510 | [0.93485, 0.93534] | 51.0% |

768 study units (96 dates × 8 pairs), 6,634,752 ten-second transitions; equal-weight asset means within date, then date means with Student intervals. Mechanical persistence $29/31 = 0.93548$.

Table S19: Asset means under forward-paired and observed flow

| Pair | Forward-paired $\hat{Q}$ | Observed $\hat{Q}$ | $\hat{Q}_{30}$ | Observed $\hat{S}$ |
|---|---|---|---|---|
| BTC/USDT | 0.0141870 | 0.0006073 | 0.0004891 | 0.93464 |
| ETH/USDT | 0.0130885 | 0.0005675 | 0.0002549 | 0.93511 |
| SOL/USDT | 0.0087763 | 0.0002129 | 0.0001100 | 0.91966 |
| ADA/USDT | 0.0067465 | 0.0003928 | 0.0003546 | 0.91452 |
| DOGE/USDT | 0.0058956 | 0.0002464 | 0.0001432 | 0.92496 |
| BNB/USDT | 0.0057420 | 0.0004479 | 0.0002762 | 0.89533 |
| XRP/USDT | 0.0057034 | 0.0002249 | 0.0001508 | 0.92362 |
| LTC/USDT | 0.0049997 | 0.0002870 | 0.0002853 | 0.91792 |

96 dates per pair. $\hat{Q}_{30}$ controls 30 ten-second bars of lagged flow. Placebo $|\hat{Q}| \leq 1.1 \times 10^{-5}$ for every pair.

Table S20: Feedback beyond filtering mechanics (Binance BTCUSDT, 72 days, 2021–2023)

| | Real prices | Placebo (independent random walk) |
|---|---|---|
| Mean loop gain $\hat{Q}$ | 0.0045 | 0.0000009 |
| 95% CI (day bootstrap) | [0.0034, 0.0056] | [−0.0000006, 0.0000023] |
| Mean price persistence $\hat{J}_{11}$ | 0.9364 | 0.9357 |
| Mechanical value $1 - 2/31$ | | 0.9355 |

Placebo 95% band across days: $[-0.0000095, 0.0000156]$. This construction pairs the state with next-second flow; it is not comparable in level with the observed-flow ten-second index of Table S18.

Table S21: Feedback estimators under order splitting (simulation, 500 replications)

| Estimator | Coverage at $\gamma = 0.2$ | Rejection at $\gamma = 0$ | Bias | RMSE |
|---|---|---|---|---|
| OLS | 0.000 | 1.000 | 0.892 | 0.893 |
| Cochrane–Orcutt, true $\rho$ | 0.944 | 0.054 | 0.001 | 0.039 |
| Cochrane–Orcutt, estimated $\rho$ | 0.916 | 0.080 | 0.002 | 0.043 |
| 2SLS | 0.938 | 0.044 | −0.002 | 0.116 |
| LIML | 0.938 | 0.044 | −0.003 | 0.117 |

Newey–West standard errors (10 lags), $T = 3{,}000$. Median first-stage $F = 224$; a placebo instrument (future $z$) has $F = 0.62$. Estimators outside 93–97% coverage give invalid inference.

Table S22: Zero fees on BTCUSDT: activity and measured feedback

| | Zero fees introduced (2022-07-08) | Zero fees removed (2023-03-22) | Pooled |
|---|---|---|---|
| Log quote volume, $\hat{\beta}$ | 0.945 (0.074) | 1.599 (0.080) | 1.272 (0.084) |
| Observed-flow $\hat{Q}$, $\hat{\beta}$ | 0.00058 (0.00032) | −0.00091 (0.00060) | −0.00016 (0.00036) |
| $p$-value | 0.078 | 0.140 | 0.650 |
| Pair-days | 320 | 320 | 640 |

$\hat{\beta}$ is the effect of zero fees on BTCUSDT relative to seven control pairs; date-clustered standard errors in parentheses. For the second event ZeroFee equals one before March 22, 2023.

Table S23: Sample requirement with real Bitcoin price noise: fitted slope of $\log N^*$ in $\log \Delta\eta$

| | Theory | Real noise | Gaussian control | Pre-specified criterion | Result |
|---|---|---|---|---|---|
| Price only, $K = 2$ (H1) | −6 | −6.01 | −6.22 | slope in $[-7.5, -4.5]$ | pass |
| Tagged, $K = 2$ (H2) | bounded | 256–724 | 256–1,024 | max/min $\leq 8$ | pass (2.8) |
| Price only, $K = 3$ (H3) | −10 | −9.81 | −9.96 | slope in $[-12.5, -7.5]$ | holds |

Synthetic groups at rates $r_c + \Delta\eta\, s$ ($s$ the $K$-point Gauss nodes; $r_c = 0.6$ and $\Delta\eta$ from 0.5 to 0.125 for $K = 2$, $r_c = 0.5$ and $\Delta\eta$ from 0.6 to 0.25 for $K = 3$), $a = 0.5$, $\kappa = 0.8$, total weight 0.08, $\sigma = 10^{-3}$, $\sigma_f = 0.02$. The test passes if H1 and H2 hold; H3 is reported either way. The tagged row gives the range of $N^*$ over the spacing grid. Requirements above 4,096 use the central-limit draw described in the text. Requirements at each spacing are in Supplement Table S3. Test opened once.

Table S24: General memory: moment-matched cohort pairs and the moment estimator

| | | | | | | |
|---|---|---|---|---|---|---|
| *A. Best approximation of* $\lvert x\rvert$ | | | | | | |
| $L$ | 2 | 4 | 6 | 10 | 20 | 30 |
| $L\,E_L(\lvert x\rvert)$ | 0.2500 | 0.2705 | 0.2756 | 0.2785 | 0.2797 | 0.2800 |
| *B. Pairs of Lemma S17.1* | | | | | | |
| Atoms per measure $n$ | 2 | 4 | 6 | 8 | 10 | 12 |
| $W_1(\mu_A,\mu_B)$, $\rho=0.9$ | 0.2250 | 0.1037 | 0.0669 | 0.0494 | 0.0391 | 0.0324 |
| $n\,W_1$, $\rho=0.9$ | 0.450 | 0.415 | 0.402 | 0.395 | 0.391 | 0.388 |
| $\lVert k^A-k^B\rVert_2$, $\rho=0.9$ | $4.41\times10^{-2}$ | $1.03\times10^{-3}$ | $4.08\times10^{-5}$ | $1.96\times10^{-6}$ | $1.05\times10^{-7}$ | $5.95\times10^{-9}$ |
| $\lVert k^A-k^B\rVert_2$, $\rho=0.99$ | $7.41\times10^{-2}$ | $5.84\times10^{-3}$ | $1.07\times10^{-3}$ | $2.76\times10^{-4}$ | $8.33\times10^{-5}$ | $2.76\times10^{-5}$ |
| *C. Linear-programming estimator, continuous memory distribution,* $\rho=0.9$ | | | | | | |
| $\log_{10}N$ | 4 | 8 | 12 | 16 | 20 | 24 |
| Moments used $(L+1)$ | 2 | 3 | 5 | 6 | 7 | 8 |
| Error $d(\hat\mu,\mu)$ | 0.00878 | 0.00759 | 0.00432 | 0.00445 | 0.00382 | 0.00399 |
| Error $\times\log(N/\sigma^2)$ | 0.202 | 0.245 | 0.179 | 0.226 | 0.229 | 0.276 |

A: Remez algorithm; the limit is Bernstein's constant 0.2802. B: moments of orders $0,\dots,2n-2$ agree to $10^{-61}$ (60-digit arithmetic); every $W_1$ exceeds $\rho E_{2n-2}(\lvert x\rvert)$. C: true merged measure Beta(2,3) on $[0,0.9]$ with mass 0.1; kernel values observed with noise $\sigma N^{-1/2}$, $\sigma=10^{-3}$; $L$ from the rule in Theorem S17.2(ii); measure fitted on a 401-point grid; 20 replications.

Table S25: Fitted persistence of trader-class positions, CFTC Commitments of Traders, eight regulated futures markets

| Market | Weeks | Fast $\hat r$ [95% CI] | Slow $\hat r$ | $\hat r_{\text{fast}}-\hat r_{\text{slow}}$ [95% CI] | $\hat w_{\text{fast}}$ | Ratio |
|---|---|---|---|---|---|---|
| Bitcoin (CME) | 434 | 0.951 [0.928, 0.969] | 0.986 | −0.035 [−0.062, −0.010] | − | 23.7 |
| Ether (CME) | 282 | 0.890 [0.826, 0.932] | 0.960 | −0.070 [−0.143, −0.011] | − | 9.1 |
| E-mini S&P 500 | 1,039 | 0.914 [0.884, 0.937] | 0.954 | −0.040 [−0.071, −0.012] | − | 18.3 |
| 10-year T-note | 1,039 | 0.992 [0.985, 0.998] | 0.993 | −0.000 [−0.006, +0.007] | (0) | 71.4 |
| Euro | 1,035 | 0.961 [0.947, 0.972] | 0.996 | −0.034 [−0.051, −0.022] | + | 37.3 |
| Yen | 1,035 | 0.955 [0.939, 0.968] | 0.984 | −0.029 [−0.047, −0.013] | + | 34.0 |
| WTI crude oil | 1,039 | 0.961 [0.946, 0.974] | 0.999 | −0.037 [−0.053, −0.023] | + | 35.2 |
| Natural gas | 1,039 | 0.981 [0.969, 0.990] | 0.980 | +0.001 [−0.014, +0.017] | + | 44.0 |

*Pre-specified hypotheses*: H1 (primary) 8 of 8, pass; H2 4 of 8, fail; H3 6 of 8, pass (two indistinguishable).

"Fast" (designated speculative) class: leveraged funds (financial futures) or managed money (commodities); "slow" (designated comparison) class: asset managers or producers/merchants. The labels are fixed by class role before estimation, not by the estimates; in natural gas the designated fast class has the (insignificantly) higher persistence. $\hat w_{\text{fast}}$: sign of the response to returns, + or − when the 95% interval excludes zero, (0) otherwise. Ratio: mean width of the two price-only rate intervals over the width of the fast-class interval; the price-only intervals span 0.95–0.96 of the admissible range $[0,0.99]$ in every market. Weekly returns: Binance spot (Bitcoin, Ether); S&P 500 index, euro and yen spot, minus the change in the 10-year yield, and front-month crude-oil and natural-gas futures from Yahoo Finance. For the six non-futures series the spot–futures basis is ignored; a post hoc rerun with CME futures prices gives the same counts and the same signs of $\hat w_{\text{fast}}$ (see text). Test opened once.

Table S26: Price data needed to match the precision of the CFTC class reports under standard root-$T$ scaling (post hoc)

| Market | Weeks | Ratio | Factor | Years | Market | Weeks | Ratio | Factor | Years |
|---|---|---|---|---|---|---|---|---|---|
| CME Bitcoin | 434 | 23.7 | 562 | 4,688 | Gold | 1041 | 32.1 | 1,032 | 20,652 |
| CME Ether | 282 | 9.1 | 82 | 446 | Copper | 1041 | 36.3 | 1,315 | 26,318 |
| E-mini S&P 500 | 1039 | 18.3 | 336 | 6,718 | Coffee | 1041 | 68.8 | 4,735 | 94,796 |
| 10-year Treasury | 1039 | 71.4 | 5,105 | 102,000 | Live cattle | 1041 | 41.0 | 1,684 | 33,712 |
| Euro | 1035 | 37.3 | 1,391 | 27,684 | Nasdaq-100 | 1041 | 23.3 | 541 | 10,837 |
| Yen | 1035 | 34.0 | 1,155 | 22,997 | Sugar | 1041 | 52.1 | 2,719 | 54,427 |
| Crude oil | 1039 | 35.2 | 1,239 | 24,753 | Silver | 1041 | 23.1 | 532 | 10,658 |
| Natural gas | 1039 | 44.0 | 1,939 | 38,750 | Corn | 1041 | 48.1 | 2,313 | 46,302 |
| Cotton | 1041 | 50.9 | 2,587 | 51,793 | Soybeans | 1041 | 42.0 | 1,762 | 35,266 |
| British pound | 1039 | 29.9 | 896 | 17,910 | Wheat | 1041 | 44.1 | 1,943 | 38,893 |

Ratio: price-only over tagged width of the 95% interval for the speculative class's persistence, from the registered tests (first eight: original markets; last twelve: the replication). Factor = Ratio$^2$, the growth of the price sample needed to reach the tagged precision if both intervals shrink as $T^{-1/2}$; Years = Weeks × Factor/52. The price-only intervals are cut off by the admissible range $[0, 0.99]$, so every entry is a lower bound. Median factor 1,353, median 27,001 years.

Table S27: Fitted persistence rates with and without class-tagged data

| Symbol | Days | Top position $\hat{r}$ [95% CI] | All accounts $\hat{r}$ [95% CI] | Width ratio | Classes differ |
|---|---|---|---|---|---|
| BTCUSDT | 48 | 0.9889 [0.9842, 0.9935] | 0.9986 [0.9955, 1.0005] | 121.7 | yes |
| ETHUSDT | 48 | 0.9793 [0.9616, 0.9907] | 0.9972 [0.9934, 0.9997] | 55.6 | yes |
| CELRUSDT | 48 | 0.9884 [0.9830, 0.9922] | 0.9882 [0.9804, 0.9914] | 81.6 | no |
| CVXUSDT | 33 | 0.9822 [0.9732, 0.9908] | 0.9829 [0.9694, 0.9900] | 45.9 | no |

| *Pre-specified test on fresh days, 30 USD-M perpetuals, 2025–2026* | |
|---|---|
| (i) width ratio $> 10$ | 30 of 30 (range 27.6–124.0) |
| (ii) price-only rate interval wider than 0.25 | 30 of 30 (range 0.65–0.99) |
| (iii) merged response $\sum_m k_m$ identified | 5 of 30 |
| Classes differ | 11 of 30 |

Top panel: training days 2023–2024; width ratio is the mean price-only rate-interval width over the mean tagged width. Estimates are from day-demeaned regressions, which bias persistence down; corrected by a split-panel jackknife, the median tagged rate is 1.0008 (Supplement S19). The price-only intervals for both rates cover most of $[0, 0.99]$. Bottom panel: specification pre-specified before the fresh days were downloaded; a symbol passes if all three of the first rows hold, and the test required 80% of symbols to pass.

Table S28: Pre-specified out-of-sample cohort-state trajectory test on untouched 2nd-and-9th-of-month days, 2025–2026

| 30-min post-shock target (19 thin) | $R^2_{\text{pf}}$ | $R^2_{\text{tag}}$ | $\Delta R^2$ (pts) [95% CI] | CW $t$ ($p$) | Res. |
|---|---|---|---|---|---|
| Prim. 1: Divergence $d_{t+6}$ (9,875 shocks) | −4.766 | 45.962 | +50.728 [+45.291, +55.918] | 21.27 ($< 10^{-16}$) | pass |
| Prim. 2: Top state $n_{\text{top},t+6}$ (9,875 shocks) | 4.441 | 54.845 | +50.404 [+47.226, +53.724] | 21.85 ($< 10^{-16}$) | pass |
| Sec.: Divergence $d_{t+6}$, all (210,144 bars) | −0.586 | 37.053 | +37.639 [+36.232, +39.107] | 72.75 ($< 10^{-16}$) | — |
| Sec.: Top state $n_{\text{top},t+6}$, all (210,144 bars) | 0.186 | 46.779 | +46.593 [+45.555, +47.643] | 80.44 ($< 10^{-16}$) | — |

Out-of-sample $R^2$ in percent. Trained on 24 dates in 2024 (12,046 shocks, 120,384 bars); evaluated once on 42 untouched test dates (2nd and 9th of each month, Jan 2025–Sep 2026); 2,000-draw date-block bootstrap 95% CIs; Holm $p < 10^{-16}$ across both primaries.

Table S29: What the empirical tests found: passes and failures

| ID | Test | P/S | Prediction | Result (threshold where pre-specified) |
|---|---|---|---|---|
| E1 | CFTC, 8 markets, H1 | P | Class positions pin down a persistence that weekly prices leave open | Pass, 8 of 8 |
| E2 | CFTC, H2 | P | The designated speculative class trend-follows | Fail, 4 of 8 (three markets have the opposite sign) |
| E3 | CFTC, H3 | P | That class has the less persistent positions | Pass, 6 of 8 (two indistinguishable) |
| E4 | CFTC flows (weekly changes), H1 | P | Class flows pin down their own persistence (95% interval narrower than 0.5) | Fail, 3 of 8 (6 required); flow response nonzero in 6 of 8 |
| E5 | CFTC positions, two-root test, H1 | P | Positions carry a second, faster root (memory) distinct from the slow turnover root | Fail, 2 of 8 (6 required); where positive, the second root has a half-life under half a week |
| E6 | Real-noise injection, $K = 2$ | P | Price-only requirement grows as $(\Delta\eta)^{-6}$; tagged requirement stays bounded | Pass: slope $-6.01$ (large-$N$ points use a normal approximation); tagged max/min 2.8 ($\leq 8$ required) |
| E7 | Same test, $K = 3$ (secondary) | S | Price-only requirement grows as $(\Delta\eta)^{-10}$ | Holds: slope $-9.81$ |
| E8 | Binance, 30 fresh markets, (i)–(ii) | P | Class data restrict persistence; price-only intervals wide | Pass, 30 of 30 (one-hour price kernel) |
| E9 | Binance, same test, (iii) | P | Prices pin down the combined response | Fail, 5 of 30; the test failed its 80% criterion |
| E10 | Binance, 30 new markets | P | Horizon-matched: class interval narrower | Pass, 29 of 29 evaluable (median factor 7.3) |
| E11 | Binance, same test, secondary | S | Class persistence is a stable memory | Not supported: close to a random walk after bias correction |
| E12 | Binance, 105-day follow-up | P | Combined response far cheaper to learn than the separate rates | Fail, 10 of 30 (23 required) |
| E13 | Binance, 58-day follow-up, P2 | P | Tagging advantage grows with the sample | Pass, 22 of 29 (22 required) |
| E14 | Same test, P1 | P | Tagged interval shrinks at the parametric rate | Fail, 20 of 29 (22 required) |
| E15 | Binance, return forecasts | P | Class memory states improve 30-minute return forecasts | Fail ($\Delta R^2 = -1.09$ points, inconclusive) |
| E16 | Binance, state tracking | P | Class data track class positions better than prices | Pass, but explained by persistence of the target |
| E17 | CFTC, 12 new markets, H1 (replication) | P | Class positions pin down a persistence that weekly prices leave open | Pass, 12 of 12 (tagged widths 0.014–0.042, price-only 0.85–0.96) |
| E18 | Same, H2 | P | The managed-money or leveraged class trend-follows | Pass, 11 of 12 (Nasdaq-100 has the opposite sign) |
| E19 | Same, H3 | P | That class has the less persistent positions | Fail, 3 of 12 (6 required) |
| E20 | Binance, Oct. 2026, 30 markets, N2 | P | Prices improve out-of-sample forecasts of the top-trader position beyond its own persistence | Fail, 1 of 30 (18 required); median $R^2$ gain $-0.01$ points |
| E21 | Same, N3 | P | Taker flow pins down the combined response to past returns | Fail, 5 of 30 (24 required); interval excludes zero in 17 |
| E22 | Same, N3 secondary | S | The sign of that response is stable within the month | Fail, 21 of 30 (24 required) |
| E23 | Same, N3 secondary | S | Its interval shrinks at the root-$T$ rate | Pass, 30 of 30 (median ratio 1.69, theory 1.73) |

Each row is one recorded outcome. P marks a primary hypothesis or the single criterion of a test; S marks a secondary prediction reported with it. Of the 23 outcomes, 11 passed or held, 11 failed and 1 was not supported (E11); among the 19 primary outcomes, 9 passed and 10 failed; of the 4 secondary predictions, 2 held, 1 failed and 1 was not supported. Counts in the text use this table. The table reports every outcome of the registered and follow-up tests on CFTC and Binance data; none is excluded.

Table S30: Two-point sample complexity for two-cohort weight recovery

| $\Delta\eta$ | $\Delta M_3$ | $\|\Delta h\|_2$ | $N^*_{\mathrm{LeCam}}$ (95%) | $N_{\mathrm{CR}}$ |
|---|---|---|---|---|
| 0.30 | $2.67 \times 10^{-4}$ | $1.98 \times 10^{-3}$ | 7.1 | 47.8 |
| 0.20 | $7.92 \times 10^{-5}$ | $5.60 \times 10^{-4}$ | 88.2 | 597.0 |
| 0.14 | $2.72 \times 10^{-5}$ | $1.89 \times 10^{-4}$ | 775.3 | 5,267.9 |
| 0.10 | $9.90 \times 10^{-6}$ | $6.84 \times 10^{-5}$ | 5,927.6 | 40,374.2 |
| 0.08 | $5.07 \times 10^{-6}$ | $3.49 \times 10^{-5}$ | 22,740.5 | 155,048.9 |
| 0.06 | $2.14 \times 10^{-6}$ | $1.47 \times 10^{-5}$ | 128,333.0 | 875,732.3 |
| 0.05 | $1.24 \times 10^{-6}$ | $8.50 \times 10^{-6}$ | 383,859.1 | 2,620,321.2 |
| 0.04 | $6.34 \times 10^{-7}$ | $4.35 \times 10^{-6}$ | 1,466,361.5 | 10,012,662.4 |

*Worst-case failure at* $\Delta\eta = 0.10$*,* $N = 50$*, 1,000 trials per hypothesis*: bound 44.1%; ML 94.5%, ridge 47.5%, positive-part shrinkage 100%, heavy prior 73.9%.

$\tau = 0.01$, $\sigma_h = 1.6 \times 10^{-3}$, $a = 0.5$, $\kappa = 0.8$, 121 taps. Estimators: ML is least squares from four starts; ridge and heavy prior are posterior modes with prior standard deviations $(0.05, 0.10)$ and $(0.005, 0.02)$ for (weights, rates) around the symmetric centre of the two hypotheses; positive-part shrinkage pulls the ML weights toward their mean. Failure means some weight is off by at least $\tau$.

Table S31: The main symbols in plain words

| Symbol | Meaning |
|---|---|
| $K$ | Number of trader groups ("cohorts") |
| $\eta_i$ | Group $i$'s learning rate: the weight it puts on the newest price. Large $\eta_i$ means a short memory and fast reactions; small $\eta_i$ means a long memory |
| $r_i = 1 - \eta_i$ | Group $i$'s memory rate: the share of last period's view that survives |
| $q_i$ | Group $i$'s trading intensity: how strongly it trades on its view. Positive $q_i$ means it buys when the price is above value, as trend followers do |
| $\Delta\eta$, $2\delta$ | How far apart the groups' learning rates are ($\Delta\eta$ for unknown groups, $2\delta$ for a known pair) |
| $a$, $\kappa$ | How persistent pricing errors are, and how far one unit of net order flow moves the price |
| $M$ (merged measure) | The combined summary of all groups that prices reveal: where the memories are and how much trading sits at each |
| $N$ | Number of experiments (test trades from a fresh start) or observations |
| Par units | The model's own unit of quantity and money, before conversion to USDT (Table S14) |

Table S32: Exact known-pair error: formula versus simulation

| Pulse $V$ | Horizon $T$ | $\Phi(-V\|d_T\|_2/2\sigma)$ | Simulated error, $H_0$ | Simulated error, $H_\delta$ |
|---|---|---|---|---|
| 20 | 3 | 0.3745 | 0.3734 | 0.3762 |
| 40 | 3 | 0.2611 | 0.2599 | 0.2603 |
| 40 | 10 | 0.0029 | 0.0030 | 0.0027 |
| 80 | 20 | $1.8 \times 10^{-10}$ | 0 | 0 |

200,000 paths per hypothesis; $a = 0.5$, $\kappa = 0.8$, $q = 0.2$, $\eta = 0.4$.

Table S33: Gauss and Gauss–Radau rules on $[-1, 1]$ used in Theorem 6.1

| $K$ | Gauss nodes / Gauss–Radau nodes | Max gap, orders $< 2K-1$ | Gap at order $2K-1$ |
|---|---|---|---|
| 2 | $\pm 0.5774$ / $-1$, 0.3333 | $6 \times 10^{-17}$ | 0.2222 |
| 3 | 0, $\pm 0.7746$ / $-1$, $-0.2899$, 0.6899 | $2 \times 10^{-16}$ | 0.0533 |
| 4 | $\pm 0.3400$, $\pm 0.8611$ / $-1$, $-0.5753$, 0.1811, 0.8228 | $4 \times 10^{-16}$ | 0.0131 |

All Gauss–Radau weights are positive.

Table S34: Numerical check of Theorem 6.1

| | $K=2$ | $K=3$ | $K=4$ |
|---|---|---|---|
| Theoretical exponent of $N^*$ | $-6$ | $-10$ | $-14$ |
| Fitted log–log slope (small $\Delta\eta$) | $-5.96$ | $-9.96$ | $-14.01$ |

Le Cam bound $N^* \propto \sigma^2/\|h^A - h^B\|_2^2$ with $\sigma = 1$, $a = 0.5$, $\kappa = 1$, total merged weight 0.10, rates centered at 0.70, responses over 400 lags.

Table S35: Sample requirement of Prony's method against the lower-bound order

| | $K=2$ | $K=3$ | $K=4$ | $K=5$ |
|---|---|---|---|---|
| Lower-bound order $-(4K-2)$ | $-6$ | $-10$ | $-14$ | $-18$ |
| Fitted slope of $N^*$ in $\Delta\eta$ (Prony) | $-5.94$ | $-10.00$ | $-13.93$ | $-18.34$ |

Rates $0.7 + \Delta\eta\, s$ with $s$ the $K$-point Gauss nodes, total weight 0.1, $\Delta\eta \in [0.03, 0.2]$; $N^*$ is the number of experiments for a weight RMSE of 10% of the smallest weight, from 400 simulated estimates per point. At $K = 5$ the fit is numerically sensitive: rerunning the same seeded script with a different numerical library gives slopes $-5.94$, $-10.00$, $-13.92$ and $-17.89$.

Table S36: Targeted versus market-wide intervention: two economies that prices barely separate

| $K$ | $\Delta\eta$ | $\lvert S^A - S^B \rvert$ | $\kappa q^G_{\max}$ | $\kappa q^R_{\max}$ | Window for $\bar{s}$ | $N$ for a correct choice |
|---|---|---|---|---|---|---|
| 2 | 0.20 | $3.9 \times 10^{-3}$ | 0.112 | 0.144 | [0.527, 0.562] | 5 |
| 2 | 0.10 | $5.0 \times 10^{-4}$ | 0.093 | 0.131 | [0.532, 0.570] | 306 |
| 2 | 0.05 | $6.5 \times 10^{-5}$ | 0.086 | 0.125 | [0.536, 0.575] | 18,095 |
| 3 | 0.20 | $2.8 \times 10^{-4}$ | 0.073 | 0.092 | [0.583, 0.603] | 1,092 |
| 3 | 0.10 | $7.8 \times 10^{-6}$ | 0.071 | 0.076 | [0.587, 0.592] | $1.4 \times 10^6$ |

Calibration $a = 0.5$, $\kappa = 0.8$, $W = 0.08$, $r_c = 0.60$, observation noise $\sigma = 0.01$ per tap; G and R are the Gauss and Gauss–Radau economies. Both economies have $S \approx 0.66$–$0.68$. "Window" is the range of thresholds in which halting the largest group restores $S \le \bar{s}$ in one economy and not in the other. The last column is the exact Gaussian-shift requirement for a choice that is correct with probability 3/4 in both.

# S33 A finite-horizon decision that needs only the combined response

A separate note, `paper1_finite_horizon_supplement.pdf` in the public repository, works out one financial decision in full. An unlevered desk that warehouses inventory for $H$ periods under a hard stress budget buys $N$ reset experiments, pays for them out of the same capital, and sizes its position from an upper confidence bound on the summed price response over the horizon. With probability at least $1 - \gamma$ the position meets the stress budget for every bounded shock path; its profit shortfall is at most $AN^{-1/2} + Dc_H N$ for explicit constants $A$ and $D$ and cost $c_H$ per experiment; and a stated inequality guarantees a positive expected net value of the information. These guarantees hold uniformly over the positive cohort class, whatever the number and spacing of the cohorts. Under the same observations, telling apart two particular two-cohort economies whose rates are of order $\delta$ apart requires a number of experiments of order $\delta^{-6}$, so at a fixed information price cohort identification becomes unaffordable while the decision keeps its positive value.

The lesson cuts both ways. Many risk decisions need only the combined response, which public data identify at the ordinary rate; class-level records are valuable only for decisions that depend on attribution, such as the targeted halt of Proposition 7.4. The note's proofs have not been independently reviewed, its numerical illustration is exploratory, and its priority is unconfirmed.

# S34 Reproducibility

Every table, figure and number in this paper is mapped to the notebook or script and the result file that produce it in the reproducibility map, `ceiling_suite/papers/paper1/REPRODUCIBILITY_MAP.md` (also as `paper1_reproducibility_map.pdf`) in the public repository https://github.com/Pratikn03/Fianince, release tag `paper1-ssrn-v1`. The map also lists the replication notebooks in run order and the specification files of the pre-specified tests.